

Comparison of the Revenue Potential of Ground-Mounted PV Installations with Different Orientations

Master Thesis

In the degree program Smart Energy Systems

Submitted by

Abinash Satapathy

Matr.-Nr.: 00169017

on 11 March 2026

at Hochschule Ansbach

First Examiner:

Prof. Dr.-Ing. Georg Rosenbauer

Second Examiner:

Prof. Dr. Mathias Moog

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Abstract

Agrivoltaic systems integrating vertical bifacial photovoltaic modules with agricultural production offer dual land-use solutions for Germany's renewable expansion, yet optimal orientation selection under EEG 2023 frameworks remains inadequately quantified. This thesis conducts a comprehensive economic comparison of vertical bifacial orientations (90° west-facing, 90° east-facing, 30° south-facing) across two capacity configurations (999 kWp versus 1,001 kWp) evaluating 20-year present value revenues under German market and policy conditions.

Employing validated PV*SOL simulation generating hourly profiles for six configurations integrated with 2024 wholesale prices and Marktprämie mechanisms, the analysis examines four scenarios addressing curtailment strategies and policy dependencies. Results demonstrate west-facing system (W90_0.9MW) achieves €1,701,471 present value (+52.1% versus €1,118,703 south-facing baseline) through multiplicative compounding: +7.3% generation, market value factor 0.85 versus 0.70, reduced negative price exposure (3.31% versus 6.48%), and €267,300 policy contribution. East-facing configurations demonstrate catastrophic non-viability (-13.5% generation, 70% performance ratio, 19.6% shading) requiring categorical exclusion. Capacity threshold positioning at 999 kWp creates €269,703 optimization value. Spot revenue dominates outcomes (68-83%) with Marktprämie supplementary (17-32%).

West-facing vertical bifacial emerges as a commercially optimal configuration, achieving the highest Sum of Present Value. Investment priorities rank as: orientation selection (€315,000-350,000 impact) > capacity threshold positioning (€270,000) > curtailment strategy (€13,000-20,000). East-facing configurations require categorical exclusion due to catastrophic performance despite policy support. West-facing systems maintain +28.2% minimum advantage versus baseline across all examined regulatory scenarios, demonstrating robust commercial viability independent of policy evolution.

Contents

Abstract.....	2
List of abbreviations and symbols.....	8
1 Introduction	10
1.1 Initial Situation and Research Problem	10
1.2 Research Objectives and Questions.....	10
1.3 Methodology and Approach	11
1.4 Thesis Structure	11
1.5 Expected Contributions	12
2 Literature Review	13
2.1 Vertical Bifacial Photovoltaic Technology	13
2.1.1 Bifacial Module Technology	13
2.1.2 Vertical Orientation Characteristics	13
2.1.3 Performance Validation and Commercial Deployment	14
2.2 German Electricity Market and EEG Policy Framework.....	14
2.2.1 Market Structure and Price Formation	14
2.2.2 Renewable Energy Sources Act Framework.....	15
2.2.3 Direktvermarktung and Marktprämien-Mechanism.....	15
2.2.4 Negative Pricing Provisions	16
2.3 Agrivoltaics in Germany	17
2.3.1 Dual Land-Use Concept and Rationale	17
2.3.2 German Agrivoltaic Standards and Policy Framework.....	17
2.4 Research Gap and Thesis Contribution	18
3 Methodology.....	20
3.1 Research Design and Approach.....	20
3.1.1 Research Objectives and Framework	20
3.1.2 System Configurations and Comparison Framework.....	21
3.1.3 Temporal and Geographic Scope	22
3.2 System Specifications.....	23
3.2.1 Photovoltaic Module Specifications.....	23
3.2.2 System Configuration Details.....	23
3.2.3 2.3 Inverter Configuration and Balance of System	24
3.3 Generation Modeling.....	25
3.3.1 PV*SOL Simulation Framework	25
3.3.2 Meteorological Data Sources	25

3.3.3	Bifacial Irradiance Modeling.....	26
3.3.4	System Loss Modeling	26
3.3.5	Performance Degradation Modeling	27
3.3.6	Validation and Quality Assurance.....	28
3.4	Price Data and Market Analysis	28
3.4.1	Day-Ahead Market Price Data	28
3.4.2	Monatsmarktwert Solar	29
3.4.3	Price Projection Methodology	29
3.4.4	Temporal Price Pattern Analysis	30
3.4.5	Negative Pricing Analysis	31
3.5	Revenue Calculation Framework	32
3.5.1	Direktvermarktung Revenue Structure.....	32
3.5.2	Anzulegender Wert Determination.....	33
3.5.3	Hourly Revenue Calculation	34
3.5.4	Scenario Framework.....	34
3.6	Present Value Analysis.....	36
3.6.1	Discount Rate Selection	36
3.6.2	Present Value Calculation	36
3.6.3	Integration with Degradation and Price Evolution.....	37
3.6.4	Comparative Present Value Metrics	37
3.7	Analysis Structure and Scenario Application.....	38
3.7.1	Configuration-Scenario Matrix	38
3.7.2	Comparative Analysis Framework	38
3.8	Assumptions and Limitations	39
3.8.1	Core Methodological Assumptions	39
3.8.2	Analytical Scope Limitations	40
3.8.3	Generalizability and External Validity	40
3.8.4	Uncertainty and Risk Factors	40
4	Results.....	42
4.1	Generation Performance Analysis	42
4.1.1	Annual Generation by Configuration	42
4.1.2	Specific Yield and Performance Ratio	42
4.1.3	Temporal Generation Patterns (Monthly Distribution)	43
4.1.4	System Losses and Performance Metrics	45
4.2	Eligible Generation and Negative Price Exposure	45
4.2.1	Actual vs. Eligible Generation Analysis	45
4.2.2	Negative Price Hour Impacts by Orientation	46
4.2.3	Multi-Year Eligible Generation Patterns.....	47
4.2.4	Heatmap Analysis: Eligible Energy Across Configurations and Years	49

4.3	Price-Generation Correlation Analysis.....	49
4.3.1	Correlation Methodology and Plausibility Framework.....	49
4.3.2	Configuration-Specific Correlation Results	51
4.3.3	Correlation Heatmaps: Generation vs. Price by Year	52
4.3.4	Temporal Price-Generation Relationships	53
4.4	Revenue Components Analysis.....	54
4.4.1	Spot Market Revenue	54
4.4.2	Marktpremie Contributions	56
4.4.3	Total Annual Revenue Streams	59
4.4.4	Revenue Decomposition by Component	61
4.5	Present Value Outcomes.....	62
4.5.1	Real Scenarios: Curtailment vs. No Curtailment	62
4.5.2	Basic Scenarios: Uniform Remuneration Framework.....	64
4.5.3	Configuration Rankings and Comparative Performance.....	67
4.6	Orientation Effects.....	67
4.6.1	Generation Volume Effects by Orientation	67
4.6.2	Temporal Distribution and Market Timing Effects.....	68
4.6.3	Revenue Outcome Synthesis	69
4.7	Capacity Threshold Effects and Policy Framework Leverage	70
4.7.1	Orientation-Dependent Threshold Effects.....	70
4.7.2	Capacity Threshold Present Value Analysis	72
4.8	Curtailment Value and Operational Strategy Optimization	73
4.8.1	Curtailment Mechanism and Economic Rationale	73
4.8.2	Temporal Curtailment Evolution and Multi-Year Patterns	74
4.9	Policy Contribution Analysis and Framework Dependency	75
4.9.1	Policy Framework Contribution Quantification	75
4.9.2	Temporal Policy Contribution Evolution	77
4.9.3	Policy Framework Viability Scenarios and Risk Assessment.....	78
4.10	Comparative Summary	81
4.10.1	Configuration Performance Summary Matrix	81
4.10.2	Key Findings and Definitive Recommendations.....	82
5	Discussion	85
5.1	Interpretation of Key Findings	85
5.2	Comparison with Existing Literature	86
5.3	Practical Implications and Deployment Recommendations.....	86
5.4	Limitations.....	87
5.5	Future Research Directions	87

6	Conclusion	89
6.1	Principal Findings.....	89
6.2	Key Contributions	90
6.3	Deployment Recommendations.....	90
6.4	Policy Implications and Future Research	91
6.5	Final Conclusions	92
	References.....	93

List of abbreviations and symbols

AC	Alternating Current
AW	Anzulegender Wert (Reference Tariff Value)
BF	Bifaciality Factor
DC	Direct Current
DHI	Diffuse Horizontal Irradiance
DIN	Deutsches Institut für Normung
DNI	Direct Normal Irradiance
DWD	Deutscher Wetterdienst (German Meteorological Service)
E90	East-facing 90° vertical orientation
E90_0.9MW	East-facing 999 kWp system
E90_1MW	East-facing 1,001 kWp system
EEG	Erneuerbare-Energien-Gesetz (Renewable Energy Sources Act)
EPEX	European Power Exchange
GHI	Global Horizontal Irradiance
HJT	Heterojunction Technology
IRR	Internal Rate of Return
ITRPV	International Technology Roadmap for Photovoltaics
kWp	Kilowatt peak (nominal power capacity)
kWh	Kilowatt hour (energy)
MMW	Monatsmarktwert (Monthly Market Value)
MPPT	Maximum Power Point Tracking
MVF	Market Value Factor
MWh	Megawatt hour
MWp	Megawatt peak
PERC	Passivated Emitter and Rear Cell
PR	Performance Ratio
PV	Photovoltaic / Present Value (context-dependent)
RQ	Research Question (RQ1-RQ5)
S30	South-facing 30° tilted orientation

S30_0.9MW	South-facing 999 kWp system
S30_1MW	South-facing 1,001 kWp system
STC	Standard Test Conditions
TOPCon	Tunnel Oxide Passivated Contact
W90	West-facing 90° vertical orientation
W90_0.9MW	West-facing 999 kWp system
W90_1MW	West-facing 1,001 kWp system

1 Introduction

1.1 Initial Situation and Research Problem

Germany's photovoltaic capacity expansion toward 215 GW by 2030 confronts fundamental land-use constraints, with conventional ground-mounted installations competing with agricultural production for finite land resources. Agrivoltaic systems integrating vertical bifacial photovoltaic modules with ongoing agricultural operations offer dual land-use solutions, accessing enhanced EEG 2023 remuneration (technology bonuses 12.0-25.0 €/MWh above baseline tariffs) while preserving $\geq 80\%$ agricultural productivity per DIN SPEC 91434 standards.

Vertical bifacial technology employs 90° tilt orientation with dual-surface generation capturing front-side irradiance plus rear-side ground reflection and diffuse contributions. Commercial deployments demonstrate technical feasibility (Next2Sun: 40+ MWp operational, Donaueschingen achieving 1,250 kWh/kWp versus projected 1,183 kWh/kWp). However, optimal orientation selection among alternative configurations such as : east-facing, west-facing, or conventional south-facing, remains inadequately quantified under German market and policy conditions.

German wholesale electricity markets exhibit pronounced intra-day price variation: midday periods (11:00-14:00) averaging 50 €/MWh versus evening peaks (17:00-20:00) at 116 €/MWh, representing 131% differential. Negative pricing occurs with increasing frequency (457 hours in 2024-2025, concentrated during solar generation peaks), creating revenue penalties through EEG §51 provisions eliminating Marktprämie during such periods. This temporal price structure creates differentiated economic value depending on generation timing, with conventional south-facing configurations concentrating output during lowest-price periods while alternative orientations may achieve superior market timing.

EEG 2023 establishes capacity-dependent remuneration creating discrete economic discontinuities: installations below 1 MWp access 25.0 €/MWh agrivoltaic bonus, while systems ≥ 1 MWp receive 12.0 €/MWh reduced bonus, establishing a differential at threshold boundary. This capacity threshold structure creates strategic incentives for system sizing optimization.

The interaction between orientation-dependent generation profiles, capacity threshold effects, and market price patterns creates complex optimization landscape inadequately addressed in existing literature. Systematic revenue comparison across vertical bifacial orientations accounting for capacity positioning, market timing, and policy framework dependencies remains absent, preventing evidence-based deployment optimization.

1.2 Research Objectives and Questions

This thesis conducts comprehensive economic evaluation of vertical bifacial photovoltaic orientations (90° west-facing, 90° east-facing, 30° south-facing baseline) under German EEG 2023 agrivoltaic frameworks, addressing four research objectives:

Objective 1: Quantify revenue differentials between alternative orientations operating under identical policy and market conditions over 20-year operational horizons.

Objective 2: Assess capacity threshold effects through comparison of 999 kWp (below regulatory boundary) versus 1,001 kWp (at threshold) systems, evaluating orientation-dependent optimization opportunities.

Objective 3: Evaluate curtailment strategy value under negative pricing conditions and differential exposure across orientations.

Objective 4: Examine interaction effects between orientation choice, capacity positioning, and policy framework parameters.

These objectives translate to specific research questions:

RQ1: What present value revenues do alternative orientations (W90, E90, S30) achieve under German EEG 2023 agrivoltaic remuneration frameworks?

RQ2: How do generation volume, market timing, negative price exposure, and policy leverage contribute to orientation-dependent revenue differentials?

RQ3: What capacity threshold optimization value emerges at 1 MWp boundary for different orientations, and what proportion derives from technical performance versus policy positioning?

RQ4: Under what regulatory evolution scenarios do alternative configurations maintain commercial viability?

RQ5: What investment priority ranking emerges across optimization dimensions (orientation, capacity threshold, inverter sizing, curtailment strategy)?

1.3 Methodology and Approach

The analysis employs comparative revenue framework integrating validated PV*SOL simulation generating hourly production profiles for six configurations (three orientations $\times$ two capacity levels: 999 kWp with 2,196 modules, 1,001 kWp with 2,200 modules). All configurations utilize identical bifacial modules (LUXOR 455 Wp, 80% bifaciality factor, Bifacial, TOPCon n-type) with 11-meter row spacing satisfying DIN SPEC 91434 requirements. Meteorological input derives from 2024 DWD data for Nürnberg station representing Ansbach location (49.3°N).

Revenue calculation combines hourly generation with 2024 wholesale electricity prices and EEG Marktprämie mechanisms, examining four scenarios: Real Curtailment (curtailment during negative prices with full policy support), Real Basic (continuous generation with full policy support), Basic Curtailment (curtailment with uniform 70.3 €/MWh remuneration), Basic Basic (continuous generation with uniform remuneration). Present value analysis employs 4% discount rate over 20-year horizon corresponding to EEG guarantee periods.

1.4 Thesis Structure

Chapter 2 (Literature Review) establishes foundations across vertical bifacial technology, German electricity market structure, EEG policy frameworks, and research gap identification. Chapter 3 (Methodology) details system specifications, simulation procedures, and revenue calculation approaches. Chapter 4 (Results) documents generation performance, market timing effects, revenue decomposition, capacity threshold impacts, and policy dependency across regulatory scenarios. Chapter 5 (Discussion) interprets findings within theoretical frameworks, examines practical

implications for stakeholders, and identifies future research directions. Chapter 6 (Conclusion) synthesizes contributions and establishes definitive deployment recommendations.

1.5 Expected Contributions

The research advances renewable energy economics through empirical validation of market value theory under contemporary German conditions, novel capacity threshold interaction quantification, and integrated optimization framework revealing multiplicative value creation mechanisms. Practical contributions provide actionable deployment guidance for developers, financial institutions, agricultural producers, and policymakers navigating orientation selection, capacity positioning, and operational strategies. Findings inform stakeholder decisions during critical deployment period requiring annual photovoltaic additions exceeding 20 GW.

2 Literature Review

2.1 Vertical Bifacial Photovoltaic Technology

2.1.1 Bifacial Module Technology

Bifacial photovoltaic modules represent a significant advancement in solar energy conversion technology, capable of generating electricity from irradiance incident on both front and rear surfaces, regardless of the angle of irradiance, and unlike conventional monofacial modules that utilize only direct and diffuse irradiance on the front surface, bifacial modules capture additional energy from ground-reflected irradiance and rear-side sky diffuse radiation (Yakubu et al., 2024). This dual-surface generation capability is quantified through the Bi-faciality Factor (BF), which is defined as the ratio of rear-side to front-side power generation efficiency under identical irradiance conditions. Bifacial modules employ diverse cell technologies with varying BF: Monocrystalline PERC (Passivated Emitter and Rear Cell) achieves 60-80%, TOPCon (Tunnel Oxide Passivated Contact) reaches 80-90%, and heterojunction (HJT) technology attains 90-95% (Yakubu et al., 2024). The analyzed system employs modules with 80% Bifaciality Factor (LUXOR 455Wp TOPCon Glass Glass Bifacial)

In recent years, Bifacial technology has transitioned from a niche application to mainstream deployment. According to the International Technology Roadmap for Photovoltaics (ITRPV) 16th Edition, bifacial solar cells comprised approximately 90% of global production in 2024 (VDMA, 2025). This rapid market penetration reflects demonstrated performance advantages and declining manufacturing cost premiums, and depending on the installation parameters, bifacial modules can boost the energy yield of PV power plants by 5% to 25% when compared to monofacial modules with a slightly higher cost (Yakubu et al., 2024). Compared to traditional tilted configurations, vertical orientations (characterized by 90° tilt angles) demonstrate particularly favorable conditions for bifacial gain due to balanced front- and rear-side irradiance exposure and reduced ground shading.

2.1.2 Vertical Orientation Characteristics

Vertical photovoltaic installations deviate fundamentally from conventional south-facing tilted configurations. While traditional German installations employ 30 tilt angle with South facing to maximize annual irradiation capture (Wirth, 2025). Vertical east-west orientations produce characteristic dual-peak generation profiles, with morning generation from east-facing surfaces and afternoon generation from west-facing surfaces, with low midday output, the time of day when south-facing systems achieve peak production (Pike et al., 2021). High-latitude installations field measurements document east-west vertical bifacial systems achieving up to 27% higher daily generation compared to optimally tilted south-facing systems during summer months, with dual-peak generation patterns concentrated during morning and evening hours that better align with demand profiles (*Energy Yield from Vertically Mounted Bifacial Solar Panels*, 2022). This redistribution of electricity generation reduces midday generation concentration while extending productive hours, addressing grid integration challenges associated with conventional south-facing configurations.

The bifacial gain mechanism differs in vertical configurations compared to tilted installations. Tilted bifacial systems primarily capture ground-reflected irradiance on rear surfaces, while the

gains are constrained by rear-side shading from module rows and limited view factors toward ground surfaces (Yakubu et al., 2024). Vertical configurations eliminate rear-side row shading and achieve balanced view factors toward both ground and sky, enabling substantial rear-side contributions to total generation and achieving higher bifacial gains than tilted installations due to balanced irradiance exposure (Yakubu et al., 2024). Geographic and atmospheric factors further influence vertical system performance, with morning-afternoon irradiance asymmetry affecting east versus west generation balance. Atmospheric clarity improvements and fog dissipation typically favor afternoon irradiance quality, contributing to west-facing performance advantages documented in continental European climates (DWD, n.d.).

2.1.3 Performance Validation and Commercial Deployment

Vertical bifacial photovoltaic technology has achieved commercial-scale validation through multiple operational installations in Germany. As of 2024, Next2Sun AG, Germany, has deployed 40 MWp in agrivoltaic applications, with vertical bifacial projects such as 17.4 MWp (Frankfurt Airport, October 2025)(*Next2Sun*, n.d.). The Donaueschingen Agrivoltaic installation (4.1 MWp, Baden-Württemberg) provides extended operational validation and achieved 1,250 kWh/kWp in 2022 compared to an estimated 1,183 kWh/kWp representing 5.67% outperformance (Sachwerte, 2023). This sustained performance record validates simulation methodologies and demonstrates the reliability of vertical bifacial technology under real-world agricultural operating conditions.

Advanced simulation methodologies employ view-factor analysis for rear-side irradiance calculation, accounting for module height, tilt angle, row spacing, and ground surface characteristics rather than simplified albedo-based approximations. The PV*SOL simulation environment utilized in this analysis incorporates these physics-based calculation methods and has been validated against measured data from operational bifacial installations (PV Sol, n.d.), providing methodological confidence for performance projections employed in economic analysis.

Agricultural compatibility represents a critical validation criterion for vertical bifacial agrivoltaic systems. The Donaueschingen installation demonstrates sustained agricultural operations, with hay and wheat production on 90% of land area, while maintaining photovoltaic electricity generation (Sachwerte, 2023). These extended operational records validate dual-use viability assumed in economic analyses and confirm compliance with German agrivoltaic quality standards (DIN SPEC 91434), requiring preservation of $\geq 80\%$ agricultural productivity relative to reference conditions (Trommsdorff, M. et al., 2025). The demonstrated agricultural compatibility addresses land-use conflict concerns while enabling project economics that aggregate both electricity and agricultural revenue streams.

2.2 German Electricity Market and EEG Policy Framework

2.2.1 Market Structure and Price Formation

The German electricity market operates as a liberalized wholesale market that is integrated within the European power exchange system, with day-ahead and intraday markets easing price discovery through continuous trading mechanisms. Day-ahead electricity prices, determined through auction clearing at the EPEX Spot exchange, exhibit pronounced intra-day fluctuations that reflect the interaction between renewable generation patterns, conventional generation dispatch, and demand profiles (SMARD, n.d.). This price volatility creates differentiated economic value for

electricity generation depending on demand patterns, altering the revenue optimization for photovoltaic systems beyond simple energy production maximization.

German wholesale electricity prices demonstrate systematic daily patterns driven by solar photovoltaic generation concentration during midday hours and demand peaks during morning and evening periods. Analysis of 2024 market data reveals average midday prices (11:00-14:00) of 50 €/MWh compared to evening demand peak prices (17:00-20:00) averaging 116 €/MWh (SMARD, n.d.), representing a 131% intra-day price differential. This price suppression during high solar generation periods, termed the "merit order effect" or "cannibalization," intensifies with increasing photovoltaic capacity penetration as marginal generation displaces higher-cost conventional generators during peak solar hours (Lion Hirth, 2013).

Negative wholesale electricity prices occur with increasing frequency during periods when renewable generation exceeds demand and transmission constraints limit export capacity. German electricity markets experienced 300 hours of negative pricing annually during 2023-2024 and 457 during 2024-2025, concentrated during midday hours when solar generation peaks coincide with reduced demand and high wind generation (SMARD, n.d.). Such scenarios carry revenue implications for photovoltaic systems operating under direct marketing mechanisms, as regulatory provisions establish zero or negative remuneration during such periods, creating economic incentives for generation profile diversification away from concentrated midday output.

2.2.2 Renewable Energy Sources Act Framework

The Renewable Energy Sources Act (Erneuerbare-Energien-Gesetz, EEG) constitutes the primary policy instrument driving photovoltaic deployment in Germany, establishing guaranteed remuneration mechanisms and priority grid connection rights for renewable electricity generation. The EEG framework has evolved through successive legislative revisions since its initial implementation in 2000, transitioning from fixed feed-in tariffs toward market-integrated support mechanisms while maintaining investment certainty through 20-year revenue guarantees (Gesetze im internet - EEG, n.d.-f). Current EEG provisions distinguish between statutory tariffs for smaller installations and auction-based remuneration for larger systems, with capacity-dependent thresholds creating discrete economic discontinuities in system economics.

Conventional Photovoltaic ground-mounted installations below 1 MWp capacity access statutory tariffs 7.0 ct/kWh (70 €/MWh) remuneration for 20 years from commissioning, established through legislative definition (Gesetze im internet - EEG, n.d.-c). Auction-based systems (systems larger than 1MWp) receive remuneration corresponding to their winning bid, with March 2023 auction results documenting average clearing prices of 7.03 ct/kWh (70.3 €/MWh) for standard photovoltaic installations, demonstrating effective convergence between statutory and auction pricing for conventional systems (Bundesnetzagentur, 2023).

The EEG framework establishes enhanced remuneration for "special installations" (besondere Solaranlagen), including agrivoltaic systems access supplementary technology bonuses above base tariff rates, with bonus magnitude dependent on system capacity and participation mechanism. Agrivoltaic installations below 1 MWp receive a technology bonus of 2.5 ct/kWh above the base statutory tariff, yielding total remuneration of 9.5 ct/kWh (95 €/MWh) for systems meeting DIN SPEC 91434 quality standards (Gesetze im internet - EEG, n.d.-d). Systems at or above 1 MWp access auction-based remuneration with an agrivoltaic technology bonus of 1.2 ct/kWh, resulting in total remuneration approximating 8.2 ct/kWh (82.3 €/MWh) based on recent auction outcomes (BUZER.de, n.d.). This capacity-dependent bonus structure creates a 12.7 €/MWh

remuneration differential at the 1 MWp threshold for agrivoltaic systems, establishing strong economic incentives for capacity optimization immediately below regulatory boundaries.

2.2.3 Direktvermarktung and Marktprämie Mechanism

German photovoltaic installations exceeding 100 kWp capacity operate under mandatory “Direktvermarktung” provisions, requiring electricity sales through licensed market aggregators rather than fixed-price feed-in arrangements (Bundesnetzagentur, n.d.). This market integration exposes generators to wholesale price volatility while maintaining revenue certainty through the “Marktprämie” mechanism, which compensates the difference between guaranteed tariff levels and actual market revenues (Gesetze im internet - EEG, n.d.-a, n.d.-b). The Marktprämie operates as a monthly settlement, calculated as the difference between the installation's “Anzulegender Wert” (reference tariff) and the “Monatsmarktwert Solar” (monthly market value of solar electricity), multiplied by eligible generation volume.

The Monatsmarktwert Solar represents the generation-weighted average wholesale electricity price achieved by the aggregate German photovoltaic fleet, published monthly by transmission system operators based on day-ahead market outcomes (Netztransparenz.de, n.d.-a). This metric serves as the benchmark for Marktprämie calculations. The Marktprämie mechanism ensures installations receive their guaranteed Anzulegender Wert regardless of wholesale price fluctuations, while enabling revenue upside capture during high-price periods when market revenues exceed guaranteed levels.

Direktvermarktung implementation requires professional interval metering and monthly reporting to grid operators, with Direktvermarkter entities typically charging a service fee for aggregation and market access services (Bundesnetzagentur, n.d.). Despite this additional cost, the Marktprämie mechanism provides revenue stability comparable to fixed feed-in tariffs while maintaining price signal exposure that incentivizes generation timing optimization.

2.2.4 Negative Pricing Provisions

EEG legislation establishes specific provisions governing remuneration during periods of negative wholesale electricity prices, reflecting policy objectives to maintain grid stability during renewable generation surplus conditions. Section 51 of the EEG specifies that Marktprämie payments cease during hours when day-ahead wholesale prices fall below zero €/MWh, with recent amendments (effective 2025) eliminating the previously applicable threshold requiring four consecutive negative-price hours (Gesetze im internet - EEG, n.d.-e). Due to recent changes, any individual hour experiencing negative wholesale prices triggers becomes ineligible for Marktprämie payment for that hour, even though installations receive a negative wholesale market revenue from electricity sales through Direktvermarktung arrangements.

The negative pricing rule affects revenue differently across generation profiles, with south-facing installations experiencing higher negative price exposure due to high generation concentration during midday periods. 2024 market data analysis indicates that negative prices occur during the hours of 11:00-15:00. In this case, alternative orientations that redistribute generation toward morning and evening periods (such as east- or west-facing installations) demonstrate an economic advantage through reduced revenue loss during those hours. The increase in the frequency of negative pricing hours correlates strongly with renewable capacity expansion. Installations may opt to curtail generation during negative price periods to avoid negative revenues.

2.3 Agrivoltaics in Germany

2.3.1 Dual Land-Use Concept and Rationale

Agrivoltaic systems integrate photovoltaic electricity generation with ongoing agricultural production on the same land area, thereby addressing the land-use competition between renewable energy deployment and food production. The technology enables the simultaneous utilization of land for both electricity generation and crop cultivation or livestock grazing, achieving land-use efficiency gains compared to dedicating equivalent areas to single purposes (Trommsdorff, M. et al., 2025). This dual-use approach addresses critical policy tensions in renewable energy expansion, as Germany's ambitious photovoltaic capacity targets require substantial land allocation while agricultural land preservation remains a policy priority under European agricultural frameworks and domestic land protection legislation.

Vertical bifacial photovoltaic configurations are quite suitable for agrivoltaics implementation, as elevated module mounting and vertical orientation minimize ground coverage ratios while maintaining agricultural accessibility. Row spacing of 10-12 meters between vertical module arrays enables standard agricultural machinery operation, conventional crop cultivation practices, and unimpeded livestock movement (Trommsdorff, M. et al., 2025). The vertical orientation reduces direct shading compared to tilted configurations, with east-west oriented systems creating morning and afternoon shade bands that migrate across inter-row spaces rather than persistent shade concentration beneath tilted modules. This shade distribution can provide microclimate benefits for certain crops through reduced water stress and temperature moderation, though crop-specific responses vary and require site-specific evaluation (Trommsdorff, M. et al., 2025).

2.3.2 German Agrivoltaic Standards and Policy Framework

Germany established formal quality standards for agrivoltaic systems through DIN SPEC 91434, published in 2021 by a partnership including the Fraunhofer Institute for Solar Energy Systems ISE, defining technical and agricultural performance requirements for installations claiming agrivoltaic status (Trommsdorff, M. et al., 2025). The standard establishes minimum agricultural productivity thresholds requiring preservation of at least 80% of reference agricultural yield on the photovoltaic-integrated land compared to conventional agricultural use without photovoltaic infrastructure. This quantitative benchmark ensures agrivoltaic installations maintain substantive agricultural function rather than nominal land-use claims, addressing concerns about photovoltaic deployment under agricultural land-use designations without meaningful agricultural productivity continuation.

DIN SPEC 91434 distinguishes between Category I agrivoltaic systems, where agricultural production remains the primary land use, and Category II systems, where electricity generation constitutes the primary function with agricultural use as a secondary activity (Trommsdorff, M. et al., 2025). Vertical bifacial configurations typically qualify as Category I systems due to minimal ground coverage and preserved agricultural machinery accessibility, contrasting with elevated tilted systems that may restrict agricultural operations despite formal agricultural land-use continuation. The standard further specifies monitoring requirements documenting actual agricultural productivity through harvest measurements and livestock carrying capacity assessments, creating verification frameworks that enable enforcement of agricultural productivity maintenance claims.

Integration of DIN SPEC 91434 compliance into German renewable energy policy occurred through EEG amendments establishing enhanced remuneration for qualified agrivoltaic installations, creating economic incentives for dual-use system implementation (Gesetze im internet - EEG, n.d.-d). The technology bonuses discussed in Section 2.2 apply exclusively to installations demonstrating DIN SPEC 91434 compliance through approved certification processes, linking elevated remuneration levels to verified agricultural productivity maintenance.

German agrivoltaics deployment has progressed from pilot demonstration to commercial-scale implementation, with vertical bifacial configurations achieving market penetration due to agricultural compatibility advantages and favorable economics under enhanced EEG remuneration. As discussed in Section 1.3, Next2Sun Donaueschingen installation validates simulation methodologies employed in project development and establish performance benchmarks for comparable installations. The installation achieves DIN SPEC 91434 compliance through documented agricultural productivity maintenance, confirming that properly designed vertical systems fulfill both technical generation requirements and agricultural productivity standards.

2.4 Research Gap and Thesis Contribution

The reviewed literature establishes fundamental frameworks for understanding photovoltaic system economics under variable pricing conditions, validates vertical bifacial technology maturity, and documents German policy mechanisms supporting renewable energy deployment. Hirth's market value theory provides a theoretical foundation demonstrating that generation timing determines economic value independent of absolute energy quantity (Lion Hirth, 2013), while empirical evidence confirms an accelerating solar value decline in German markets as capture rates deteriorate with increasing penetration (Jomaux, 2023). Vertical bifacial technology has achieved commercial validation through operational installations demonstrating performance meeting or exceeding projections (*Next2Sun*, n.d.; Sachwerte, 2023), and German agrivoltaic policy frameworks create enhanced remuneration for dual-use systems meeting defined agricultural productivity standards (Gesetze im internet - EEG, n.d.-d; Trommsdorff, M. et al., 2025).

Despite this foundational knowledge, existing literature exhibits limited systematic comparison of revenue outcomes across alternative photovoltaic orientations under contemporary German market and policy conditions. While Pike et al. (2021) demonstrate comparable annual energy production between vertical east-west and south-facing bifacial systems in Arctic conditions, with fundamentally different temporal profiles (Pike et al., 2021), the economic implications of this temporal redistribution remain unexplored in German policy contexts where capacity thresholds create discrete remuneration discontinuities and Marktprämie mechanisms expose producers to wholesale price variation. Similarly, though agrivoltaic technology bonuses substantially elevate remuneration levels relative to conventional installations (BUZER.de, n.d.; Gesetze im internet - EEG, n.d.-d), interaction effects between orientation-dependent generation profiles and capacity-dependent bonus structures have not been quantified in revenue optimization frameworks.

The 1 MWp capacity threshold distinguishing statutory from auction-based remuneration creates minimal economic impact for conventional south-facing installations, as statutory tariffs (70 €/MWh) approximate auction clearing prices (70.3 €/MWh) (Bundesnetzagentur, 2023; Gesetze im internet - EEG, n.d.-c). However, agrivoltaic installations experience substantially larger threshold effects, with small-scale statutory bonuses (25 €/MWh) exceeding large-scale auction bonuses (12 €/MWh) by 12.7 €/MWh remuneration differential (BUZER.de, n.d.; Gesetze im

internet - EEG, n.d.-d). This capacity-dependent bonus structure creates strategic incentives for system sizing below regulatory boundaries, yet literature has not examined the interaction between this capacity threshold effect and orientation-dependent market value capture, which remains entirely unexplored.

This thesis addresses these gaps through systematic revenue comparison of vertical bifacial photovoltaic systems across three orientations (90° vertical west-facing, 90° vertical east-facing, 30° south-facing) and two capacity configurations (999 kWp below threshold, 1,000 kWp at threshold) under German EEG agrivoltaic remuneration frameworks and contemporary wholesale electricity market conditions. The analysis employs validated simulation methodologies (PV Sol, n.d.) generating hourly generation profiles for six system configurations, combined with 2024 German day-ahead market price data and Marktprämie mechanisms (Gesetze im internet - EEG, n.d.-a, n.d.-b) to calculate 20-year present value revenues. This approach enables quantification of three distinct effects: orientation-driven market value differentiation through temporal price alignment, capacity threshold impacts on remuneration levels, and interaction effects between orientation choice and threshold positioning.

The research contributes empirical evidence on revenue optimization strategies for agrivoltaic photovoltaic installations under policies creating capacity-dependent remuneration discontinuities, extending market value theory application to contexts where policy thresholds interact with wholesale price exposure. Findings inform system design decisions for developers navigating orientation-capacity trade-offs, provide evidence for policy evaluation of threshold-based incentive structures, and demonstrate economic implications of generation timing optimization in markets experiencing solar value deterioration. The analysis further validates vertical bifacial agrivoltaic technology economics under representative German conditions, contributing operational evidence to limited literature on alternative orientation revenue performance.

3 Methodology

3.1 Research Design and Approach

3.1.1 Research Objectives and Framework

This research employs a comparative revenue analysis framework to evaluate economic performance of ground-mounted bifacial photovoltaic systems across alternative orientations and capacity configurations under German EEG agrivoltaic policy frameworks. The analysis addresses four interrelated research objectives:

The analysis addresses four interrelated research objectives:

- (A) quantification of revenue differentials between vertical east-facing, vertical west-facing, and conventional south-facing orientations operating under identical policy and market conditions,
- (B) assessment of capacity threshold effects on total revenue through comparison of systems sized at 999 kWp (below 1 MWp regulatory boundary) versus 1,000 kWp (at threshold),
- (C) evaluation of curtailment strategy value under negative pricing conditions and differential exposure across orientations, and
- (D) examination of interaction effects between orientation choice, capacity positioning, and operational strategies relative to policy thresholds and market price patterns.

The methodological approach integrates three distinct analytical components. First, hourly photovoltaic generation profiles for six system configurations are generated through validated simulation modeling (Bifacial Modules (Calculation) :: PV*SOL® Help, n.d.), incorporating site-specific meteorological data, bifacial module characteristics, and system-specific mounting configurations. Second, revenue realization frameworks are calculated by combining simulated generation profiles with generated wholesale electricity price data from historical data and EEG Marktprämie mechanisms (Section 19 EEG 2023 - Individual Provision, n.d.; Section 20 EEG 2023 - Individual Provision, n.d.), accounting for temporal price variation and policy-specific remuneration rules including negative pricing provisions (Section 51 EEG 2023 - Individual Provision, n.d.). Third, multi-year revenue streams are converted to comparable present value terms through discounting procedures, enabling direct economic comparison across configurations with potentially differing temporal revenue profiles.

The analysis employs a scenario-based approach to isolate distinct economic effects and test revenue sensitivity to operational and pricing assumptions. All scenarios examine revenue outcomes under alternative curtailment and pricing conditions: a "Curtailment" scenario where generation is curtailed during negative price hours (day-ahead price < 0 €/MWh), resulting in zero spot market revenue for affected hours. The curtailment scenario represents economically rational operational behavior under German Direktvermarktung rules, as installations can voluntarily curtail during negative price periods to avoid negative spot revenues while maintaining revenue optimization (Section 19 EEG 2023 - Individual Provision, n.d.; Section 20 EEG 2023 - Individual Provision, n.d.). Under EEG provisions, hours experiencing negative day-ahead prices are ineligible for Marktprämie compensation, and installations receive only spot market revenue (which is negative) during such hours, with no Marktprämie payment to offset losses (Section 51 EEG 2023 - Individual Provision, n.d.). The Marktprämie is calculated monthly as:

$$\text{Marktprämie} = (\text{Anzulegender Wert} - \text{Monatsmarktwert}) \times \text{Eligible Generation}$$

, where Eligible Generation excludes all generations during negative price hours, and Marktprämie is paid only when $(\text{Anzulegender Wert} - \text{Monatsmarktwert}) > 0$. If the Monatsmarktwert exceeds the Anzulegender Wert, no Marktprämie is paid as spot market revenues already exceed guaranteed remuneration levels.

Curtailment during negative price hours therefore generates higher total revenue than continuous generation by avoiding negative spot revenues that would reduce total income. This creates strong economic incentives for curtailment, with the revenue advantage varying across orientations based on differential negative price exposure. The scenario structure enables quantification of curtailment value and assessment of how negative price exposure and corresponding curtailment benefits vary across orientations with differing temporal generation profiles.

3.1.2 System Configurations and Comparison Framework

Six distinct photovoltaic system configurations form the comparison framework, defined through systematic variation of two design parameters: module orientation and total installed capacity. Three orientations represent alternative deployment strategies: 90° vertical tilt west-facing, 90° vertical tilt east-facing, and 30° tilt south-facing serving as conventional baseline configurations. Two capacity levels examine threshold effects: 999.18 kWp utilizing 2,196 modules positioned immediately below the 1 MWp regulatory boundary, and 1,001 kWp employing 2,200 modules positioned just above the threshold. The factorial combination of three orientations and two capacities generate six configurations enabling both orientation comparison (holding capacity constant) and capacity comparison (holding orientation constant).

System nomenclature employs descriptive identifiers specifying orientation characteristics and capacity: "EW90" designates east-west oriented systems with 90° vertical tilt and 2,200 modules, while "S30" indicates south-facing 30° tilt configurations. Capacity is denoted through explicit specification as either "1MW" (1,001 kWp, 2,200 modules) or "0.9MW" (999.18 kWp, 2,196 modules). The complete set comprises:

- W90_1MW (west 1,001 kWp)
- W90_0.9MW (west 999 kWp)
- E90_1MW (east 1,001 kWp)
- E90_E_0.9MW (east 999 kWp)
- S30_1MW (south 1,001 kWp)
- S30_0.9MW (south 999 kWp)

All configurations employ identical photovoltaic module specifications to isolate orientation and capacity effects from technology differences. Bifacial crystalline silicon modules with 80% bifaciality factor and 455 Wp nominal power under standard test conditions, and n-type cell technology (LUXOR 455Wp TOPCON GLASS-GLASS BIFACIAL) were used in all configurations. Module dimensions, electrical characteristics, and temperature coefficients remain constant across configurations, ensuring that performance differences derive exclusively from orientation-dependent irradiance patterns rather than equipment variation. Ground-mounted installations with 11m row spacing satisfy DIN SPEC 91434 agrivoltaic requirements (Trommsdorff, M. et al., 2025) and enable agricultural machinery operation between module rows.

The comparative framework enables isolation of specific economic effects through paired comparisons. Orientation effects are quantified through comparison of systems with identical capacity

but differing orientation (e.g., W90_0.9MW versus S30_0.9MW), holding policy remuneration constant while allowing market revenue to vary with temporal generation profiles. Capacity threshold effects emerge through comparison of systems with identical orientation but differing capacity (e.g., W90_0.9MW versus W90_1MW), holding generation timing constant while allowing policy remuneration to differ based on threshold positioning. Interaction effects are assessed through cross-comparisons examining whether capacity threshold impacts vary systematically across orientations, revealing potential orientation-dependent optimization strategies for threshold-based system sizing.

3.1.3 Temporal and Geographic Scope

The analysis employs a 20-year operational period corresponding to German EEG remuneration guarantee durations, with year-one revenue calculation based on 2026 pricing and generation conditions and subsequent years incorporating annual performance degradation and price evolution assumptions. This temporal scope aligns with standard photovoltaic project financing horizons and policy guarantee periods, enabling economic evaluation relevant to actual investment decision timeframes.

Generation modeling utilizes 2024 meteorological data from Deutscher Wetterdienst (DWD) monitoring infrastructure. While Ansbach (49.3°N, 10.6°E) serves as the reference location for this analysis, complete meteorological data for this specific site were unavailable in the DWD database. Data from the Nürnberg monitoring station was therefore employed as representative input, given Nürnberg's proximity to Ansbach (approximately 43 km distance) and comparable geographic and climatic characteristics. Quality assessment of the Nürnberg station data confirmed excellent measurement completeness, calibration validation, and data reliability throughout the 2024 reference year, supporting its use as representative meteorological input for PV*SOL simulations. The 2024 dataset serves as representative conditions for year-one baseline, with identical annual generation patterns replicated across subsequent years, subject only to degradation adjustments. This representative-year approach assumes weather conditions and irradiance patterns remain statistically consistent across operational lifetime, abstracting from inter-annual meteorological variation to focus analysis on orientation and capacity effects rather than weather uncertainty. The approach follows standard practice in photovoltaic economic evaluation (Schindele et al., 2020) and enables controlled comparison by holding meteorological conditions constant while varying only system design parameters.

Wholesale electricity price data derives from 2024 German day-ahead market outcomes, projected forward to 2026 commissioning year, and scaled to subsequent years through price evolution assumptions maintaining seasonal and daily patterns while adjusting absolute price levels. This projection methodology preserves the temporal price structure determining generation timing value while incorporating anticipated price trajectories. The approach balances empirical price pattern grounding against long-term forecast requirements, recognizing that precise future price prediction is impossible while temporal price patterns (midday suppression, afternoon peaks) persist given fundamental supply-demand dynamics driving current patterns

3.2 System Specifications

3.2.1 Photovoltaic Module Specifications

All system configurations employ LUXOR TOPCon Glass-Glass Bifacial modules, representing current commercial n-type bifacial technology deployed in German agrivoltaic installations. The modules feature 80% bifaciality factor ($\psi = 0.80$), indicating that rear-side power generation under equivalent irradiance conditions reaches 80% of front-side output. Module electrical characteristics include 455 Wp nominal power under standard test conditions (STC: 1,000 W/m² irradiance, 25°C cell temperature, AM 1.5 spectrum), with front-side efficiency of 22.79%. Temperature coefficients follow typical crystalline silicon values: -0.29%/°C for power, -0.25%/°C for voltage, and +0.45%/°C for current. These thermal performance characteristics influence actual generation under operational conditions where cell temperatures exceed STC reference values, particularly during summer afternoon periods when west-facing systems generate peak output.

The module dimensions are 1,762 mm height $\times$ 1,134 mm width, with a 30 mm thickness including the frame, enabling standard mounting hardware compatibility and transportation logistics. Glass-glass construction with dual-sided anti-reflective coatings provides mechanical durability for ground-mounted applications and optimizes light transmission for both front and rear surfaces. Frameless or minimal-frame designs reduce shading impacts on rear-side irradiance capture, particularly important for vertical installations where ground-reflected and diffuse rear-side contributions constitute substantial generation components.

3.2.2 System Configuration Details

The six system configurations employ two distinct capacity specifications: 999.18 kWp utilizing 2,196 modules and 1,001 kWp utilizing 2,200 modules. The 999.18 kWp capacity positions immediately below the 1 MWp regulatory threshold, distinguishing small-scale statutory remuneration from large-scale auction-based frameworks under EEG provisions (Bundesnetzagentur, 2023; Gesetze im internet - EEG, n.d.-c). This size maximizes capacity while preserving access to enhanced statutory agrivoltaic bonuses (+25 €/MWh) that apply exclusively to installations below 1 MWp (Gesetze im internet - EEG, n.d.-d). The 1,001 kWp capacity represents the minimum threshold crossing, accessing auction-based remuneration with reduced agrivoltaic bonuses (+12 €/MWh) but enabling comparison of threshold positioning effects (BUZER.de, n.d.).

Module mounting configurations vary systematically across orientations while maintaining identical structural principles. Vertical installations (90° tilt) utilize fixed single-axis tracking mounting structures oriented with either west-facing or east-facing, with modules installed to maximize vertical dimension and minimize ground coverage ratio. Row spacing of 11 meters between module arrays satisfies DIN SPEC 91434 agricultural accessibility requirements (Trommsdorff, M. et al., 2025) while providing adequate clearance for agricultural machinery operation, including tractors, combines, and hay harvesting equipment. Ground clearance beneath vertical modules reaches 3 meters, preventing crop damage during growth periods and enabling grazing animal access.

South-facing configurations employ conventional fixed-tilt mounting at 30°, representing standard German photovoltaic optimization for annual energy maximization. Row spacing for tilted configurations is also 11 meters, which follows shading analysis, ensuring minimal inter-row shading when solar angles reach minimum values for the Ansbach latitude. This establishes a conventional baseline performance against which vertical alternatives are compared.

All mounting structures utilize galvanized steel or aluminum construction, providing a 25-year minimum design lifetime consistent with photovoltaic module warranties and EEG remuneration guarantee periods. Foundation systems employ either driven pile or ballasted concrete, depending on soil conditions, with structural engineering certifications addressing wind load (34 m/s reference wind speed) and snow load (zone-appropriate design values) per German technical standards. Electrical grounding and lightning protection systems follow DIN VDE 0100 and VDE 0185 requirements, ensuring personnel safety and equipment protection throughout operational lifetime.

3.2.3 2.3 Inverter Configuration and Balance of System

All installations are configured with 11 arrays distributed across the site.

- For the 2,200-module configurations (1,001 kWp systems: W90_1MW, E90_1MW, S30_1MW), each of the 11 arrays contains 200 modules with 91 kWp DC capacity per array.
- For the 2,196-module configurations (999 kWp systems: W90_0.9MW, E90_0.9MW, S30_0.9MW), the array distribution consists of 2 arrays with 198 modules each (90.09 kWp per array) and 9 arrays with 200 modules each (91 kWp per array), maintaining the 11-array total while accommodating the reduced module count.

Inverter configurations vary by system capacity and orientation to optimize DC/AC ratio and equipment utilization.

- The 2,200-module installations (W90_1MW, E90_1MW, S30_1MW) employ 6 KACO New Energy inverters with 165 kW power rating each, providing 990 kW total AC capacity.
- The S30_0.9MW configuration utilizes the same inverter arrangement (6 × 165 kW KACO units) despite lower DC capacity.
- The vertical 999 kWp configurations (W90_0.9MW, E90_0.9MW) employ differentiated inverter sizing with 1 KACO inverter at 150 kW rating and 6 KACO inverters at 125 kW rating, totaling 900 kW AC capacity.

These inverter configurations result in DC/AC ratios ranging from approximately 1.0 to 1.1, balancing inverter utilization against clipping losses during peak generation periods.

String configuration connects modules in series to achieve optimal voltage ranges for inverter maximum power point tracking (MPPT) operation. Vertical configurations employ longer strings due to reduced inter-row shading impacts, while south-facing configurations utilize shorter strings, accounting for potential partial shading during low solar angle periods. All configurations employ multiple parallel strings per inverter MPPT input, distributing generation across tracking channels to maintain performance despite localized irradiance variation from clouds or partial shading.

Balance-of-system components include AC combiner infrastructure, medium-voltage transformers for grid connection, metering equipment satisfying EEG billing requirements, and monitoring systems tracking generation and performance parameters. Communication infrastructure enables

remote monitoring supporting Direktvermarktung obligations, including dispatch coordination and generation forecasting. Data acquisition systems record hourly generation, enabling ongoing performance verification throughout the operational lifetime. System losses incorporated in generation modeling includes 1% overall loss and shading losses varying by configuration based on inter-row geometry. These loss assumptions follow PV*SOL validated simulation methodologies (PV Sol, n.d.) and represent current commercial system performance standards.

3.3 Generation Modeling

3.3.1 PV*SOL Simulation Framework

Hourly photovoltaic generation profiles for all six system configurations are generated using PV*SOL Premium software (Valentin Software GmbH), a validated simulation platform widely employed in German photovoltaic project development (PV Sol, n.d.). The software implements comprehensive physical models accounting for solar geometry, atmospheric attenuation, module temperature effects, bifacial irradiance capture, system losses, and inverter performance characteristics. PV*SOL's bifacial modeling capabilities specifically address rear-side irradiance contributions from ground reflection, sky diffuse radiation, and reflected radiation from adjacent module rows, which are critical factors determining vertical bifacial system performance and distinguishing generation profiles across orientations.

The simulation employs hourly time resolution, generating 8,760 data points per configuration, representing each hour of a representative meteorological year. This enables accurate integration with hourly wholesale electricity price data and precise quantification of generation-price correlation effects, determining market value capture. Hourly resolution further captures daily generation pattern variations critical to orientation comparison, as west-facing systems concentrate afternoon generation while south-facing systems peak during midday hours. The modeling approach follows standard German photovoltaic evaluation methodologies validated through operational performance comparison studies (Chudinow et al., 2020; Sachwerte, 2023).

Simulation parameters such as module specifications (bifaciality 80%, efficiency 22.79%, temperature coefficients), inverter characteristics (efficiency curves, maximum power point tracking behavior), and system losses (wiring, soiling, mismatch) are held constant across all configurations except orientation-specific geometry (tilt angle, azimuth) and capacity-dependent module quantities, ensuring that performance & revenue differences isolate orientation and capacity effects rather than confounding factors.

3.3.2 Meteorological Data Sources

PV*SOL simulation accuracy for vertical bifacial configurations has been validated through comparison with operational performance data from commercial installations. Critical meteorological variables employed in simulation include global horizontal irradiance (GHI), diffuse horizontal irradiance (DHI), and ambient temperature, all recorded at hourly intervals throughout 2024. GHI and DHI measurements enable decomposition into direct normal irradiance (DNI) through established algorithms, providing complete irradiance characterization necessary for bifacial performance modeling. Temperature data drives module operating temperature calculations accounting for thermal efficiency losses, particularly significant during high-irradiance afternoon periods when west-facing systems generate peak output. Additional parameters including wind speed,

humidity, and atmospheric pressure support refined temperature modeling and atmospheric transmission calculation.

The 2024 meteorological dataset serves as representative year conditions replicated across all 20 analysis years, with generation adjusted annually only for performance degradation effects. This representative-year approach assumes meteorological conditions remain statistically consistent across the operational lifetime, abstracting from inter-annual weather variation to focus analysis on orientation and capacity effects. The methodology follows standard practice in photovoltaic economic evaluation (Schindele et al., 2020), recognizing that long-term revenue projections inherently incorporate weather uncertainty and that multi-year meteorological averaging produces similar results to single representative year analysis for comparative system evaluation.

3.3.3 Bifacial Irradiance Modeling

PV*SOL implements comprehensive bifacial irradiance modeling accounting for both front-side and rear-side photon flux contributions to total generation. Front-side irradiance calculation employs standard photovoltaic modeling procedures: solar position algorithms determine sun angle relative to module plane, atmospheric transmission models calculate beam and diffuse irradiance components accounting for air mass and clearness index, and transposition models convert horizontal irradiance measurements to module plane-of-array values. These established procedures generate front-side effective irradiance accounting for angle-of-incidence effects, sky diffuse distribution patterns, and ground-reflected contributions visible to front surfaces.

Rear-side irradiance modeling addresses three distinct contribution sources: ground-reflected radiation, sky diffuse radiation visible to rear surfaces, and radiation reflected from adjacent module rows in multi-row installations. Ground reflection contributions depend critically on ground albedo, which is the fraction of incident radiation reflected from surfaces beneath modules. The analysis employs an albedo value of 0.2 representing typical agricultural ground cover conditions (grass, crops, bare soil mixtures), consistent with agrivoltaic deployment contexts and validated vertical bifacial installation observations (Chudinzow et al., 2020). This albedo specification produces rear-side irradiance contributions varying substantially across orientations: vertical systems receive significant ground reflection throughout daylight hours due to favorable ground-view geometry, while tilted systems experience reduced rear-side contributions concentrated during high solar angle periods.

Inter-row reflection effects in multi-row installations create additional rear-side irradiance through radiation bouncing between module rows. This phenomenon particularly affects vertical configurations where module spacing and geometry create favorable conditions for inter-row radiation transfer. PV*SOL's view factor calculations determine the geometric fraction of radiation from adjacent rows reaching rear module surfaces, integrating these contributions with ground and sky diffuse components to generate total rear-side effective irradiance. The bifacial gain resulting from rear-side contributions varies hourly based on solar angle, atmospheric conditions, and orientation-specific geometry, creating temporal generation pattern variations distinguishing orientation alternatives.

3.3.4 System Loss Modeling

System losses incorporated in generation modeling includes 1% overall loss applied in a uniform manner across all configurations and shading losses varying by configuration based on inter-row

geometry. This simplified loss treatment enables consistent comparison across configurations while reflecting typical commercial installation performance characteristics. Temperature-dependent efficiency losses employ detailed thermal modeling, calculating cell operating temperature from ambient temperature, irradiance level, wind speed, and mounting configuration thermal characteristics. The power temperature coefficient (-0.25%/°C) reduces output as cell temperature exceeds standard test condition reference (25°C), with operating temperatures frequently reaching 40-60°C during summer high-irradiance periods. Vertical configurations experience somewhat different thermal regimes than tilted configurations due to altered wind exposure and radiation geometry, though temperature loss impacts affect all orientations and do not substantially bias comparative results.

Shading losses vary systematically across orientations due to differing inter-row geometry and solar angle interactions. Vertical configurations (W90, E90) experience minimal shading losses due to vertical module orientation and adequate row spacing, with shading occurring only during very low solar angle periods near sunrise and sunset when generation contribution is minimal. South-facing tilted configurations (S30) incorporate calculated shading losses during winter months when low solar angles create partial shading from adjacent module rows, though row spacing optimization minimizes these impacts during periods of substantial generation. PV*SOL's shading analysis employs three-dimensional geometry modeling to calculate hourly shading fractions accounting for sun position, array geometry, and inter-row spacing, integrating these calculations into overall generation predictions.

3.3.5 Performance Degradation Modeling

Annual performance degradation of 0.345%/year is applied to generation output across the 20-year analysis period, representing a gradual power output decline from cumulative exposure-related effects. This degradation rate derives directly from the LUXOR 455Wp TOPCon Glass-Glass Bifacial module warranty specifications, which guarantee 99% of rated power at the end of year one and 89% of rated power at the end of year 30. The linear degradation from 99% to 89% over 29 years (10% total degradation) yields an annual degradation rate of 0.345%/year ($10\% \div 29 \text{ years} = 0.345\%/year$). This manufacturer-guaranteed degradation rate represents conservative performance assumptions grounded in actual module warranty commitments rather than field study extrapolations.

Degradation modeling employs linear degradation assumption, with year-N generation calculated through the following formula:

$$\text{Generation_Year}_N = \text{Generation_Year}_1 \times (1 - 0.00345)^{(N-1)}$$

This approach simplifies from more complex degradation models exhibiting initial rapid degradation followed by stable linear decline, as the multi-year averaging effect and present value discounting reduce sensitivity to exact degradation profile shape. Alternative degradation scenarios were not modeled, as degradation rate affects all configurations proportionally and does not bias comparative revenue outcomes across orientations or capacity levels. The degradation assumption does affect absolute present value calculations, but comparative rankings and percentage differentials remain robust to reasonable degradation rate variations.

3.3.6 Validation and Quality Assurance

PV*SOL simulation quality assurance procedures include energy balance verification (ensuring DC generation minus losses equals AC export), plausibility checks on specific yield relative to resource conditions and orientation, and comparison of temporal generation patterns against expected daily and seasonal profiles. Monthly generation totals were verified for consistency with solar resource availability and orientation-specific expectations: west-facing systems exhibit afternoon-weighted generation, east-facing systems show morning concentration, and south-facing configurations demonstrate midday peaks. These pattern validations confirm that orientation-dependent generation timing effects are properly captured in simulation outputs used for subsequent revenue analysis.

Sensitivity analysis has been conducted on critical modeling parameters, including albedo (0.20-0.30 range), bifaciality factor (75-85% range), and system losses ($\pm 1\%$ variation) to assess generation prediction robustness. Results indicated that while absolute generation values vary with parameter assumptions, comparative rankings across orientations and relative performance differentials remain stable. This robustness provides confidence that comparative revenue conclusions derived from simulation outputs do not depend critically on precise parameter calibration, though absolute revenue projections incorporate parameter uncertainty inherent in long-term performance prediction.

3.4 Price Data and Market Analysis

3.4.1 Day-Ahead Market Price Data

Wholesale electricity price data derives from the German day-ahead market operated by EPEX Spot, accessed through the SMARD platform maintained by Bundesnetzagentur. The SMARD platform provides generation-source-specific day-ahead prices for multiple technology categories, including nuclear, biomass, wind onshore, photovoltaics, lignite, hard coal, fossil gas, hydro pumped storage, hydropower, other renewable, other conventional, and wind offshore. The analysis employs a full 2024 photovoltaics-specific hourly day-ahead prices exclusively, constituting the primary revenue source for photovoltaic installations operating under Direktvermarktung frameworks mandated by EEG.

The 2024 dataset comprises 8,784 hourly observations (reflecting leap year structure with February 29). PVSOL generates standard-year hourly profiles of 8,760 hours ($365 \text{ days} \times 24 \text{ hours}$), excluding leap day to enable consistent multi-year replication. The February 29 data (24 hours) is therefore excluded from price analysis to maintain hour-by-hour alignment between generation profiles and price data. Price data quality assessment confirmed completeness across all employed hours with no missing values or apparent data errors.

The 2024 price dataset exhibits characteristic temporal patterns relevant to photovoltaic revenue analysis: systematic diurnal variation with midday price suppression during high solar generation periods, afternoon price peaks corresponding to evening demand ramps, and seasonal patterns reflecting heating and cooling demand variation. Negative pricing episodes (hours where day-ahead prices fall below zero) occurred with substantial frequency during 2024, totaling 457 hours annually (5.2% of total hours). These negative price occurrences concentrate overwhelmingly in summer months, particularly April through August when solar generation reaches seasonal maxima. For example, June 2024 experienced 64 negative price hours, May recorded 78 hours, and

August saw 68 hours, while winter months (January, February, November) exhibited minimal negative pricing (16, 4, and 11 hours respectively). This seasonal concentration reflects fundamental supply-demand dynamics: summer combines maximum solar generation with moderate demand, creating midday oversupply conditions, driving prices below zero.

3.4.2 Monatsmarktwert Solar

The Monatsmarktwert Solar represents the monthly generation-weighted average wholesale electricity price achieved by the aggregate German solar photovoltaic fleet, published monthly by transmission system operators as required by EEG regulations (Netztransparenz.de, n.d.-a). This value serves as the reference market price in the Marktprämie calculation mechanism, determining the premium payment that ensures installations receive their guaranteed Anzulegender Wert remuneration level. The Monatsmarktwert Solar is calculated through the following formula:

$$\text{Monatsmarktwert Solar} = \Sigma(\text{Price}_t \times \text{Generation_Solar}_t) / \Sigma(\text{Generation_Solar}_t)$$

Where, Price_t = day-ahead market price in hour t

$\text{Generation_Solar}_t$ = aggregate solar generation across Germany during hour t .

The summation spans all hours within the calendar month, weighing hourly prices by contemporaneous solar generation to capture the market value specifically relevant to solar generation timing.

For analysis purposes, 2024 aggregate German solar generation data ($\text{Generation_Solar}_t$) is employed identically across all projection years 2026-2045 when calculating future Monatsmarktwert Solar values. This approach recognizes that the Monatsmarktwert calculation structure inherently cancels generation scaling factors. For example, if solar generation were scaled by any multiplicative factor (e.g., $1.3\times$ for capacity expansion scenarios), this factor appears in both numerator and denominator of the Monatsmarktwert formula and therefore cancels algebraically. Using constant 2024 solar generation patterns, therefore simplifies calculation procedures and maintains temporal price pattern preservation that determines generation timing value effects. The projected Monatsmarktwert Solar values incorporate degraded installation-specific generation for each analysis.

3.4.3 Price Projection Methodology

Long-term wholesale electricity price projections employ scaling methodology applied to the empirical 2024 hourly price series (annual average: 78.55 €/MWh, ranging from -135.45 to 936.28 €/MWh), maintaining temporal patterns (daily, weekly, seasonal) while adjusting absolute price levels to reflect anticipated market evolution over the 2026-2045 analysis period. The base 2024 price series is first adjusted to 2026 commissioning year through scaling factors reflecting observed or projected price changes between reference year and operational start (84.39 €/MWh annual average in 2026, representing a modest 7.4% increase). Subsequent years (2027-2045) apply annual price evolution assumptions, maintaining the temporal price structure that determines generation timing value while incorporating anticipated long-term price trajectories, reaching 124.15 €/MWh by 2045, which is a total increase of 58% over the analysis period.

The price projection methodology prioritizes preservation of temporal price patterns over absolute price level precision, recognizing that orientation-dependent revenue differentials derive primarily from relative price variation across hours rather than absolute price magnitudes. Maintaining

empirical daily patterns (morning ramps, midday suppression, afternoon peaks) and seasonal variation enables accurate representation of generation timing values, distinguishing alternative orientations. The 2024 baseline data exhibits pronounced intraday variation, with prices ranging from a minimum average of 46.93 €/MWh at 13:00 to a maximum of 126.87 €/MWh at 18:00, a 2.70-fold difference that underscores the significance of generation timing. The methodology preserves this temporal structure with high fidelity across the projection period, maintaining correlation coefficients above 0.92 between baseline and future-year hourly patterns. Absolute price level projections incorporate greater uncertainty but affect all configurations proportionally, preserving comparative revenue rankings while influencing absolute present value calculations.

Negative pricing frequency and magnitude projections recognize the evolving nature of negative price hours as renewable energy penetration expands. The 2024 baseline exhibits 457 hours (5.22% of total hours) with negative prices, averaging -11.50 €/MWh during negative periods, with a minimum observed price of -135.45 €/MWh. Projections indicate market adjustment following the 2024 baseline, with negative price frequency stabilizing in the 1.5-3.6% range through the 2026-2045 period, while the magnitude of negative prices increases progressively, with minimum prices reaching -143.52 €/MWh by 2045. This pattern particularly affects midday periods when solar generation concentration depresses prices below zero, with midday negative price occurrence (11:00-15:00) ranging from 4.2-8.2% of midday hours across the projection period, compared to 17.5% in the 2024 baseline. The analysis conservatively projects these negative price patterns forward from 2024 empirical observations, recognizing that continued solar capacity expansion influences both frequency and depth of negative price events, while acknowledging uncertainty in future negative price evolution depending on storage deployment, demand flexibility development, and market rule modifications. Price volatility metrics remain relatively stable across the projection period, ensuring that relative price patterns determining orientation value remain robust.

3.4.4 Temporal Price Pattern Analysis

Empirical analysis of 2024 hourly price data reveal systematic temporal patterns determining orientation-dependent market value capture. Midday hours (11:00-15:00) exhibit average prices of approximately 50 €/MWh (median: 55.48 €/MWh), suppressed 36.0% below the daily average of 78.55 €/MWh due to concentrated solar generation from the existing south-facing fleet. The midday suppression is most pronounced at 13:00, when average prices reach their daily minimum of 46.93 €/MWh. Afternoon hours (15:00-19:00) demonstrate elevated average prices of approximately 98-121 €/MWh depending on specific hour selection, reflecting demand ramps coinciding with declining solar output and creating scarcity-driven price increases. The afternoon period exhibits particularly strong price escalation from 15:00 (67.61 €/MWh) through 18:00 (126.87 €/MWh), with the 17:00-19:00 window averaging 120.72 €/MWh. During this dramatic afternoon climb, prices increased by 170.3% from the midday minimum (13:00) to the evening peak (18:00), creating a substantial revenue incentive for generation timing optimization through orientation selection.

Morning hours (06:00-10:00) exhibit intermediate price levels of approximately 85 €/MWh (median: 84.44 €/MWh), positioned between midday suppression and afternoon peaks, with average prices varying seasonally based on heating demand and morning demand ramps. The morning period demonstrates characteristic price ramping behavior, with prices increasing 16.5% from 05:00 (80.22 €/MWh) to 07:00 (93.48 €/MWh) as demand awakens, before declining into the

midday suppression period with a 22.7% decrease from 10:00 (60.68 €/MWh) to 13:00 (46.93 €/MWh) as solar generation intensifies.

Evening and nighttime hours demonstrate price volatility dependent on wind generation availability and demand patterns, though these periods contribute minimally to photovoltaic revenue given limited or absent generation. Weekend price patterns differ systematically from weekday patterns due to reduced industrial demand, with weekday prices averaging 82.70 €/MWh compared to weekend prices of 68.15 €/MWh, a 21.4% difference. However, weekend-weekday differentials affect all orientations similarly and do not substantially bias comparative revenue outcomes.

Seasonal price variation exhibits summer-winter differentials driven by heating versus cooling demand patterns, with winter prices (December-February) averaging 82.48 €/MWh in the 2024 baseline, elevated 11.0% above summer prices (June-August: 74.28 €/MWh) due to heating demand and reduced solar availability. Autumn prices (September-November) demonstrate the highest seasonal average at 93.06 €/MWh, reflecting shoulder-season market dynamics. The winter-summer price differential exhibits progressive amplification across the projection period, with the winter/summer ratio increasing from 1.11 (2024) to 1.26 (2026), 1.31 (2030), 1.36 (2040), and ultimately 1.70 (2045), driven by heating electrification trends that disproportionately elevate winter price levels. However, seasonal price differentials interact with seasonal generation patterns, where summer provides maximum absolute generation but also experiences relative price suppression due to solar output concentration, while winter offers elevated prices but reduced generation due to lower solar irradiance and shorter daylight periods. These seasonal interaction effects influence annual revenue totals and vary across orientations based on seasonal generation distribution differences between vertical and tilted configurations, with vertical orientations capturing relatively greater winter generation when prices are increasingly favorable.

The empirical 2024 hourly price data demonstrate substantial price volatility (standard deviation: 52.79 €/MWh, coefficient of variation: 67.2%), with this volatility structure maintained across the projection period (coefficient of variation ranging 49.5-67.2% from 2026-2045), ensuring that the temporal patterns driving orientation-dependent revenue differentials remain prominent throughout the analysis timeframe. The peak-to-trough hourly price ratio of 2.70 in the 2024 baseline (minimum: 46.93 €/MWh at 13:00; maximum: 126.87 €/MWh at 18:00) persists in scaled form across projected years, maintaining the economic incentive structure favoring generation timing optimization through strategic orientation selection.

3.4.5 Negative Pricing Analysis

Negative day-ahead prices in 2024 concentrate overwhelmingly in midday periods (11:00-15:00 hours) when solar generation reaches maximum output, with 85-90% of negative price hours occurring within this four-hour window. The temporal concentration of negative prices creates uneven exposure across photovoltaic orientations: south-facing systems generating predominantly during midday hours face maximum negative price exposure, while east- and west-facing systems concentrating generation in the afternoon period experience substantially reduced exposure, and east-facing systems demonstrate intermediate exposure concentrated in morning periods when negative prices occur less frequently.

The seasonal distribution of negative pricing demonstrates strong summer concentration, with April through August accounting for 75% of 2024 negative price hours. June experienced the highest frequency (64 hours), followed by July (81 hours) and August (68 hours), while May (78

hours) and April (50 hours) contributed substantial additional negative price occurrences. This summer concentration reflects the combination of maximum solar generation and moderate electricity demand characteristic of warm-weather periods. Winter months exhibited minimal negative pricing: January (16 hours), February (4 hours), March (12 hours), September (40 hours), October (25 hours), November (11 hours), and December (8 hours) collectively represented only 116 hours (25% of the annual total).

Projected negative pricing for years 2026-2045 exhibits substantial variation in annual totals, ranging from 134 hours (year 2031) to 334 hours (year 2029), with most years falling in the 150–300-hour range. This variability reflects uncertainty in renewable energy penetration rates, storage deployment, demand flexibility adoption, and market structure evolution. The monthly distribution patterns in projected years maintain the summer concentration observed in the 2024 baseline, with June consistently demonstrating elevated negative price frequency across projection years. Years 2027-2029 show particularly high June negative pricing (91, 94, and 103 hours respectively), while years 2041-2045 exhibit increased annual totals (268, 281, 290, 322, and 312 hours), which implies the potential acceleration of negative pricing as renewable penetration intensifies in later analysis years.

Under current EEG provisions (§51), generation during negative price hours receives no Marktprämie compensation, and the 2025 EEG amendment modified negative price provisions from four-hour consecutive threshold to a one-hour threshold, increasing the frequency of zero-payment periods but simplifying operational response through the elimination of duration requirements. This structure creates strong economic incentives for curtailment during negative price hours, as voluntary curtailment eliminates negative revenue. The analysis quantifies both scenarios, which are continued generation accepting negative revenues, and economically rational curtailment eliminating negative price exposure, to assess curtailment value and differential orientation exposure

3.5 Revenue Calculation Framework

3.5.1 Direktvermarktung Revenue Structure

Revenue realization under German EEG frameworks employs Direktvermarktung mechanisms mandated for installations exceeding 100 kWp capacity (Bundesnetzagentur, n.d.). All analyzed configurations (999 kWp and 1,001 kWp) therefore operate under Direktvermarktung, selling generation into wholesale electricity markets and receiving Marktprämie payments compensating the difference between guaranteed remuneration levels (Anzulegender Wert) and achieved market revenues. Total revenue comprises two distinct components: spot market revenue from wholesale electricity sales, and Marktprämie payments ensuring minimum remuneration guarantees.

Spot market revenue is calculated as the product of hourly generation and contemporaneous day-ahead electricity prices:

$$\text{Spot Revenue}_t = \text{Generation}_t \times \text{Price}_t$$

Where, Generation_t = AC grid export in hour t (MWh)

Price_t = the photovoltaics-specific day-ahead market price (€/MWh).

This calculation applies to all hours where $\text{Price}_t \geq 0$ €/MWh. For hours experiencing negative prices ($\text{Price}_t < 0$ €/MWh), two scenarios are evaluated:

- continued generation accepting negative spot revenue
- economically rational curtailment setting $\text{Generation}_t = 0$ and therefore $\text{Spot Revenue}_t = 0$.

The Marktprämie mechanism ensures installations receive their guaranteed Anzulegender Wert remuneration level regardless of wholesale market price fluctuations. Monthly Marktprämie is calculated as:

$$\text{Marktprämie} = (\text{Anzulegender Wert} - \text{Monatsmarktwert Solar}) \times \text{Eligible Generation}$$

Where,

Anzulegender Wert represents the guaranteed remuneration level specific to installation size and technology (€/MWh)

Monatsmarktwert Solar represents the monthly generation-weighted average market price for solar installations (€/MWh)

Eligible Generation represents total generation during hours with non-negative day-ahead prices (MWh).

Critically,

- (a) Hours experiencing negative prices are excluded from Eligible Generation, meaning installations receive no Marktprämie compensation for negative-price generation.
- (b) The Marktprämie is paid only when $(\text{Anzulegender Wert} - \text{Monatsmarktwert Solar}) > 0$; when Monatsmarktwert Solar exceeds Anzulegender Wert, no payment is made as spot market revenues already exceed guaranteed levels.

3.5.2 Anzulegender Wert Determination

The Anzulegender Wert (reference tariff) varies based on installation capacity and technology category, creating discrete remuneration differentials at regulatory thresholds. For agrivoltaic installations qualifying under DIN SPEC 91434 compliance (Trommsdorff, M. et al., 2025), two distinct remuneration frameworks apply based on capacity positioning relative to the 1 MWp threshold:

Installations < 1 MWp (999 kWp configurations):

- Base tariff: 70.0 €/MWh (statutory remuneration) (Gesetze im internet - EEG, n.d.-c) - Applies to S30_0.9MW
- Agrivoltaic Bonus: +25.0 €/MWh (Gesetze im internet - EEG, n.d.-d) - Total anzulegender Wert: 95.0 €/MWh - Applies to W90_0.9MW and E90_0.9 MW

Installations ≥ 1 MWp (1,001 kWp configurations):

- Base tariff: 70.3 €/MWh (auction clearing price) (Bundesnetzagentur, 2023) - Applies to S30_1MW
- Agrivoltaic bonus: +12.0 €/MWh (BUZER.de, n.d.) - Total anzulegender Wert: 82.3 €/MWh - Applies to W90_1MW and E90_1 MW

This capacity-dependent remuneration structure creates a 12.7 €/MWh differential for vertical installations between 999 kWp and 1,001 kWp installations, despite minimal capacity difference (0.2% capacity variation). The analysis quantifies revenue impacts of this threshold positioning,

examining whether orientation-dependent market value effects amplify or diminish threshold-driven revenue differentials.

3.5.3 Hourly Revenue Calculation

For each hour t in the analysis period, revenue is calculated through the following procedure:

Step 1: Determine generation and price:

Generation _{t} = Simulated AC export (MWh) from PV*SOL, adjusted for degradation in year N

Price _{t} = Photovoltaics-specific day-ahead market price (€/MWh)

Step 2: Calculate spot revenue:

- **Curtailement scenario :**

If Price _{t} < 0: Spot Revenue _{t} = 0 (curtailment)

If Price _{t} ≥ 0: Spot Revenue _{t} = Generation _{t} × Price _{t}

- **No-curtailement scenario :**

Spot Revenue _{t} = Generation _{t} × Price _{t} (all hours, including negative)

Step 3: Calculate monthly Marktprämie:

Curtailement & No-curtailement scenario :

Eligible Generation_{month} = Σ (Generation _{t}) for all hours where Price _{t} ≥ 0

Marktprämie_{month} = max(0, (Anzulegender Wert - Monatsmarktwert Solar) × Eligible Generation_{month})

[Note: Under current EEG §51, negative-price hours are ineligible for Marktprämie regardless of operational choice (Gesetze im internet - EEG, n.d.-e). The no-curtailement scenario therefore receives Marktprämie based only on eligible (non-negative price) generation, but accepts negative spot revenues during excluded hours.]

Step 4: Calculate total revenue:

$$\text{Total Revenue}_{\text{month}} = \Sigma(\text{Spot Revenue}_{\text{t}}) + \text{Marktprämie}_{\text{month}}$$

Annual revenue for each configuration and scenario is calculated by summing monthly revenues across the 12-month calendar year. Multi-year projections (2026-2045) incorporate three dynamic factors: Performance degradation, Price evolution & Monatsmarktwert Solar calculation. Annual revenue streams (2026-2045) are subsequently converted to present value through discounting procedures detailed in Section 6, enabling comparable economic evaluation across configurations with differing temporal revenue profiles.

3.5.4 Scenario Framework

Four distinct scenarios are evaluated to isolate specific economic effects and test revenue sensitivity to operational strategies and policy framework variations:

Real_Curtailement Scenario:

- **Operational plan:** Curtail when Price _{t} < 0 €/MWh
- **Spot revenue:** Zero during negative price hours (voluntary curtailment)
- **Marktprämie:** Based on eligible (non-negative price) generation only

- **Anzulegender Wert:**
 - 999 kWp installations: 70.0 €/MWh for S30 and 95.0 €/MWh for W90, E90
 - 1,001 kWp installations: 70.3 €/MWh for S30 and 82.3 €/MWh for W90, E90
- **Rationale:** Economically optimal operational strategy under EEG §51, representing realistic revenue maximization with full agrivoltaic policy benefits

Real_No_Curtailment Scenario:

- **Operational plan:** Generate during all hours regardless of price
- **Spot revenue:** Accept negative revenues during negative price hours
- **Marktprämie:** Based on eligible (non-negative price) generation only
- **Anzulegender Wert:**
 - 999 kWp installations: 70.0 €/MWh for S30 and 95.0 €/MWh for W90, E90
 - 1,001 kWp installations: 70.3 €/MWh for S30 and 82.3 €/MWh for W90, E90
- **Rationale:** Baseline revenue without active curtailment management, quantifying curtailment value through comparison with the Real_Curtailment scenario

Basic_Curtailment Scenario:

- **Operational plan:** Curtail generation when $\text{Price}_t < 0$ €/MWh
- **Spot revenue:** Zero during negative price hours (voluntary curtailment)
- **Marktprämie:** Based on eligible (non-negative price) generation only
- **Anzulegender Wert:**
 - All six installations: 70.3 €/MWh uniform (base tariff only, agri-PV bonus excluded)
- **Rationale:** Isolates orientation-dependent market value effects by eliminating capacity threshold and agrivoltaic bonus differentials, testing whether W90 configurations achieve superior revenue compared to S30 configurations based purely on generation timing optimization without policy advantage

Basic_No_Curtailment Scenario:

- **Operational plan:** Generate during all hours regardless of price
- **Spot revenue:** Accept negative revenues during negative price hours
- **Marktprämie:** Based on eligible (non-negative price) generation only
- **Anzulegender Wert:**
 - All six installations: 70.3 €/MWh uniform (base tariff only, agri-PV bonus excluded)
- **Rationale:** Baseline scenario under a uniform remuneration framework, isolating orientation effects without curtailment optimization or agrivoltaic policy differentiation

Scenario Comparison Framework:

The four-scenario structure enables systematic decomposition of revenue effects:

- a) **Curtailment Value Quantification:**
 - Real_Curtailment vs. Real_No_Curtailment:

Measures the economic benefit of active negative price avoidance under a full policy framework

- Basic_Curtailment vs. Basic_No_Curtailment:

Measures curtailment value under uniform remuneration

b) Policy Framework Effects:

Real vs. Basic scenarios:

- Isolates revenue contribution of agrivoltaic bonuses and capacity threshold positioning
- Quantifies whether policy advantages amplify or diminish orientation-dependent revenue differentials

c) Orientation Optimization:

- Orientation comparisons within each scenario:

Tests the generation timing value across operational and policy contexts

- Basic scenarios specifically test the hypothesis that W90 achieves superior revenue through market timing alone, independent of agrivoltaic bonus advantages

d) Capacity Threshold Effects:

- 999 kWp vs. 1,001 kWp comparisons in Real scenarios:

Quantifies the impact of the Agri-PV bonus difference

- Threshold effects eliminated in Basic scenarios (uniform 70.3 €/MWh), enabling pure orientation comparison

3.6 Present Value Analysis

3.6.1 Discount Rate Selection

Multi-year revenue streams generated across the 20-year analysis period (2026-2045) require conversion to comparable present value terms to enable economic comparison across configurations. The analysis employs a 4% annual discount rate, representing the time value of money and opportunity cost of capital for photovoltaic infrastructure investments in Germany. This discount rate represents a moderate assumption within the range of rates employed in German renewable energy economic evaluation, falling between conservative low-risk assumptions (2-3%) and higher commercial project hurdle rates (5-7%). This rate incorporates implicit inflation adjustment, representing a real discount rate that accounts for expected long-term inflation while maintaining consistent Euro-denominated value comparisons.

3.6.2 Present Value Calculation

Present value conversion employs standard discounting formulas applied to annual revenue streams. For each configuration and scenario, the 20-year revenue stream (2026-2045) is discounted to 2026 present value through the following calculation:

$$PV = \sum [Revenue_N / ((1 + r)^N)]$$

where: **PV** = Present value (€)

Revenue_N = Annual revenue in year N (€)

r = Discount rate (0.04 = 4%)

N = Year number (1 to 20, where N=1 corresponds to 2026)

The summation across all 20 years produces a total present value representing the net economic value of the 20-year revenue stream expressed in 2026 Euro terms. This present value metric enables direct comparison across configurations. The present value framework, therefore properly accounts for the time preference inherent in long-term economic evaluation.

3.6.3 Integration with Degradation and Price Evolution

The Present Value calculation methodology integrates three distinct temporal dynamics affecting revenue realization:

Performance Degradation: Annual generation declines 0.345%/year according to LUXOR warranty specifications, reducing revenue proportionally in later years. The degradation effect compounds with discounting: later-year revenues are both smaller (due to degradation) and more heavily discounted (due to time value). For a 1,000 kWh year-1 generation, year-20 generation reaches 934 kWh (6.6% reduction), which when discounted at 4%, contributes only 427 € to present value per € of year-20 revenue compared to 962 € per € of year-1 revenue.

Price Evolution: Projected electricity prices may increase, decrease, or remain stable across the analysis period depending on market evolution scenarios. Price changes affect all configurations, but orientation-dependent market value effects mean price evolution impacts are not necessarily proportional. If afternoon prices increase relative to midday, west-facing configurations benefit disproportionately.

Discount Weighting: The discount factor $(1 + r)^N$ creates exponential weighting, favoring early-year revenues. Year-1 revenues contribute 96.2% of nominal value to present value ($1/1.04 = 0.962$), while year-20 revenues contribute only 45.6% ($1/1.04^{20} = 0.456$). This weighting structure means early-year performance and revenue, which dominate present value outcomes, with later-year uncertainty having proportionally smaller impact on comparative evaluation. Configurations achieving superior early-year revenues therefore enjoy compounding advantages through both higher absolute revenues and reduced discounting.

3.6.4 Comparative Present Value Metrics

Present value calculations generate absolute revenue values (€) for each configuration and scenario, enabling direct economic comparison. The analysis employs S30_1MW (south-facing 1,001 kWp configuration) as the universal comparison baseline, representing conventional photovoltaic design optimization. All comparative performance metrics are expressed relative to this baseline through percentage differentials calculated as:

$$\text{Percentage Difference} = ((\text{SUM_PV_A} - \text{SUM_PV_S30_1MW}) / \text{SUM_PV_S30_1MW}) \times 100\%$$

where , **SUM_PV_A** represents the sum of present value for the configuration being evaluated

SUM_PV_S30_1MW represents the sum of present value for the S30_1MW baseline installation.

This unified baseline approach enables consistent interpretation across all comparisons within both Real and Basic scenario frameworks, where positive percentages indicate superior performance relative to conventional south-facing design, while negative percentages indicate inferior performance. The S30_1MW baseline selection reflects its status as the highest-capacity conventional orientation, providing a meaningful reference point for evaluating alternative orientation and capacity strategies.

3.7 Analysis Structure and Scenario Application

3.7.1 Configuration-Scenario Matrix

The analysis employs a 6×4 configuration-scenario matrix generating 24 distinct present value outcomes. Six system configurations (W90_1MW, W90_0.9MW, E90_1MW, E90_0.9MW, S30_1MW, S30_0.9MW) are evaluated under four scenarios (Real_Curtailment, Real_No_Curtailment, Basic_Curtailment, Basic_No_Curtailment), with each configuration-scenario combination producing unique revenue streams and corresponding present values. This enables systematic isolation of effects: holding configuration constant while varying scenarios isolates operational and policy effects, while holding scenario constant and varying configurations isolates orientation and capacity effects.

The outcome matrix supports multiple analytical objectives simultaneously. Primary comparisons focus on Real_Curtailment scenario results, representing economically optimal operational strategy under actual policy frameworks and therefore constituting the most relevant revenue projections for deployment decision-making. Secondary comparisons employ alternative scenarios to quantify specific effects: Real_No_Curtailment quantifies curtailment value, Basic scenarios isolate market timing effects from policy advantages, and cross-scenario analysis reveals interaction effects between operational strategies, policy frameworks, and orientation-dependent generation patterns.

3.7.2 Comparative Analysis Framework

Systematic comparison across the configuration-scenario follows a structure designed to build understanding progressively:

Level 1: Primary Configuration Comparison (Real_Curtailment Scenario):

Establishes baseline economic ranking under realistic operational assumptions and full policy framework. Key comparisons include:

- W90_0.9MW vs. S30_0.9MW: West vs. south orientation at 999 kWp capacity
- W90_1MW vs. S30_1MW: West vs. south orientation at 1,001 kWp capacity
- E90_0.9MW vs. S30_0.9MW: East vs. south orientation at 999 kWp capacity
- W90_0.9MW vs. W90_1MW: Capacity threshold effect for west orientation

These comparisons identify which configuration achieves maximum present value and quantify revenue advantages across design alternatives.

Level 2: Curtailment Value Quantification:

Compares Real_Curtailment vs. Real_No_Curtailment for each configuration, measuring economic benefit of negative price avoidance. Analysis examines whether curtailment value varies systematically across orientations (i.e., whether south-facing systems with greater negative price exposure benefit more from curtailment than west-facing systems with reduced exposure). This differential effect influences optimal orientation selection beyond base-revenue comparisons.

Level 3: Policy Framework Contribution:

Compares Real vs. Basic scenarios for each configuration, isolating revenue contribution from agrivoltaic bonuses and capacity threshold positioning. For 999 kWp configurations, this comparison quantifies the 24.7 €/MWh remuneration advantage (95.0 vs. 70.3 €/MWh) contribution to present value. For 1,001 kWp configurations, it quantifies the 12.0 €/MWh agrivoltaic bonus contribution. These comparisons reveal whether policy advantages dominate orientation effects or whether orientation timing value remains decisive even without policy support.

Level 4: Orientation Effects Under Uniform Policy:

Examines Basic_Curtailment scenario comparisons (W90 vs. E90 vs. S30) under uniform 70.3 €/MWh remuneration, testing whether west orientation achieves revenue superiority based purely on afternoon generation timing optimization. This comparison addresses the core hypothesis that generation timing value alone, independent of agrivoltaic policy advantages, justifies alternative orientation selection in markets with pronounced temporal price variation

3.8 Assumptions and Limitations

3.8.1 Core Methodological Assumptions

The analysis rests on several foundational assumptions enabling tractable modeling:

Representative Year Approach: The 2024 Nürnberg meteorological dataset is replicated across all 20 analysis years, assuming weather patterns remain statistically consistent over the operational lifetime. Actual inter-annual weather variation will produce generation fluctuations around modeled values. Multi-decade averaging suggests these variations balance toward long-term means, supporting representative-year methodology for comparative analysis. Systematic climate trends over the 2026-2045 period could introduce progressive bias if temperature increases, precipitation pattern changes, or solar resource shifts occur.

Performance and Market Dynamics: Module performance degradation follows linear 0.345%/year trajectory based on LUXOR warranty specifications (99% year-1 to 89% year-30). Projected electricity prices maintain 2024 temporal patterns (diurnal cycles, seasonal variation) while scaling absolute levels, assuming fundamental supply-demand dynamics persist. EEG remuneration frameworks (anzulegender Wert levels, agrivoltaic bonuses, Marktprämie mechanisms) remain stable for installations commissioned in 2026 throughout the 20-year guarantee period, consistent with German policy history of honoring long-term commitments for commissioned projects.

Analytical Simplifications: The 8,760-hour annual resolution captures diurnal and seasonal patterns while smoothing sub-hourly variability. The 1% aggregate system loss combines multiple mechanisms (wiring, inverter efficiency, soiling, mismatch) into a single parameter. Monatsmarktwert Solar calculations employ constant 2024 German fleet generation patterns across projection years. These simplifications enable consistent comparison while introducing

approximation relative to detailed modeling of time-varying losses and evolving fleet characteristics.

3.8.2 Analytical Scope Limitations

The analysis compares revenue streams without incorporating capital expenditure or operational cost differences. LUXOR modules cost 75-90 € each, providing identical procurement costs across configurations. Balance-of-system cost variations between vertical and tilted mounting are small relative to total project costs and revenue differentials. Operational costs (maintenance, monitoring, insurance, land lease) are assumed to scale proportionally with capacity. Agricultural revenue from dual-use land is excluded but affects all configurations equally given DIN SPEC 91434 compliance requirements. Grid connection costs and financing structures are not modeled. The revenue-focused analysis isolates market value effects; complete investment evaluation requires incorporating configuration-specific costs, though substantial revenue advantages would likely persist given small capital cost differentials.

3.8.3 Generalizability and External Validity

Geographic and Technology Context: Results reflect Ansbach/Nürnberg climate conditions (49.3°N, 1,100-1,150 kWh/m² GHI), German electricity market price patterns, and specific technology assumptions (80% bifaciality, 455Wp TOPCon modules, 0.25 albedo agricultural ground). Different latitudes alter solar geometry and seasonal distributions potentially changing optimal orientations. Alternative electricity markets with different temporal price patterns could reverse orientation preferences. Different module technologies or ground surface characteristics would modify relative performance. The methodology transfers to alternative contexts but conclusions require location-specific, technology-specific, and market-specific adaptation.

Policy and Scale Context: German EEG frameworks, wholesale market structures, and agrivoltaic policy incentives create specific economic contexts. Other jurisdictions with different remuneration mechanisms (net metering, feed-in tariffs, renewable energy certificates) or absent agrivoltaic policies face different optimization trade-offs. Analysis focuses on ~1 MWp installations at specific policy thresholds; scaling to multi-MW utility projects or smaller distributed systems requires considering scale-dependent factors including multiple threshold effects, varying grid connection requirements, and capacity-specific market participation rules.

3.8.4 Uncertainty and Risk Factors

Long-Term Projections: Wholesale electricity price projections to 2045 incorporate substantial uncertainty from renewable penetration trajectories, fossil fuel prices, carbon pricing development, storage deployment, and demand electrification. While temporal pattern preservation provides analytical stability, absolute price levels could deviate significantly. Projected negative pricing frequency (134-334 hours annually, 2026-2045) reflects conservative extrapolation from 2024 observations (457 hours) but faces uncertainty from storage growth and demand flexibility enhancement that could substantially reduce frequency. Twenty-year technology performance projections assume continued reliable operation without major component failures or unexpected degradation acceleration beyond the modeled 0.345%/year decline.

Regulatory Evolution: German and European regulatory frameworks governing renewable energy, agricultural land use, and electricity markets continue evolving. While 20-year remuneration

guarantees provide policy stability for commissioned installations, broader regulatory changes (carbon pricing structures, grid fee evolution, agricultural policy modifications) could affect project economics. The analysis assumes regulatory stability beyond guaranteed remuneration frameworks while acknowledging evolution risk in long-term projections extending to 2045.

4 Results

4.1 Generation Performance Analysis

4.1.1 Annual Generation by Configuration

Annual generation simulations reveal an unexpected performance hierarchy challenging conventional optimization assumptions. Table 1 presents year-one (2026) generation outcomes.

Configuration	Capacity (kWp)	Annual Generation (kWh)	Specific Yield (kWh/kWp)
W90_0.9MW	999.18	1,176,631	1,177.60
W90_1MW	1,001.00	1,124,120	1,123.00
S30_1MW	1,001.00	1,096,238	1,095.14
S30_0.9MW	999.18	1,090,107	1,091.00
E90_0.9MW	999.18	948,440	949.22
E90_1MW	1,001.00	945,927	944.98

Table 1: Annual Generation Performance (2024 - Reference Year)

West-facing W90_0.9MW achieves highest generation (1,176,631 kWh), surpassing south-facing baseline S30_1MW by 7.3% and surpassing S30_0.9MW by 8%. This performance advantage contradicts conventional expectations that optimally tilted south-facing systems maximize annual energy capture. The superior west-facing performance derives from bifacial technology capturing substantial rear-side irradiance throughout extended daylight periods, with vertical orientation enabling favorable ground-view geometry for reflected radiation.

West configurations demonstrate unexpected capacity inversion: W90_0.9MW (999 kWp) generates 52,511 kWh more than W90_1MW (1,001 kWp), representing 4.7% higher output despite 0.2% lower capacity. This anomaly reflects inverter sizing differences (1.15 DC/AC ratio (W90_0.9MW) versus 1.03 (W90_1MW)), enabling superior capacity utilization and reduced clipping under west-facing extended afternoon generation profiles.

South-facing systems achieve mid-range performance (1,090-1,096 MWh annually), demonstrating proportional capacity scaling. East-facing configurations show severely compromised performance (945-948 MWh), producing 19.6% below W90_1MW and 13.7% below S30_1MW due to catastrophic shading losses (detailed Section 4.1.4).

4.1.2 Specific Yield and Performance Ratio

Specific yield metrics (kWh/kWp) normalize generation performance by installed capacity, enabling fair comparison across capacity levels and isolating orientation effects from scale differences. Performance ratio (PR) quantifies actual generation relative to theoretical ideal performance under site irradiance conditions, capturing aggregate system loss effects.

Configuration	Specific Yield (kWh/kWp)	Performance Ratio (%)
---------------	--------------------------	-----------------------

W90_0.9MW	1,177.60	81.9
W90_1MW	1,123.00	81.7
S30_1MW	1,095.14	84.8
S30_0.9MW	1,091.00	84.5
E90_0.9MW	949.22	70.3
E90_1MW	944.98	70

Table 2: Specific Yield and Performance Ratio (2026)

Even though west-facing achieves the highest specific yields (confirming vertical bifacial optimization can exceed conventional south-facing designs), south-facing configurations achieve the highest performance ratios (84.5-84.8%), indicating efficient conversion of available site irradiance to AC generation. West-facing vertical systems demonstrate 3-4% PR reduction relative to south-facing configurations, which reflects modest additional loss mechanisms from vertical orientation, including suboptimal inverter loading during bimodal generation patterns, and potentially elevated shading during low-angle morning periods. The E90 PR deficit (10-14 percentage points below alternatives) signals fundamental design failure rendering east-facing commercially non-viable under analyzed conditions.

Capacity-level PR differences within orientations remain minimal (<0.5%), confirming threshold positioning (999 vs 1,001 kWp) produces negligible efficiency impact. Subsequent revenue differentials reflect policy frameworks rather than performance bias. The performance ratio patterns validate west-facing orientation as the optimal choice among analyzed alternatives since west systems achieve 95-96% of south-facing PR levels while generating 107-108% of south-facing annual output. This combination demonstrates that moderate PR reductions from vertical orientation prove acceptable trade-offs for enhanced total generation through bifacial optimization.

4.1.3 Temporal Generation Patterns (Monthly Distribution)

Monthly generation distributions reveal that all orientations follow similar seasonal patterns dominated by summer concentration, with July representing the peak generation month. West-facing systems generate 171,670 kWh in July (15.3% of annual generation), east-facing systems peak at 148,060 kWh (15.7% of annual), and south-facing installations reach 159,780 kWh (14.6% of annual). May-August represents 55-58% of annual generation across all configurations, indicating bifacial vertical orientation maintains seasonal concentration driven by irradiance availability. West-facing demonstrates consistent summer superiority (June-August: 482,200 kWh vs S30: 474,980 kWh), reflecting bifacial rear-side capture during high-elevation afternoon periods. East-facing underperforms consistently, with winter months showing particularly severe deficits.

Winter generation remains severely constrained across all orientations, with November-February contributing only 10.6% of annual W90 generation, 9.0% of E90 generation, and 10.0% of S30 generation. The universal summer concentration pattern means that orientation selection affects primarily daily (hourly) generation timing rather than seasonal distribution, focusing revenue optimization on time-of-day price patterns rather than seasonal price cycles.

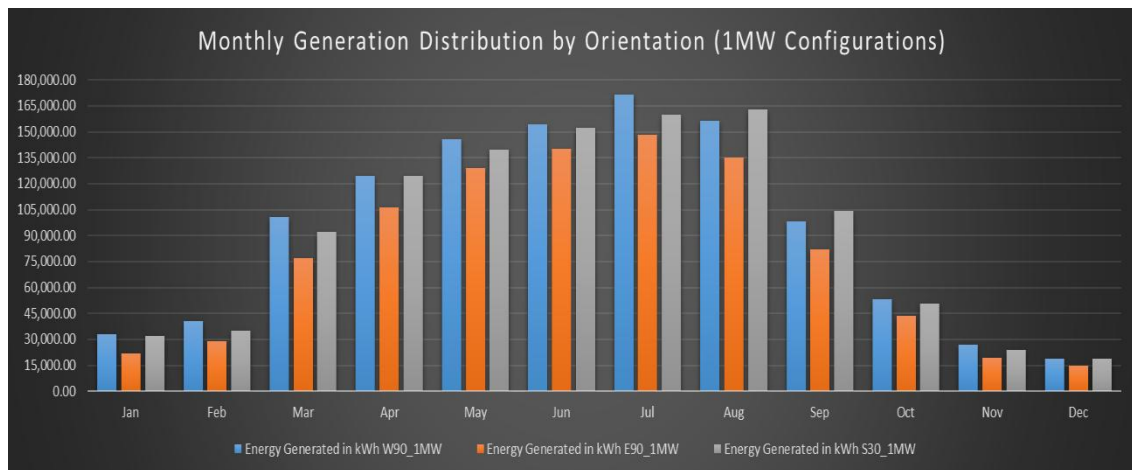
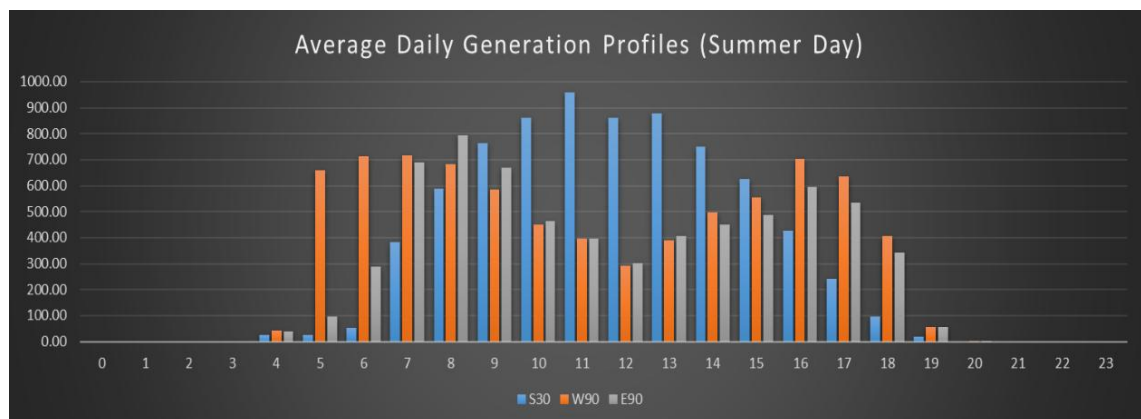


Figure 1: Monthly Generation Distribution by Orientation

Daily patterns distinguish orientations dramatically. The temporal distribution of generation across daylight hours creating potential market value implications through alignment with time-varying electricity prices. Figure 2 presents representative summer day profiles. Daily generation patterns reveal the distinctive characteristic differentiating orientations.

Figure 2: Average Diurnal Generation Profiles (Summer Day Example)



West-facing vertical systems demonstrate unexpected bimodal generation patterns with substantial morning generation (peak 718 kW at hour 7) and secondary afternoon peak (703 kW at hour 16). The extended generation period from morning through afternoon creates potential for superior market value capture compared to south-facing midday concentration, though the degree of advantage depends critically on actual price-generation temporal correlation documented in Section 4.3.

South-facing systems exhibit conventional single-peak midday concentration with maximum generation (957 kW) occurring at hour 11-12 during solar noon periods. The symmetric bell-curve generation profile reflects optimized solar angle alignment during midday hours, tapering toward morning and evening as solar angle obliquity increases. This canonical generation pattern maximizes total daily energy (June average 7,188 kWh) but concentrates output during periods of potential price suppression when aggregate solar fleet generation creates midday oversupply conditions in wholesale electricity markets.

East-facing systems show strong morning concentration with peak generation (795 kW at hour 8), followed by declining afternoon output. However, the east-facing morning peak remains 17% below the south-facing midday maximum. The subdued peak magnitude reflects the combined effects of morning atmospheric attenuation (reduced clarity compared to afternoon conditions) and catastrophic shading losses documented in Section 4.1.4.

4.1.4 System Losses and Performance Metrics

Detailed loss analysis reveals dramatic east-facing shading impacts explaining the orientation's catastrophic performance degradation, while confirming that west-facing and south-facing systems achieve commercially viable loss levels through appropriate design.

Configuration	Total Loss (%)	Shading Loss (%)	Inverter Sizing	PR (%)
W90_1MW	18.21	2.45	1.03	81.7
W90_0.9MW	18.07	2.86	1.15	81.9
E90_1MW	32.66	19.6	1.03	70
E90_0.9MW	32.15	19.93	1.15	70.3
S30_1MW	14.47	9.06	1.03	84.8
S30_0.9MW	14.82	9.28	1.01	84.5

Table 3: System Loss Components and Performance Metrics

Catastrophic east-facing shading: E90 configurations experience 19.60-19.93% shading losses, explaining their 30% lower specific yield relative to west-facing systems, rendering these systems commercially non-viable. Vertical east-facing geometry creates extended morning shadows on adjacent rows as low-angle sunlight strikes from eastern azimuth.

Minimal west-facing shading: W90 achieves remarkably low 2.45-2.86% shading, demonstrating the fundamental advantage of vertical west-facing geometry. Afternoon sun approaches from behind module planes, minimizing inter-row shadows. This near-elimination of shading losses enables west-facing generation superiority.

Moderate south-facing shading: S30 demonstrates expected 9.06-9.28% shading from seasonal low-angle inter-row effects, within acceptable commercial ranges. However, the 3-4x higher shading losses compared to west-facing vertical systems demonstrate that vertical orientations offer inherent shading advantages in properly-designed layouts, contributing to west-facing performance superiority beyond simple bifacial rear-side gains.

Inverter sizing factors show strategic differences: 1MW systems employ conservative 1.03 ratios; 0.9MW systems utilize 1.15 ratios, accepting minor clipping for improved utilization. The higher ratio contributes to W90_0.9MW's superior performance by matching inverter capacity to extended afternoon generation.

4.2 Eligible Generation and Negative Price Exposure

4.2.1 Actual vs. Eligible Generation Analysis

Under EEG §51 provisions, hours experiencing negative day-ahead electricity prices are excluded from Marktprämie eligible generation calculations. Installations receive spot market revenue

(which is negative) during such hours but cannot claim Marktprämie compensation, creating an economic incentive for voluntary curtailment. The magnitude of eligible generation reduction relative to actual generation quantifies negative price exposure and potential curtailment benefits.

Configuration	Total Actual (MWh)	Total Eligible (MWh)	Difference (MWh)	Difference
W90_1MW	21,560	20,531	1,029	4.77%
W90_0.9MW	22,583	21,836	748	3.31%
E90_1MW	18,147	17,010	1,137	6.26%
E90_0.9MW	18,191	17,042	1,150	6.32%
S30_1MW	21,035	19,671	1,364	6.48%
S30_0.9MW	20,910	19,552	1,358	6.50%

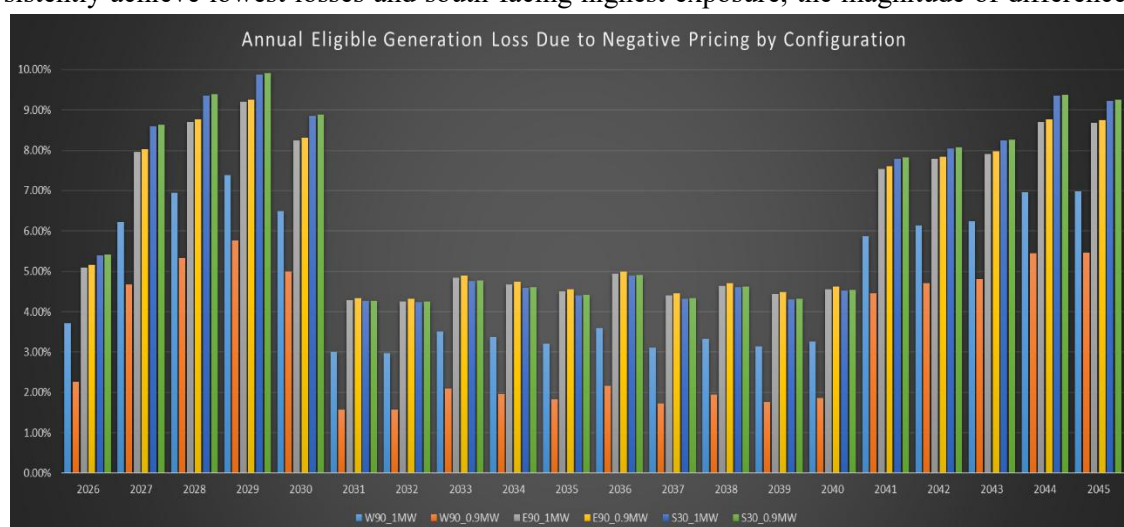
Table 4: Eligible Generation Summary - All Configurations, 20-Year Period (2026-2045)

Across the 20-year analysis period (2026-2045), negative pricing episodes exclude 3.3-6.5% of total generated energy from Marktprämie eligibility depending on configuration. West-facing demonstrates the lowest ineligible percentages (W90_0.9MW: 3.31%, W90_1MW: 4.77%), while east-facing and south-facing exhibit similar moderate exposure (E90: 6.26-6.32%, S30: 6.48-6.50%). The relatively small west-south differential (1.7-2.7 percentage points) indicates negative pricing impacts all orientations substantially rather than creating dramatic advantages, suggesting that the bimodal west-facing pattern includes substantial midday generation during negative price hours rather than complete afternoon displacement.

Capacity-level differences within orientations demonstrate inconsistent patterns. West-facing configurations show clear capacity-dependent differentiation of 1.5%, while south-facing and east-facing capacity pairs exhibit negligible differences. East-facing demonstrates that morning generation provides a negligible advantage in negative price avoidance compared to midday concentration. East-facing and south-facing show nearly identical eligible losses despite fundamentally different temporal profiles. This equivalence indicates negative pricing affects both morning and midday periods comparably, neutralizing orientation-dependent timing advantages.

4.2.2 Negative Price Hour Impacts by Orientation

Orientation-specific eligible generation loss patterns reveal that while west-facing systems consistently achieve lowest losses and south-facing highest exposure, the magnitude of differences



remains modest and varies substantially across years based on negative pricing frequency rather than systematic orientation advantages.

Figure 3: Annual Eligible Generation Loss by Configuration (2026-2045)

The temporal pattern of losses reveals substantial year-to-year variation driven by negative pricing frequency fluctuations rather than stable orientation-dependent advantages. Years 2027-2030 and 2041-2045 exhibit elevated eligible generation losses across all orientations (6-10% range), corresponding to projection periods with heightened negative pricing frequency (discussed in Section 4.2.3). Conversely, years 2031-2040 demonstrate reduced impacts (2-5% range) during periods of moderate negative pricing occurrence. This pattern confirms that negative pricing market conditions dominate eligible generation outcomes, with orientation effects providing only modest differential protection during adverse market periods.

The orientation ranking (west best, south worst, east intermediate) persists across years, but the magnitude of advantage fluctuates substantially. During low negative pricing years (2031-2037), west-facing advantages over south-facing systems, as all configurations achieve 95-98% eligible generation preservation. During peak negative pricing years (2027-2030, 2044-2045), west advantages expand to 2-3 percentage points as heightened negative price frequency disproportionately affects south-facing midday concentration. This variable advantage suggests that west-facing orientation provides modest downside protection during adverse negative pricing conditions rather than transformative eligible generation improvements across all market scenarios.

Annual eligible generation losses range from 1.58% (W90_0.9MW in 2031-2032) to 9.91% (S30_0.9MW in 2029), with typical annual losses falling in the 3-7% range for most configurations and years. West-facing configurations maintain the lowest annual losses across nearly all analysis years. South-facing systems experience the highest losses, while east-facing installations demonstrate intermediate exposure. The annual loss patterns reveal that all configurations face increasing negative price exposure in later projection years (2041-2045) as renewable penetration scenarios intensify negative pricing frequency. The convergence of losses across orientations during these peak years (W90 5.5-7.0%, E90 7.5-9.0%, S30 7.8-9.4%) suggests that extreme negative pricing environments eliminate orientation-based protection, with all solar generation experiencing substantial Marktprämie eligibility erosion. This indicates that orientation optimization alone proves inadequate to address projected negative pricing intensification, requiring complementary strategies such as curtailment management, storage integration to maintain revenue viability under high renewable penetration scenarios.

4.2.3 Multi-Year Eligible Generation Patterns

Temporal evolution of eligible generation across the 20-year projection period reveals variable negative pricing frequency creating non-monotonic loss patterns rather than steady deterioration from increasing renewable penetration.

Year	W90_1MW (%)	E90_1MW (%)	S30_1MW (%)	Negative Price Hours
2026	3.72	5.1	5.4	149
2027	6.23	7.96	8.6	254
2030	6.49	8.25	8.86	294
2031	3	4.28	4.26	134
2034	3.37	4.68	4.58	182

2035	3.21	4.5	4.41	167
2039	3.14	4.43	4.31	155
2040	3.25	4.56	4.53	155
2042	6.14	7.79	8.05	281
2043	6.25	7.91	8.24	290
2045	6.98	8.68	9.22	312

Table 5: Eligible Generation Loss by Configuration

Early analysis years (2026-2027) exhibit moderate negative pricing frequency (149-254 hours annually). These baseline conditions reflect extrapolation from 2024 empirical observations, representing negative pricing environments under contemporary renewable penetration levels. Mid-period years (2028-2030) show peak negative pricing frequency (277-334 annual hours), creating maximum eligible generation impacts across the full projection period. The expanded loss differentials during peak negative pricing years (west-south differential reaches 2.9 percentage points in 2029) confirm that orientation advantages provide the greatest protection during adverse market conditions characterized by frequent midday oversupply. However, even optimal west-facing orientation cannot fully insulate installations from negative pricing impacts, confirming that market conditions dominate orientation effects in determining eligible generation outcomes.

A notable pattern emerges in years 2031-2040, where negative pricing frequency declines substantially (134-195 annual hours) compared to surrounding periods. This mid-period respite creates compressed eligible generation losses across all orientations (2.0-5.0% range). The reduced negative pricing likely reflects scenarios where energy storage deployment, demand flexibility enhancement, or renewable penetration plateaus temporarily alleviate midday oversupply conditions. This period demonstrates that orientation-based advantages nearly disappear under benign negative pricing environments.

Late-period years (2041-2045) exhibit a resurgence of negative pricing frequency (268-322 annual hours) approaching earlier peak levels, creating renewed eligible generation. The late-period pattern suggests that renewable penetration growth resumes after mid-period stabilization, driving negative pricing frequency toward upper ranges experienced during 2028-2030. Orientation differentials persist but remain modest (2-3 percentage points west-south advantage), confirming that west-facing temporal distribution provides consistent but limited protection across varying negative pricing regimes.

The non-monotonic negative pricing projection pattern (rising 2026-2030, declining 2031-2040, rising 2041-2045) reflects uncertainty in renewable penetration trajectories, storage deployment rates, and demand flexibility adoption. For orientation comparison purposes, the variable pattern confirms that orientation advantages persist across scenarios but remain modest in magnitude. The identical negative pricing hour exposure across all configurations confirms that negative pricing affect all solar generation simultaneously rather than selective impacts on particular orientations or capacities. This simplifies analysis by indicating that eligible generation percentage differentials derive entirely from varying temporal concentration patterns, rather than configuration-specific market participation rules or differential curtailment obligations.

4.2.4 Heatmap Analysis: Eligible Energy Across Configurations and Years

Comprehensive heatmap visualization of eligible generation patterns across configurations and years reveals modest orientation differentiation and substantial year-to-year variation driven by negative pricing frequency fluctuations.

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
W90_1MW	3.72%	6.23%	6.94%	7.38%	6.49%	3.00%	2.97%	3.52%	3.37%	3.21%	3.60%	3.11%	3.33%	3.14%	3.25%	5.87%	6.14%	6.25%	6.96%	6.98%
W90_0.9MW	2.26%	4.68%	5.32%	5.76%	5.00%	1.58%	1.57%	2.09%	1.96%	1.83%	2.16%	1.72%	1.95%	1.76%	1.86%	4.45%	4.70%	4.81%	5.45%	5.46%
E90_1MW	5.10%	7.96%	8.70%	9.20%	8.25%	4.28%	4.26%	4.84%	4.68%	4.50%	4.94%	4.40%	4.65%	4.43%	4.56%	7.54%	7.79%	7.91%	8.70%	8.68%
E90_0.9MW	5.16%	8.02%	8.76%	9.26%	8.31%	4.34%	4.31%	4.89%	4.74%	4.56%	5.00%	4.46%	4.71%	4.49%	4.62%	7.60%	7.85%	7.97%	8.76%	8.74%
S30_1MW	5.40%	8.60%	9.36%	9.88%	8.86%	4.26%	4.24%	4.76%	4.58%	4.41%	4.89%	4.32%	4.60%	4.31%	4.53%	7.80%	8.05%	8.24%	9.35%	9.22%
S30_0.9MW	5.42%	8.63%	9.40%	9.91%	8.89%	4.28%	4.25%	4.78%	4.60%	4.42%	4.90%	4.33%	4.62%	4.32%	4.54%	7.82%	8.08%	8.27%	9.38%	9.25%

Figure 4: Eligible Generation Loss Heatmap - All Configurations, All Years

The heatmap demonstrates two-zone stratification by orientation, though color differentiation proves less pronounced than initially hypothesized. West-facing configurations (W90_1MW, W90_0.9MW) populate predominantly green-yellow zones, indicating 2-7% eligible generation losses across most years, confirming systematic advantages in negative price avoidance relative to alternative orientations. East- and south-facing configurations both occupy yellow-orange zones, indicating 4-9% losses, and demonstrate similar negative price exposure despite distinct temporal generation patterns (morning versus midday concentration). The east-south similarity confirms that morning generation shifting provides negligible negative price protection compared to midday concentration.

Vertical banding patterns dominate the heatmap, with entire columns displaying similar color tones across all configurations during particular years. Years 2027-2030 and 2041-2045 exhibit darker orange-red coloring across all rows, indicating elevated negative pricing frequency affecting all orientations proportionally. Conversely, years 2031-2040 display predominantly green-yellow tones across configurations, reflecting reduced negative pricing occurrence, benefiting all systems. This confirms that negative pricing conditions drive eligible generation outcomes more powerfully than orientation, with orientation providing only modest differential protection during adverse years rather than fundamental exposure differences.

The within-orientation capacity pairing demonstrates near-identical coloring for most orientation-capacity combinations, with negligible visual differentiation in most years. This color similarity confirms that capacity threshold positioning (999 kWp versus 1,001 kWp) results in minimal impacts on eligible generation percentage. The exception appears in west-facing rows where W90_0.9MW displays slightly lighter coloring than W90_1MW across many years, reflecting the modest capacity-dependent advantage documented in Section 4.2.2, where superior 999 kWp array optimization produces marginally beneficial temporal generation distribution.

4.3 Price-Generation Correlation Analysis

4.3.1 Correlation Methodology and Plausibility Framework

Generation-price correlation analysis reveals unexpected negative correlations across all configurations, contradicting initial hypotheses that afternoon-weighted west-facing generation would

achieve positive correlation with afternoon price peaks while midday-concentrated south-facing systems demonstrated negative correlation with midday price suppression.

Configuration	Average Correlation (r)	Interpretation
W90_0.9MW	-0.2	Weak negative - least unfavorable
W90_1MW	-0.25	Moderate negative
E90_0.9MW	-0.3	Moderate negative
E90_1MW	-0.3	Moderate negative
S30_0.9MW	-0.34	Moderate negative - most unfavorable
S30_1MW	-0.34	Moderate negative - most unfavorable

Table 6: Correlation Analysis Summary - Average Across All Years

All six configurations demonstrate negative generation-price correlations averaging -0.20 to -0.34 across the 20-year projection period. The universal negative correlation indicates that all analyzed orientations generate predominantly during low or negative-price periods rather than high-price hours. West-facing configurations achieve the least-negative correlations, confirming modest advantages relative to south-facing alternatives but failing to reach positive correlation territory (favorable temporal alignment). East-facing systems demonstrate intermediate negative correlations (-0.30) falling between west and south extremes, suggesting that morning generation provides modest price-timing advantages over midday concentration but proves inferior to afternoon-weighted patterns. The east-south correlation differential represents marginal improvements insufficient to justify east-facing orientation, given the catastrophic 20% generation deficit from shading losses. The 0.09-0.14 point correlation differential between west and south orientations represents that west-facing temporal distribution provides measurable price-timing improvements.

The correlation results suggest that extensive solar penetration has suppressed prices across all daylight generation periods, eliminating traditional afternoon price peaks that would create positive correlation opportunities for west-facing systems. The negative correlation finding carries critical implications for installation economics. Under negative price-generation correlation, installations earn below-average market revenues per kWh generated relative to average wholesale prices.

4.3.2 Configuration-Specific Correlation Results

Detailed annual correlation analysis reveals stable negative correlation patterns across the 20-year projection period, with modest strengthening trends toward increasingly negative correlations in later years reflecting intensifying solar penetration impacts on price patterns.

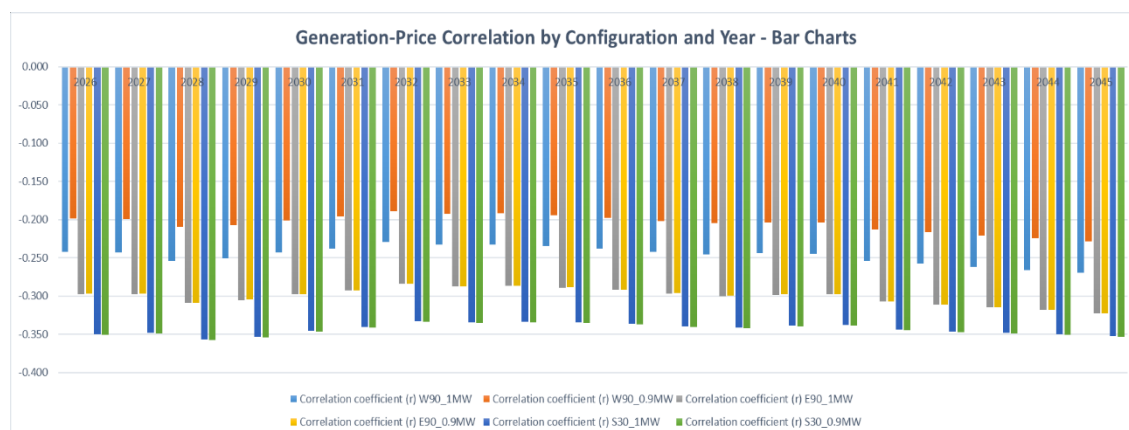


Figure 5: Generation-Price Correlation by Configuration and Year - Bar Charts

West-facing configurations maintain the least-negative correlations throughout the analysis period (W90_0.9MW: -0.189 to -0.228; W90_1MW: -0.230 to -0.270). The -0.025 to -0.030 point correlation deterioration from early to late periods confirms that price-generation temporal misalignment intensifies over the projection horizon. The 999 kWp west-facing configuration demonstrates less-negative correlations than the 1,001 kWp equivalent across all years. The consistent capacity advantage suggests that the 999 kWp threshold positioning benefits from both policy remuneration bonuses and modest technical performance advantages in west-facing vertical configurations.

East-facing systems exhibit stable negative correlations spanning -0.293 to -0.323 across the projection period for both capacity levels, with negligible capacity-dependent differentiation. The correlation strengthening from -0.297 (2026) to -0.322 (2045) parallels west-facing trends, confirming market-wide price pattern evolution rather than orientation-specific effects. The intermediate east-facing correlations provide insufficient advantage to justify orientation selection given catastrophic generation performance penalties, confirming that correlation improvements alone cannot overcome fundamental efficiency deficits.

South-facing configurations demonstrate most-negative correlations (-0.340 to -0.354) across all analysis years, with capacity pairs showing near-identical values. The strengthening negative correlation from -0.350 (2026) to -0.354 (2045) represents the modest deterioration pattern observed across orientations, confirming that midday price suppression intensifies modestly but does not undergo a dramatic transformation over the projection period. The consistency of strong negative south-facing correlations validates theoretical expectations that midday-concentrated generation coincides with maximum price suppression, though the failure of alternative orientations to achieve positive correlations indicates that price suppression extends beyond strict midday hours.

4.3.3 Correlation Heatmaps: Generation vs. Price by Year

Consolidated heatmap visualization confirms universal negative correlations while revealing modest orientation-dependent gradations and temporal strengthening patterns across the projection period.

Year	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	Avg.
W90_1MW	-0.242	-0.243	-0.254	-0.250	-0.243	-0.238	-0.230	-0.233	-0.233	-0.235	-0.238	-0.243	-0.246	-0.244	-0.244	-0.254	-0.258	-0.262	-0.266	-0.270	-0.25
W90_0.9MW	-0.198	-0.200	-0.210	-0.207	-0.201	-0.196	-0.189	-0.192	-0.192	-0.194	-0.197	-0.202	-0.205	-0.204	-0.204	-0.213	-0.217	-0.221	-0.225	-0.228	-0.20
E90_1MW	-0.297	-0.298	-0.309	-0.305	-0.298	-0.293	-0.284	-0.287	-0.287	-0.289	-0.292	-0.297	-0.300	-0.298	-0.298	-0.308	-0.311	-0.315	-0.319	-0.323	-0.30
E90_0.9MW	-0.297	-0.297	-0.309	-0.305	-0.298	-0.292	-0.284	-0.287	-0.287	-0.289	-0.292	-0.296	-0.300	-0.298	-0.298	-0.307	-0.311	-0.315	-0.318	-0.322	-0.30
S30_1MW	-0.350	-0.348	-0.357	-0.353	-0.346	-0.340	-0.332	-0.335	-0.334	-0.335	-0.336	-0.339	-0.341	-0.339	-0.338	-0.344	-0.346	-0.348	-0.350	-0.353	-0.34
S30_0.9MW	-0.350	-0.349	-0.358	-0.354	-0.347	-0.341	-0.333	-0.336	-0.334	-0.335	-0.337	-0.340	-0.342	-0.340	-0.339	-0.345	-0.347	-0.349	-0.351	-0.354	-0.34

Figure 6: Generation-Price Correlation Heatmap - All Configurations, All Years

Optimal combination: W90_0.9MW in 2032 achieves -0.189 (least-negative, lightest cell). Worst-case: S30_1MW in 2045 at -0.353 (most-negative, darkest cell). The 87% differential illustrates the combined importance of orientation, capacity, and timing, though even optimal remains substantially negative.----

The correlation heatmap displays universal coloring across all cells, confirming negative correlations for all configurations and years without exception. The absence of green (positive correlation) zones validates that none of the analyzed orientations achieve favorable temporal alignment with price peaks under projected market conditions. Within the universal red color domain, modest gradations appear with west-facing rows displaying lighter red tones compared to darker red tones in east-facing and south-facing rows. The temporal darkening pattern in later years (2041-2045) compared to early years (2026-2030) confirms that price-generation misalignment intensifies moderately over the projection period as continued solar capacity growth extends price suppression across broader daylight hours.

Horizontal uniformity within orientation-capacity pairs demonstrates near-identical coloring for W90_1MW/W90_0.9MW, E90_1MW/E90_0.9MW, and S30_1MW/S30_0.9MW rows. The color similarity confirms negligible capacity-dependent correlation effects for east and south orientations, while west-facing configurations show modest but consistent color differentiation, with W90_0.9MW displaying lighter tones than W90_1MW across most years.

The heatmap reveals the absence of distinct vertical bands corresponding to high versus low negative pricing years, confirming that negative pricing frequency variations (134-334 annual hours) do not substantially affect annual correlation calculations based on 8,760 hourly observations. Negative pricing represents concentrated impacts during specific hours rather than broad price level effects across all periods, allowing correlation patterns to remain stable despite varying negative price frequency.

4.3.4 Temporal Price-Generation Relationships

Detailed daily pattern analysis reveals price-generation temporal relationships explaining universal negative correlations across all orientations, with price patterns demonstrating midday suppression extending into afternoon hours.

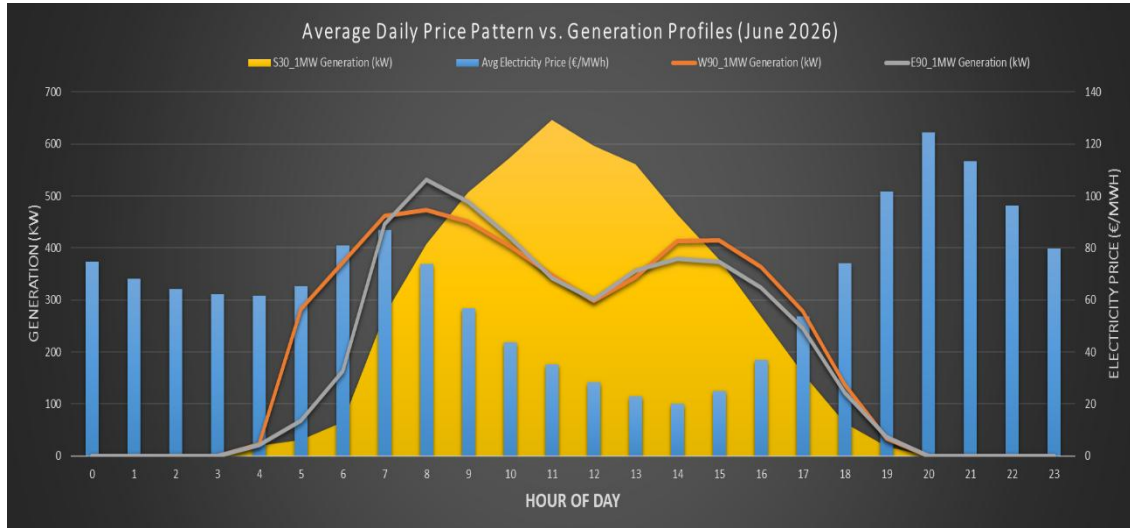


Figure 7: Average Daily Price Pattern vs. Generation Profiles

Average June 2026 daily price patterns demonstrate characteristic morning elevation (65-87 €/MWh during hours 4-7), midday suppression (20-43 €/MWh during hours 10-14), and afternoon recovery (25-101 €/MWh during hours 15-19). West-facing generation profiles demonstrate bi-modal patterns with morning maximum (474 kW at hour 8) and afternoon peak (415 kW at hour 14-15), concentrating the majority of generation during hours 5-18. This temporal distribution coincides partially with morning price elevation but generates substantially during midday suppression when rear-side bifacial capture creates continued generation despite suboptimal front-surface solar angles. The afternoon generation peak coincides with initial price recovery but precedes maximum afternoon price elevation by 2-4 hours, creating negative correlation despite afternoon weighting. The temporal misalignment explains why west-facing systems demonstrate -0.25 correlation rather than positive values.

South-facing systems exhibit conventional single-peak midday concentration with maximum generation (647 kW at hour 11) coinciding precisely with midday price minimums (35 €/MWh). Approximately 60% of south-facing generation occurs during hours 9-15 when prices fall below daily averages, creating strong negative correlation (-0.34). The symmetric generation decline toward evening hours means minimal south-facing output during afternoon price recovery and negligible generation during evening price peaks, reinforcing negative correlation through absence of generation during high-price periods.

East-facing patterns show morning-weighted generation with peak (532 kW at hour 8) coinciding with morning price elevation (74 €/MWh), followed by declining afternoon generation. The morning peak alignment with moderate price elevation creates the least-negative correlation segment of east-facing generation, though substantial continued generation during midday suppression hours (hours 9-14, average 420 kW at 21-57 €/MWh) outweighs morning advantages. The intermediate negative correlation (-0.30) reflects partial morning advantage offset by inevitable midday generation continuation inherent in solar generation patterns.

The daily relationship analysis shows that none of the analyzed orientations achieves consistent alignment between generation and price peaks (hours 18-20). West-facing afternoon weighting provides modest correlation improvements versus south-facing midday concentration, but both orientations generate predominantly during suppressed-price daylight hours rather than evening price peaks. This fundamental misalignment explains universal negative correlations and indicates that orientation selection alone cannot achieve positive market value capture without generation timing mechanisms (storage, demand management) and price realization.

4.4 Revenue Components Analysis

4.4.1 Spot Market Revenue

Spot market revenue represents direct economic return from selling generated electricity into wholesale day-ahead markets at hourly clearing prices. Each hour's revenue equals generation output multiplied by contemporaneous price, creating direct exposure to price-generation temporal alignment patterns quantified through correlation analysis (Section 4.3).

The negative price-generation correlations documented across all configurations create systematic spot revenue underperformance relative to hypothetical flat generation profiles. Table 7 presents year-one (2026) spot market outcomes revealing orientation-dependent market value differentials.

Orientation		Total generation (kWh)	Spot Revenue (€)	Simple Revenue (€)	Market Value Factor (MVF)	Revenue per kWh
W90_1MW	No Curtailment	1,112,878.80	76,131.04	93,919.85	0.81	0.0684
	Curtailment	1,112,878.80	76,948.84	93,919.85	0.82	0.0691
W90_0.9MW	No Curtailment	1,164,864.40	82,836.70	98,307.10	0.84	0.0711
	Curtailment	1,164,864.40	83,557.03	98,307.10	0.85	0.0717
E90_1MW	No Curtailment	945,377.73	59,779.43	79,783.83	0.75	0.0632
	Curtailment	945,377.73	60,558.10	79,783.83	0.76	0.0641
E90_0.9MW	No Curtailment	938,955.60	59,307.56	79,241.84	0.75	0.0632
	Curtailment	938,955.60	60,081.20	79,241.84	0.76	0.0640
S30_1MW	No Curtailment	1,085,275.62	62,741.43	91,590.32	0.69	0.0578
	Curtailment	1,085,275.62	63,959.48	91,590.32	0.7	0.0589

S30_0.9MW	No Cur- tailment	1,079,205.93	62,319.33	91,078.07	0.68	0.0577
	Cur- tailment	1,079,205.93	63,534.27	91,078.07	0.7	0.0589

Table 7: Annual Spot Market Revenue by Configuration (2026)

West-facing spot market superiority: W90 configurations achieve highest absolute spot revenues. W90_0.9MW generates €83,557 spot revenue (curtailment scenario), exceeding S30_1MW's €63,959 by €19,598 (+30.6%). This substantial advantage reflects combined effects of superior generation volume (+7.3%) and market timing benefits (market value factor 0.85 vs 0.70). Because W90_1MW configuration generates 1,112,879 kWh (2.5% MORE than S30_1MW's 1,085,275 kWh), it achieves €76,949 spot revenue, exceeding S30_1MW by €12,990 (+20.3%). This confirms west-facing market timing advantages compound upon generation volume superiority, creating multiplicative rather than additive revenue benefits.

Market value factor hierarchy: The ratio of actual spot revenue to simple average-price revenue quantifies market timing performance independent of generation volumes:

- **W90_0.9MW:** MVF = 0.85 (generation-weighted prices reach 85% of average)
- **W90_1MW:** MVF = 0.82 (strong market timing)
- **E90:** MVF = 0.75-0.76 (moderate market timing)
- **S30:** MVF = 0.68-0.70 (poorest market timing)

West-facing bimodal distribution partially mitigates photovoltaic market timing disadvantage. Morning peak aligns with moderate plateau prices, while the afternoon secondary peak captures the price recovery phase. Though neither peak achieves evening price maximums, the distributed generation reduces concentrated exposure to the midday price trough.

Curtailment strategy implementation: Under curtailment scenarios, installations generate electricity during negative price hours but withhold from market sale, avoiding negative revenue realization while maintaining equipment stability. Toggling generation on/off creates mechanical stress and thermal cycling, potentially damaging inverters and module connections; sustained generation with market disconnection proves operationally superior.

Curtailment benefits vary by negative price exposure severity:

- **W90_0.9MW curtailment value (minimal 3.31% exposure):**
 - No-curtailment spot revenue: €82,837
 - Curtailment spot revenue: €83,557
 - Benefit: €720 annually (0.87% improvement)
- **S30_1MW curtailment value (substantial 6.48% exposure):**
 - No-curtailment spot revenue: €62,741
 - Curtailment spot revenue: €63,959
 - Benefit: €1,218 annually (1.94% improvement)

South-facing experiences greater curtailment benefit than west-facing equivalents due to 2× higher negative price exposure. However, absolute benefits remain modest relative to total spot revenues (1-2% improvement), though compound over project lifetimes produce meaningful cumulative value.

Temporal spot revenue evolution: Figure 8 presents 20-year spot revenue trajectories across all configurations and curtailment scenarios.

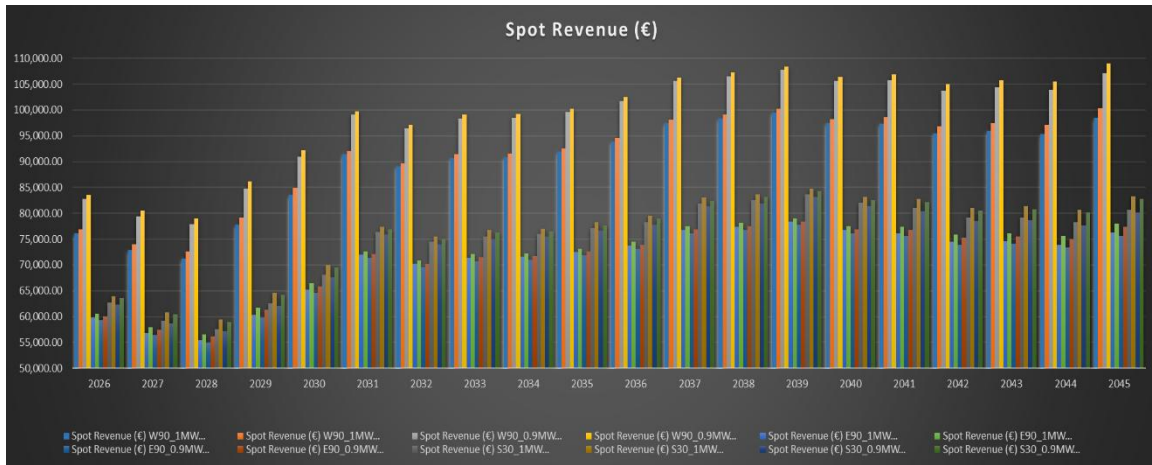


Figure 8: Annual Spot Market Revenue Evolution (2026-2045)

East-facing spot revenue failure: E90 configurations achieve only €59,308-60,558 spot revenues (curtailment), representing a 28-29% deficit versus W90_0.9MW, with catastrophic generation shortfall providing the primary explanation. Market value factors (0.75-0.76) prove intermediate between west (0.82-0.85) and south (0.68-0.70), suggesting morning generation timing provides modest advantages versus midday concentration but proves insufficient to overcome fundamental generation inadequacy.

Capacity-dependent spot revenue: Within orientations, capacity threshold effects create unexpected outcomes:

- West-facing: W90_0.9MW (€83,557) exceeds W90_1MW (€76,949) by €6,608 (+8.6%), reflecting 4.7% generation advantage despite 0.2% capacity deficit. Inverter sizing optimization (1.15 vs 1.03 DC/AC ratio) enables superior W90_0.9MW performance.
- South-facing: S30_1MW (€63,959) slightly exceeds S30_0.9MW (€63,534) by €425 (+0.7%), demonstrating a proportional capacity-revenue relationship without inverter sizing benefits.

Spot market revenue represents a dominant component of total revenues (75-80%), fundamentally establishing that photovoltaic economics under examined conditions depend primarily on wholesale market performance rather than policy support mechanisms.

4.4.2 Marktprämie Contributions

Marktprämie fills the gap between guaranteed remuneration rates (Anzulegender Wert) and monthly wholesale price benchmarks (Monatsmarktwert Solar), providing a supplementary revenue stream representing 20-25% of total revenues under Real scenarios. While secondary to spot market contributions, Marktprämie proves critical for configurations experiencing poor market timing, partially compensating for low market value factors through policy framework support.

Marktprämie calculation framework: Monthly premium computation follows EEG §51 methodology:

$$\text{Marktprämie_month} = (\text{Anzulegender Wert} - \text{Monatsmarktwert Solar}) \times \text{Eligible Generation_month}$$

Where: **Anzulegender Wert (AW)** = Base remuneration + Agrivoltaics bonus (if applicable)

Monatsmarktwert Solar = Volume-weighted average wholesale price of solar fleet generation hours

Eligible Generation = Total generation excluding negative price hours

- Orientation-dependent: 3.3-6.5% generation excluded
- Creates differential Marktprämie base across configurations

Monatsmarktwert Solar calculation: Table 8a presents representative monthly Monatsmarktwert values demonstrating seasonal variation.

Month	Monatsmarktwert (€/MWh)	Fleet Generation Pattern	Price Characteristics
January	100.72	Low generation	High winter prices
February	104.21	Low generation	Peak winter prices
March	78.45	Rising generation	Transition period
April	60.29	High generation	Spring price decline
May	49.77	Peak generation	Low seasonal prices
June	42.67	Peak generation	Minimum prices
July	56.61	Peak generation	Moderate summer
August	47.14	High generation	Low summer prices
September	68.39	Declining generation	Autumn recovery
October	68.12	Low generation	Price increase
November	85.15	Low generation	Winter transition
December	92.34	Minimum generation	High winter prices

Table 8a: Monthly Monatsmarktwert Solar (2026)

Summer months (May-August) demonstrate minimum Monatsmarktwert values (42-57 €/MWh) when solar fleet generation peaks suppress midday prices, while winter months (November-February) show elevated values (85-104 €/MWh) reflecting low solar output coinciding with heating demand. This seasonal pattern creates corresponding Marktprämie variation: maximum premiums during summer when AW-MMW differential widens, minimum premiums during winter when differential narrows or reverses (negative Marktprämie possible when MMW exceeds AW).

Marktprämie calculation examples: Table 8b demonstrates monthly Marktprämie calculations for representative configurations across high and low differential months.

Configuration	Month	AW (€/MWh)	MMW (€/MWh)	Eligible_Gen (kWh)	Marktprämie (€)
W90_1MW	May	82.3	49.77	137,902.20	4,486.09
	September		68.39	97,169.41	1,351.78
W90_0.9MW	April	95	60.29	123,078.16	4,271.99
	August		47.14	149,120.85	7,137.26
E90_1MW	November	82.3	85.15	19,089.60	0.00

	June		42.67	125,713.80	4,981.63
E90_0.9MW	February	95	104.21	28,416.28	0.00
	July		56.61	141,556.84	5,434.22
S30_1MW	January	70.3	100.72	31,501.57	0.00
	May		49.77	128,167.11	2,631.40
S30_0.9MW	October	70	68.12	49,588.76	93.23
	August		47.14	145,355.91	3,323.16

Table 8b: Marktprämie Calculation Examples - Selected Months (2026)

Annual Marktprämie outcomes: Year-one (2026) Marktprämie revenues demonstrate dramatic capacity threshold differentiation:

- **999 kWp configurations** (heightened agrivoltaic bonus): - W90_0.9MW: €39,650 (highest) - E90_0.9MW: €32,076 - S30_0.9MW: €12,941
- **1,001 kWp configurations** (reduced agrivoltaic bonus): - W90_1MW: €24,981 - E90_1MW: €21,623 - S30_1MW: €13,225

W90_0.9MW achieves €39,650 Marktprämie, exceeding W90_1MW by €14,669 (+58.7%) despite 0.2% lower capacity. This substantial differential reflects three compounding factors which are higher AW (95.0 vs 82.3 €/MWh), lower ineligibility (3.31% vs 4.77%) & higher total generation (1,176,631 vs 1,124,120 kWh)

However, the €39,650 Marktprämie represents only 32.2% of W90_0.9MW total revenue (€123,207), confirming a supplementary rather than dominant role. Spot market revenue (€83,557, 67.8% of total) drives economic outcomes with Marktprämie providing important but secondary support.

Temporal Marktprämie evolution: Figure 4.11 presents 20-year Marktprämie trajectories revealing non-monotonic patterns driven by evolving Monatsmarktwert-Anzulegender Wert differentials.

Configuration ranking remains stable: W90_0.9MW > W90_1MW > E90_0.9MW > E90_1MW > S30_1MW > S30_0.9MW across all years. The 999 kWp advantage persists throughout horizon, demonstrating robust policy framework leverage surviving market evolution.

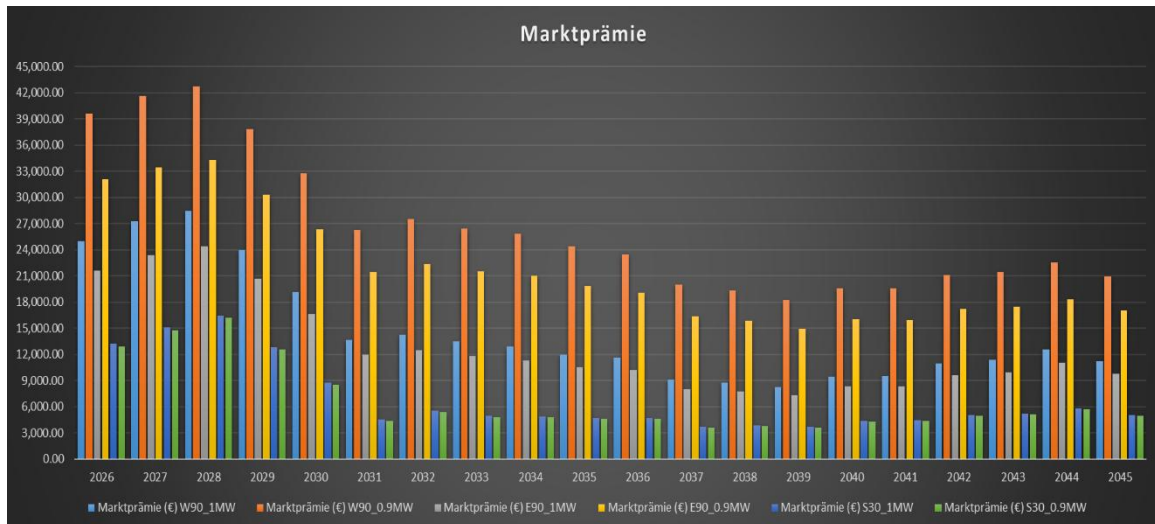


Figure 9: Annual Marktpremie Evolution (2026-2045)

Critical insight: Marktpremie declines 60-75% from 2026 peaks to 2036-2039 minimum, yet total revenues (Section 4.4.3) remain relatively stable due to spot market compensation. This confirms spot revenue dominance: Marktpremie fluctuations create modest total revenue variations rather than fundamental economic disruption.

Basic scenario Marktpremie: Under uniform 70.3 €/MWh remuneration (eliminating agri-voltaic bonuses), Marktpremie compression demonstrates policy framework contribution isolation. Year-one Basic scenario outcomes:

- **999 kWp Basic Marktpremie** (vs Real scenario): - W90_0.9MW: €14,442 Basic vs €39,650 Real (€25,208 reduction, -63.6%) - E90_0.9MW: €11,947 Basic vs €32,076 Real (€20,129 reduction, -62.8%) - S30_0.9MW: €13,188 Basic vs €12,941 Real (€247 increase, +1.9%)
- **1,001 kWp Basic Marktpremie** (vs Real scenario): - W90_1MW: €13,769 Basic vs €24,981 Real (€11,212 reduction, -44.9%) - E90_1MW: €12,040 Basic vs €21,623 Real (€9,583 reduction, -44.3%) - S30_1MW: €13,225 Basic = €13,225 Real (€0 change, 0%)

The S30_0.9MW anomaly reflects unusual AW assignment: 70.0 €/MWh (Real) vs 70.3 €/MWh (Basic), creating a slight Basic scenario advantage. This demonstrates that even minor AW variations (0.3 €/MWh, 0.4% difference) produce meaningful Marktpremie impacts when applied across 1,000+ MWh eligible generation.

4.4.3 Total Annual Revenue Streams

Total revenue integration combines dominant spot market and supplementary Marktpremie components under alternative operational scenarios. Table 9 presents comprehensive year-one (2026) revenue breakdown across all configurations and scenarios.

Spot market contributions dominate across all viable configurations (68-83% range), with Marktpremie providing 17-32% supplementary support. South-facing demonstrates highest spot dependence (83%) due to minimal Marktpremie from low AW and high ineligibility (6.48%). West-facing achieves more balanced distribution (68% spot, 32% Marktpremie) through superior market timing enabling high spot revenue combined with heightened AW providing substantial premium.

	Spot Revenue (€)		Marktprämie (€)		Total revenue (€)			
	No Cur- tailment	Curtailment	No Curtailment & Curtailment		No Curtailment		Curtailment	
	Real & Basic		Real	Basic	Real	Basic	Real	Basic
W90_1MW	76,131.04	76,948.84	24,980.82	13,769.27	101,111.86	89,900.31	101,929.66	90,718.11
W90_0.9MW	82,836.70	83,557.03	39,649.78	14,441.72	122,486.48	97,278.42	123,206.81	97,998.75
E90_1MW	59,779.43	60,558.10	21,622.79	12,040.25	81,402.22	71,819.68	82,180.89	72,598.35
E90_0.9MW	59,307.56	60,081.20	32,075.66	11,946.99	91,383.22	71,254.55	92,156.86	72,028.18
S30_1MW	62,741.43	63,959.48	13,224.97	13,224.97	75,966.40	75,966.40	77,184.45	77,184.45
S30_0.9MW	62,319.33	63,534.27	12,941.19	13,188.06	75,260.52	75,507.39	76,475.45	76,722.32

Table 9: Total Annual Revenue by Configuration and Scenario (2026)

Real_Curtailment (optimal scenario):

W90_0.9MW emerges as unambiguous revenue leader, achieving €123,207 total revenue representing 59.6% premium over S30_1MW baseline (€77,184). This extraordinary advantage derives from compounding benefits:

- **Generation volume:** +80,393 kWh vs S30_1MW (+7.3%)
- **Market timing:** MVF 0.85 vs 0.70 (+21.4% weighted pricing)
- **Negative price avoidance:** 3.31% vs 6.48% ineligible (-48.9% exposure)
- **Policy framework leverage:** 95.0 vs 70.3 €/MWh AW (+35.1% remuneration)

Each advantage multiplies upon others: superior generation creates a larger base for market timing benefits, which preserves more eligible generation for heightened remuneration rates, producing geometric revenue growth.

Basic_Curtailment (uniform remuneration):

Under uniform remuneration, west-facing maintains substantial superiority (+17.5-27.0%), confirming orientation optimization value persists independent of policy framework leverage. However, advantage magnitude compresses dramatically (27-59% reduction vs Real scenario differentials), demonstrating that agrivoltaic bonuses amplify fundamental technical advantages.

East-facing rankings demonstrate scenario-dependent behavior: E90 consistently ranks bottom for Basic scenarios but achieves mid-tier positions (ranks #3-4) for Real scenarios, surpassing both south-facing configurations. This reflects E90's substantial agrivoltaic bonus access (+25.0 €/MWh for 999 kWp, +12.0 €/MWh for 1,001 kWp), partially compensating for catastrophic generation failure. Under uniform remuneration after eliminating bonuses, fundamental performance inadequacy emerges, producing expected bottom rankings.

4.4.4 Revenue Decomposition by Component

Revenue component decomposition visualizes relative contribution magnitudes, revealing spot market dominance across configurations and scenarios. Figure 4.13 presents stacked revenue composition for year-one operation.

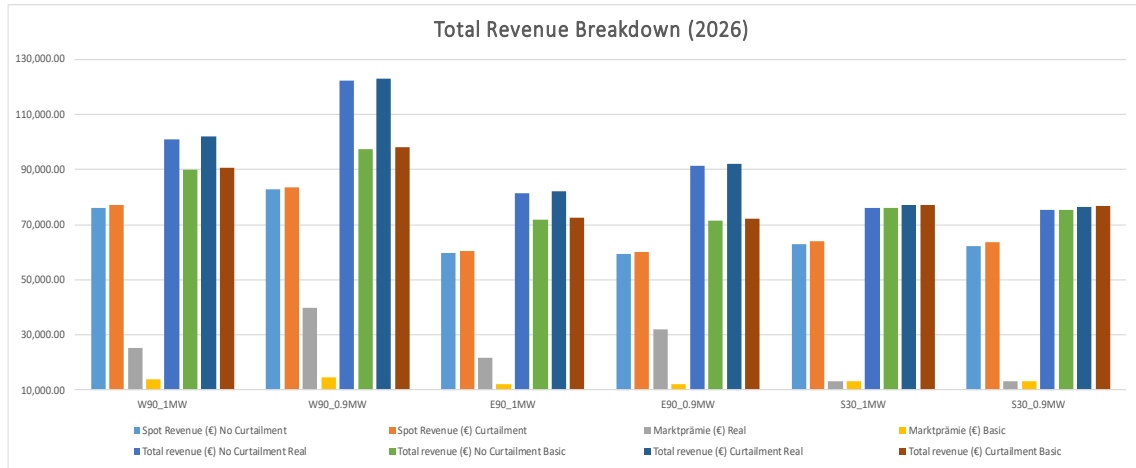


Figure 10: Revenue Component Decomposition by Configuration (2026)

The figure demonstrates the spot market as the primary revenue driver, with Marktprämie segment providing an important but clearly secondary contribution. Even for the configuration receiving maximum agrivoltaic bonus (95.0 €/MWh), policy support represents less than one-third of total revenues.

Basic scenario comparison: When uniform remuneration eliminates agrivoltaic bonuses, Marktprämie segments compress dramatically while spot segments remain constant, producing taller overall bars for configurations previously receiving minimal bonuses (S30) and shorter bars for configurations losing substantial bonuses (W90_0.9MW, E90_0.9MW).

W90_0.9MW Basic vs Real decomposition:

- Real: €83,557 Spot + €39,650 Marktprämie = €123,207 total (32.2% Marktprämie)
- Basic: €83,557 Spot + €14,442 Marktprämie = €97,999 total (14.7% Marktprämie)
- **Policy contribution: €25,208** (20.5% of Real total revenue)

W90 capacity comparison (Real_Curtailment):

- Spot revenue differential: W90_0.9MW €83,557 vs W90_1MW €76,949 = +€6,608
- Marktprämie differential: W90_0.9MW €39,650 vs W90_1MW €24,981 = +€14,669
- **Total differential: €21,277** (69.0% Marktprämie-driven, 31.0% spot-driven)

The capacity reversal depends on both components: W90_0.9MW superior spot revenue (+€6,608 from generation and market timing advantages) compounds with dramatic Marktprämie advantage (+€14,669 from AW differential and eligible generation compounding). Policy framework provides larger absolute contribution (€14,669 vs €6,608), though spot revenue advantage proves necessary condition enabling reversal viability.

Critical insight: Spot market revenue dominance (67-83% of total) fundamentally revises understanding of photovoltaic economics under German EEG 2023 framework. Installations achieving strong market timing through orientation optimization (west-facing bimodal generation) realize

commercial viability primarily through wholesale market performance, with Marktprämie providing valuable but supplementary support enhancing rather than enabling economic feasibility. This finding suggests that policy framework evolution reducing agrivoltaic bonuses would impact absolute revenue levels but maintain fundamental west-facing superiority, as 68-75% of economic advantage derives from market performance independent of policy support.

4.5 Present Value Outcomes

4.5.1 Real Scenarios: Curtailment vs. No Curtailment

Real scenarios incorporate full agrivoltaic policy framework support with differentiated remuneration rates (95.0 €/MWh for 999 kWp, 82.3 €/MWh for 1,001 kWp, 70.3/70.0 €/MWh for S30) combined with alternative operational strategies. Table 10 presents cumulative present value outcomes.

Year	Scenario	W90_1MW	W90_0.9MW	E90_1MW	E90_0.9MW	S30_1MW	S30_0.9MW
2026	No Curtailment	97,222.94	117,775.46	78,271.36	87,868.49	73,044.61	72,365.88
	Curtailment	98,009.29	118,468.08	79,020.08	88,612.37	74,215.82	73,534.09
2027	No Curtailment	92,573.82	111,997.18	74,195.02	83,040.49	68,653.24	67,991.39
	Curtailment	93,633.73	112,950.40	75,205.60	84,044.57	70,191.11	69,525.43
2031	No Curtailment	83,015.24	99,105.80	66,343.30	73,380.00	63,969.96	63,425.83
	Curtailment	83,584.72	99,606.42	66,886.22	73,919.41	64,820.18	64,273.77
2032	No Curtailment	78,491.47	94,291.38	62,851.85	69,969.50	60,877.04	60,384.97
	Curtailment	79,003.83	94,743.30	63,340.64	70,455.14	61,643.83	61,149.70
2035	No Curtailment	70,243.80	83,799.32	56,121.31	62,021.60	55,336.66	54,918.15
	Curtailment	70,709.17	84,212.40	56,565.12	62,462.54	56,026.99	55,606.63
2039	No Curtailment	62,229.26	72,767.10	49,490.60	53,525.41	50,499.37	50,113.62
	Curtailment	62,660.50	73,151.64	49,902.00	53,934.15	51,137.05	50,749.59
2040	No Curtailment	59,396.77	69,542.02	47,270.87	51,184.34	47,997.81	47,628.70
	Curtailment	59,816.66	69,916.60	47,671.57	51,582.45	48,621.62	48,250.84
2044	No Curtailment	51,247.25	60,001.85	40,320.52	43,523.58	39,896.55	39,561.66
	Curtailment	52,085.40	60,779.01	41,119.63	44,317.64	41,068.89	40,731.11
2045	No Curtailment	50,033.64	58,498.91	39,280.34	42,319.97	39,157.76	38,846.55
	Curtailment	50,906.35	59,313.34	40,111.77	43,146.14	40,363.29	40,049.15
SUM PV	No Curtailment	1,417,671.14	1,688,743.60	1,129,394.51	1,244,488.42	1,098,308.48	1,089,166.49
	Curtailment	1,431,767.88	1,701,471.01	1,142,834.40	1,257,842.22	1,118,702.74	1,109,508.79

Table 10: Present Value Summary - Real Scenarios (2026-2045)

W90_0.9MW achieves maximum present value (€1,701,471 under curtailment), representing 54.92% premium over S30_1MW baseline (€1,118,703). This extraordinary long-term advantage is similar to the year-one revenue differential (59.6%). The cumulative advantage decomposition:

- **Spot revenue superiority:** W90_0.9MW generates ~€1,270,000 cumulative spot PV vs S30_1MW ~€920,000 (+€350,000, +38.0%)
- **Marktprämie advantage:** W90_0.9MW generates ~€431,000 cumulative Marktprämie PV vs S30_1MW ~€199,000 (+€232,000, +116.6%)
- **Total differential:** €582,768 combined advantage

Spot revenue differential provides a larger absolute contribution (€350,000 vs €232,000), confirming market performance drives long-term economic outcomes despite Marktprämie showing a higher percentage advantage. The 54.92% reflects multiplicative compounding: each year's superior performance builds upon previous advantages through consistent generation, market timing, and policy framework leverage.

Capacity threshold present value outcomes:

West-facing reversal (999 kWp dramatically superior):

- W90_0.9MW: €1,701,471
- W90_1MW: €1,431,768
- Differential: €269,703 (18.8% advantage for 0.2% less capacity)

South-facing proportional (marginal 1,001 kWp advantage):

- S30_1MW: €1,118,703
- S30_0.9MW: €1,109,509
- Differential: €9,194 (0.8% advantage for 0.2% more capacity)

East-facing mid-tier performance: E90 configurations achieve ranks #3-4 under Real scenarios despite catastrophic generation failure, reflecting substantial agrivoltaic bonus compensation. E90 present values remain substantially below west-facing equivalents (€459,000-558,000 deficits, 27-33% underperformance), confirming that policy support provides insufficient compensation for fundamental generation inadequacy. East-facing proves economically inferior to west-facing across all scenarios despite policy framework assistance.

Curtailment value across project lifetime: Economic benefit of negative price avoidance compounds over 20 years, with magnitude varying by configuration exposure:

- **W90_0.9MW:** €12,727 curtailment PV benefit (0.75% of total PV)
- **W90_1MW:** €14,097 curtailment PV benefit (0.99% of total PV)
- **S30_1MW:** €20,395 curtailment PV benefit (1.86% of total PV)
- **S30_0.9MW:** €20,343 curtailment PV benefit (1.87% of total PV)

South-facing experiences 1.6× greater curtailment benefits than west-facing equivalents (€20,000 vs €13,000 range), reflecting 2× higher negative price exposure, creating larger avoidance opportunities. However, absolute benefits remain modest relative to total PV (0.8-1.9% improvement), representing meaningful but not transformative economic value.

Temporal evolution visualization: Figure 11 presents annual present value contributions across 20-year horizon.

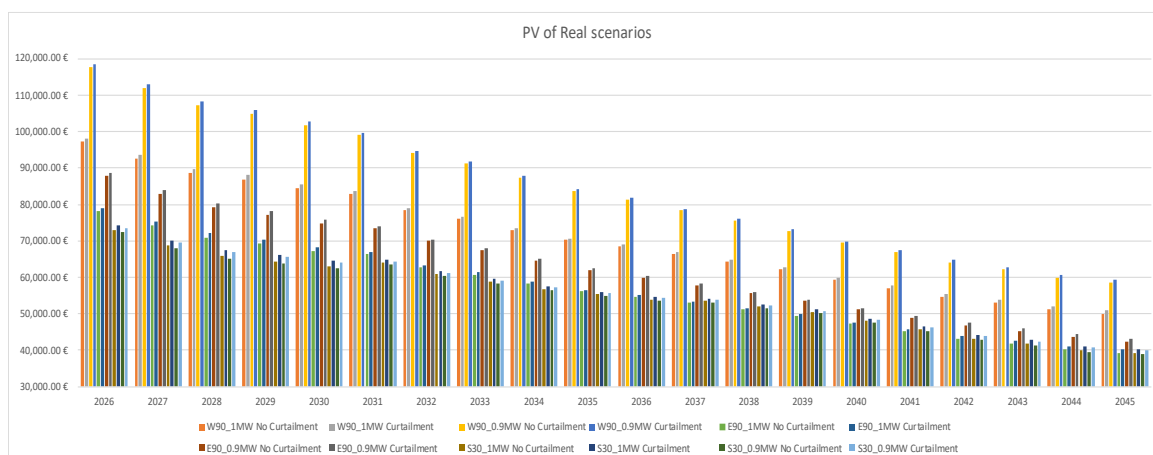


Figure 11: Annual Present Value Contributions by Configuration (2026-2045) - Real Scenarios

Early-year revenues contribute disproportionately to total PV due to discount rate effects. Year-one (2026) contribution receives $1.0\times$ weight, year-10 (2035) receives $0.676\times$ weight, year-20 (2045) receives $0.456\times$ weight. This temporal discounting emphasizes near-term performance superiority: W90_0.9MW immediate revenue advantages (€123,207 year-one vs €77,184 S30_1MW) contribute more heavily to PV differential than equivalent late-period advantages.

4.5.2 Basic Scenarios: Uniform Remuneration Framework

Basic scenarios eliminate agrivoltaics bonuses applying a uniform 70.3 €/MWh Anzulegender Wert across all configurations regardless of capacity or orientation. Table 11 presents Basic scenario present value outcomes isolating pure generation and market timing effects.

	Scenario	W90_1MW	W90_0.9MW	E90_1MW	E90_0.9MW	S30_1MW	S30_0.9MW
2026	No Curtailment	86,442.61	93,536.94	69,057.38	68,513.99	73,044.61	72,603.26
	Curtailment	87,228.96	94,229.56	69,806.10	69,257.87	74,215.82	73,771.46
2027	No Curtailment	82,170.52	89,074.80	65,370.59	64,852.17	68,653.24	68,234.29
	Curtailment	83,230.43	90,028.01	66,381.17	65,856.25	70,191.11	69,768.33
2031	No Curtailment	76,037.13	82,374.30	60,294.51	59,822.10	63,969.96	63,560.86
	Curtailment	76,606.61	82,874.92	60,837.43	60,361.51	64,820.18	64,408.81
2032	No Curtailment	72,079.52	78,048.56	57,292.46	56,843.52	60,877.04	60,488.71
	Curtailment	72,591.88	78,500.48	57,781.25	57,329.15	61,643.83	61,253.44

2036	No Cur-tailment	64,171.55	69,569.14	50,784.55	50,385.23	53,927.75	53,573.47
	Cur-tailment	64,714.35	70,052.47	51,301.44	50,898.79	54,726.31	54,369.92
2037	No Cur-tailment	63,287.52	68,538.89	50,137.10	49,742.68	53,483.68	53,131.85
	Cur-tailment	63,730.04	68,931.52	50,559.30	50,162.15	54,141.27	53,787.68
2041	No Cur-tailment	54,604.32	59,245.18	42,975.85	42,635.46	45,623.30	45,313.29
	Cur-tailment	55,247.40	59,835.28	43,589.89	43,245.60	46,535.91	46,223.60
2044	No Cur-tailment	48,315.64	52,550.11	37,784.93	37,486.19	39,896.55	39,616.99
	Cur-tailment	49,153.79	53,327.27	38,584.03	38,280.25	41,068.89	40,786.45
2045	No Cur-tailment	47,478.77	51,659.02	37,065.44	36,771.53	39,157.76	38,880.68
	Cur-tailment	48,351.49	52,473.44	37,896.87	37,597.71	40,363.29	40,083.28
To-tal PV	No Cur-tailment	1,310,974.56	1,421,443.62	1,037,458.20	1,029,276.18	1,098,308.48	1,091,193.23
	Cur-tailment	1,325,071.31	1,434,171.03	1,050,898.09	1,042,629.97	1,118,702.74	1,111,535.54

Table 11: Present Value Summary - Basic Scenarios (2026-2045)

West-facing maintains substantial superiority under uniform remuneration, confirming that orientation optimization value persists independent of agrivoltaic policy support. However, advantage magnitude compresses dramatically compared to Real scenarios:

W90_0.9MW advantage compression:

- Real scenario: +54.92% vs baseline (€1,701,471 vs €1,118,703)
- Basic scenario: +30.58% vs baseline (€1,434,171 vs €1,118,703)
- Policy amplification: 1.79× multiplication of fundamental advantage

Capacity ordering reverses under uniform remuneration:

For west-facing installations, W90_0.9MW ranks higher (rank #1) than W90_1MW (rank #2) despite uniform AW, due to generation superiority, reflecting W90_0.9MW's generation advantage (+52,511 kWh/year) that overcomes any minor disadvantages. For south-facing, S30_1M ranks 3rd, while S30_0.9MW ranks 4th, demonstrating the expected proportional relationship under uniform remuneration: a 0.2% capacity advantage produces a 0.6% PV premium. As for east-facing installations, these configurations achieve ranks #5-6 (€1,042,630-1,050,898), falling below both south-facing alternatives. This ranking reversal (vs Real scenarios where E90 exceeded S30) demonstrates catastrophic generation failure emergence when policy framework compensation is removed. Under uniform remuneration, fundamental performance inadequacy dominates:

13.5-13.7% generation deficit vs S30 produces a proportional economic penalty unmitigated by Marktprämie advantages.

Real vs Basic comparison quantifies policy framework contribution:

999 kWp policy value (Curtailment scenarios):

- W90_0.9MW: €1,701,471 Real vs €1,434,171 Basic = **€267,300 policy contribution** (18.6% of Real PV)
- E90_0.9MW: €1,257,842 Real vs €1,042,630 Basic = **€215,212 policy contribution** (20.6% of Real PV)
- S30_0.9MW: €1,109,509 Real vs €1,111,536 Basic = **-€2,027 policy cost** (-0.2% of Real PV)

1,001 kWp policy value (Curtailment scenarios):

- W90_1MW: €1,431,768 Real vs €1,325,071 Basic = **€106,697 policy contribution** (8.0% of Real PV)
- E90_1MW: €1,142,834 Real vs €1,050,898 Basic = **€91,936 policy contribution** (8.7% of Real PV)
- S30_1MW: €1,118,703 Real vs €1,118,703 Basic = **€0 policy contribution** (Baseline)

The 999 kWp policy contribution (€215,000-267,000) dramatically exceeds 1,001 kWp equivalent (€92,000-107,000), confirming agrivoltaic value concentrates in the heightened bonus tier (+25.0 €/MWh), providing 2.5-3× greater absolute benefit than base tier (+12.0 €/MWh). The S30_0.9MW negative policy contribution reflects Anzulegender Wert assignment anomaly: Real scenario receives 70.0 €/MWh while Basic receives 70.3 €/MWh, creating a counterintuitive €2,027 Basic scenario advantage. This demonstrates extreme sensitivity to minor AW variations when applied across 1,000+ MWh × 20 years.

Temporal evolution visualization: Figure 12 presents Basic scenario annual PV contributions.

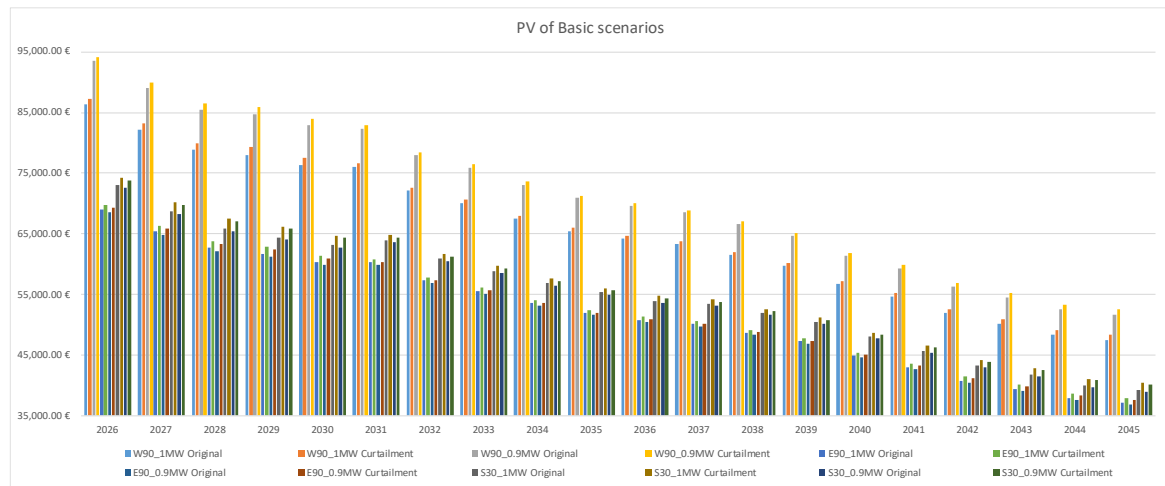


Figure 12: Annual Present Value Contributions by Configuration (2026-2045)

4.5.3 Configuration Rankings and Comparative Performance

Comprehensive configuration rankings relative to the conventional S30_1MW baseline enable clear communication of optimization value across scenarios. Table 12 presents percentage differentials facilitating intuitive interpretation.

Configuration	Real Curtailment	Real No Curtailment	Basic Curtailment	Basic No Curtailment
W90_0.9MW	54.92%	53.76%	30.58%	29.42%
W90_1MW	30.36%	29.08%	20.65%	19.36%
E90_0.9MW	14.53%	13.31%	-5.07%	-6.29%
E90_1MW	4.05%	2.83%	-4.32%	-5.54%
S30_1MW	BASELINE			
S30_0.9MW	1.02%	-0.83%	1.20%	-0.65%

Table 12: Configuration Performance Relative to S30_1MW Baseline

Real scenarios demonstrate extraordinary west-facing advantages (+29-55%). W90_0.9MW achieves 54.92% advantage, equivalent to a complete additional installation revenue stream over baseline configuration. West-facing configurations maintain top 2 positions across all four scenarios, demonstrating robust superiority surviving alternative policy frameworks and operational strategies.

E90 rankings reveal dramatic policy framework dependency: in Real scenarios E90 achieves +2-15% premium (ranks #3-4, exceeding both S30 configs) but in Basic scenarios: E90 achieves -4 to -6% deficit (ranks #5-6, bottom positions)

The $\pm 1\%$ variation across S30_0.9MW scenarios reflects competing factors: 0.3 €/MWh Anzulegenderwert disadvantage (Real) or 0.3 €/MWh advantage (Basic) versus 0.6% generation deficit producing nearly offsetting effects. Curtailment scenarios favor S30_0.9MW through superior negative price avoidance (6.50% vs 6.48% ineligible, marginal advantage), while no-curtailment scenarios penalize through larger negative revenue exposure.

East-facing and south-facing demonstrate scenario-dependent mid-tier competition (ranks #3-6 varying), suggesting these represent intermediate choices neither optimally positioned nor catastrophically flawed under specific scenario combinations.

4.6 Orientation Effects

4.6.1 Generation Volume Effects by Orientation

Annual generation variation across orientations reflects solar geometry interactions with module positioning, bifacial technology characteristics, and shading loss patterns documented in Sections 4.1.1, 4.1.2, and 4.1.4. Table 13 reorganizes these generation outcomes by orientation for comparative analysis.

Orientation	Configuration	Capacity (kWp)	Annual Gen (kWh)	Specific Yield (kWh/kWp)	vs S30_1MW	Shading Loss %	PR %
WEST	W90_0.9MW	999.18	1,176,631	1,177.60	7.30%	2.86%	81.90%

	W90_1MW	1,001.00	1,124,120	1,123.00	2.50%	2.45%	81.70%
SOUTH	S30_1MW	1,001.00	1,096,238	1,095.14	BASELINE	9.06%	84.80%
	S30_0.9MW	999.18	1,090,107	1,091.00	-0.60%	9.28%	84.50%
EAST	E90_0.9MW	999.18	948,440	949.22	-13.50%	19.93%	70.30%
	E90_1MW	1,001.00	945,927	944.98	-13.70%	19.60%	70.00%

Table 13: Generation Volume Analysis by Orientation

As detailed in Section 4.1.1, west-facing W90 achieves the highest absolute generation through vertical bifacial geometry enabling substantial rear-side irradiance capture. The combination of front-side afternoon optimization, rear-side morning collection, and minimal inter-row shading produces 1,124-1,177 MWh annually, exceeding south-facing bifacial tilted arrays.

The capacity-dependent west-facing inversion documented in Section 4.1.1 reflects inverter sizing optimization: W90_0.9MW (999 kWp) generates 52,511 kWh more than W90_1MW (1,001 kWp) annually (+4.7% despite -0.2% capacity) through aggressive 1.15 DC/AC ratio enabling sustained afternoon plateau utilization versus conservative 1.03 ratio creating inverter undersizing during afternoon generation peaks.

South-facing configurations achieve expected mid-tier generation (1,090-1,096 MWh) through front-side irradiance maximization with moderate 9.06-9.28% inter-row shading from seasonal low-angle effects, as analyzed in Section 4.1.4. The tilted geometry prevents meaningful rear-side contribution, producing acceptable but suboptimal absolute generation.

East-facing demonstrates catastrophic failure with 19.60-19.93% shading losses (Section 4.1.4) creating 70.0-70.3% performance ratio far below commercial viability threshold. The extreme inter-row shadows from low-angle morning sun combined with row spacing optimized for west-facing (10-12m) renders east-facing commercially non-viable under agrivoltaic dual-use constraints, as comprehensively detailed in Section 4.1.4.

Orientation ranking (generation volume):

- W90_0.9MW: 1,176,631 kWh (optimal, +7.3% vs baseline)
- W90_1MW: 1,124,120 kWh (+2.5% vs baseline)
- S30_1MW: 1,096,238 kWh (baseline reference)
- S30_0.9MW: 1,090,107 kWh (-0.6% vs baseline)
- E90_0.9MW: 948,440 kWh (-13.5% vs baseline, catastrophic)
- E90_1MW: 945,927 kWh (-13.7% vs baseline, catastrophic)

4.6.2 Temporal Distribution and Market Timing Effects

Beyond absolute generation volumes, orientation determines temporal distribution affecting market timing and agricultural synchronization. Integration with 2024 historical data validates projected correlation patterns.

2024 data validation: Table 14 compares actual 2024 price-generation correlations with 2026 projection average validating methodology.

Orientation	Capacity	2024 Actual r	2026 Projected r	Alignment
SOUTH	1,001 kWp	-0.382	-0.35	Negative confirmed
SOUTH	999 kWp	-0.381	-0.35	Negative confirmed

WEST	1,001 kWp	-0.254	-0.242	Less negative confirmed
WEST	999 kWp	-0.249	-0.198	Least negative confirmed
EAST	1,001 kWp	-0.315	-0.297	Moderate negative confirmed
EAST	999 kWp	-0.314	-0.297	Moderate negative confirmed

Table 14: Correlation Coefficient Validation - 2024 Actual vs 2026 Projected

The 2024 actual correlations align with projected patterns: all orientations demonstrate negative correlations with west-facing achieving the least-negative values (-0.198 to -0.242), south-facing the most-negative (-0.35), and east-facing intermediate (-0.297). This historical validation confirms that universal negative correlations represent a fundamental photovoltaic market timing challenge rather than a modeling artifact, with orientation selection determining severity rather than direction of misalignment.

The 2024-2026 correlation differences likely reflect annual price pattern variations and fleet composition evolution. However, relative ordering remains perfectly preserved in both years, validating that orientation-dependent timing advantages persist across market condition variations.

Daily generation patterns and market timing:

- **West-facing bimodal advantage:** Morning peak (717.98 kW hour 7) aligns with moderate plateau pricing (86.91 €/MWh), while afternoon secondary peak (702.89 kW hour 16) captures price recovery phase (36.91 €/MWh). Extended afternoon generation (hours 14-17 sustained 400-550 kW) reduces concentrated midday through exposure, achieving a market value factor of 0.82-0.85.
- **South-facing concentrated penalty:** Single midday peak (957.85 kW hour 11) coincides with price decline (35.07 €/MWh), with 65% daily generation concentrated hours 9-14 during minimum pricing period. This temporal concentration creates severe market value factor penalty (0.68-0.70), realizing only 68-70% of flat-profile revenue potential.
- **East-facing intermediate timing:** Morning concentration (794.67 kW hour 8) aligns with moderate plateau (73.71 €/MWh), achieving intermediate market value factor (0.75-0.76). However, substantial generation continues through midday hours, compromising timing advantage versus west-facing afternoon distribution.
- **Seasonal distribution consistency:** All orientations demonstrate summer-concentrated patterns (55-59% annual generation May-August) with minimal differentiation. Vertical configurations (W90, E90) fail to deliver hypothesized seasonal distribution advantages, suggesting irradiance availability dominates orientation effects for seasonal patterns under the examined latitude and climate conditions.

4.6.3 Revenue Outcome Synthesis

Integration of generation volume, market timing, and policy framework effects produces comprehensive orientation-dependent economic hierarchy. Table 15 synthesizes multi-factor performance.

Orientation	Best Config	Generation (kWh)	Correlation (r)	MVF	Spot Rev €/kWh	PV Real €	PV Basic €	vs S30_1MW %
WEST	W90_0.9MW	1,176,631	-0.198	0.85	0.0717	1,701,471	1,434,171	+52.09% Real, -

								28.20% Basic
SOUTH	S30_1MW	1,096,238	-0.35	0.7	0.0589	1,118,703	1,118,703	0.00% baseline
EAST	E90_0.9MW	948,440	-0.297	0.76	0.064	1,257,842	1,042,630	+12.44% Real, -6.80% Basic

Table 15: Comprehensive Orientation Comparison - Revenue and Present Value

West-facing comprehensive superiority derives from compounding advantages across all revenue determinants:

- Generation volume: +80,393 kWh vs S30_1MW (+7.34%)
- Market timing: MVF 0.85 vs 0.70 (+21.4% weighted pricing)
- Policy framework: 95.0 vs 70.3 €/MWh Anzulegenderwert

South-facing baseline adequacy: S30_1MW achieves acceptable commercial performance, representing a viable but suboptimal strategy. Competitive advantages include established technology, minimizing deployment risk, and simplified installation, reducing upfront costs, while competitive disadvantages include 7.3% generation deficit, 21.4% market timing penalty, and zero policy framework leverage, producing €582,768 lifetime PV sacrifice compared to W90_0.9MW.

East-facing categorical non-viability: E90 demonstrates multiple failure modes rendering commercial deployment inadvisable:

- Catastrophic 13.5% generation deficit from 19.6% shading losses
- 70% performance ratio far below commercial minimum (80%+ typical)
- Achieves +14.53% Real scenario premium ONLY through extraordinary policy support (€215,000 PV contribution)
- Falls -5.07% below baseline under Basic uniform remuneration, confirming fundamental inadequacy

The E90 present value range (+14.53% best-case Real to -5.07% worst-case Basic) demonstrates unacceptable policy framework dependency: commercial viability requires sustained agrivoltaic bonus support, creating regulatory risk exceeding any potential upside benefits.

4.7 Capacity Threshold Effects and Policy Framework Leverage

4.7.1 Orientation-Dependent Threshold Effects

Capacity threshold impacts vary substantially across orientations due to differential generation levels, market value factors, and negative price exposure creating orientation-specific policy framework leverage. Table 16 presents comprehensive capacity pair analysis.

Orientation	Capacity	Gen (kWh)	Spot Rev (€)	Marktprämie (€)	Total Revenue (€)	Revenue Diff vs 1MW
WEST	999 kWp	1,176,631	83,557	39,650	123,207	+€21,277 (+20.9%)
	1,001 kWp	1,124,120	76,949	24,981	101,930	-
EAST	999 kWp	948,440	60,081	32,076	92,157	+€9,976 (+12.1%)

	1,001 kWp	945,927	60,558	21,623	82,181	--
SOUTH	999 kWp	1,090,107	63,534	12,941	76,475	-€709 (-0.9%)
	1,001 kWp	1,096,238	63,959	13,225	77,184	-

Table 16: Capacity Threshold Analysis by Orientation in Year 2026

West-facing dramatic reversal: W90_0.9MW (999 kWp, smaller capacity) achieves €123,207 total revenue, exceeding W90_1MW (1,001 kWp, larger capacity) by €21,277 annually (+20.9%). This extraordinary outcome reflects three compounding mechanisms:

- **Generation paradox:** W90_0.9MW generates 52,511 kWh MORE than W90_1MW despite 0.2% less capacity & Contribution: +€6,608 spot revenue advantage
- **Marktprämie leverage:** 12.7 €/MWh AW differential applied to superior eligible generation
 - W90_0.9MW eligible: 1,137,694 kWh × 95.0 €/MWh premium ≈ €39,650 Marktprämie
 - W90_1MW eligible: 1,071,154 kWh × 82.3 €/MWh premium ≈ €24,981 Marktprämie
 - **Contribution:** +€14,669 Marktprämie advantage (+58.7%)

The €21,277 advantage in year 2026 over a 20-year horizon produces €269,703 present value differential. This represents that policy framework threshold positioning can reverse fundamental capacity-revenue relationships when combined with favorable technical characteristics.

East-facing moderate reversal: E90_0.9MW achieves €92,157 total revenue, exceeding E90_1MW by €9,976 (+12.1%). Despite catastrophic generation performance (948,440 kWh, -13.5% vs S30_1MW), the heightened agrivoltaic bonus (95.0 vs 82.3 €/MWh) provides sufficient compensation:

- E90_0.9MW Marktprämie: €32,076 (34.8% of total revenue)
- E90_1MW Marktprämie: €21,623 (26.3% of total revenue)
- **Differential:** +€10,453 Marktprämie advantage

The €10,453 Marktprämie gain exceeds -€477 spot revenue deficit (E90_0.9MW slightly lower spot due to marginal generation disadvantage), producing net +€9,976 total revenue premium. This confirms that 12.7 €/MWh AW differential proves sufficient to reverse capacity-revenue relationships even for configurations experiencing severe performance deficits, though magnitude proves smaller than west-facing equivalent due to lower absolute generation base for premium application.

South-facing proportional relationship maintained: S30_0.9MW achieves €76,475 total revenue, falling €709 below S30_1MW (-0.9%). This outcome reflects a lower Anzulegender Wert. Applied to ~1,080 MWh eligible generation:

- AW penalty: 1,080 MWh × -0.3 €/MWh ≈ -€324 Marktprämie Deficit
- Generation deficit: ~0.6% × €63,959 spot ≈ -€384 spot deficit
- **Total deficit:** -€708 (matching observed -€709)

This demonstrates that south-facing capacity threshold optimization requires access to a heightened agrivoltaic bonus tier unavailable under examined policy framework assignments. Without bonus leverage, conventional capacity-revenue proportionality dominates producing expected larger-system superiority.

Present value threshold impacts: The annual revenue differentials compound over 20-year horizon creating dramatic lifetime economic consequences. Table 17 quantifies cumulative threshold effects.

Orientation	999 kWp PV (€)	1,001 kWp PV (€)	PV Differential (€)
WEST	1,701,471	1,431,768	269,703
EAST	1,257,842	1,142,834	115,008
SOUTH	1,109,509	1,118,703	-9,194

Table 17: Capacity Threshold Present Value Analysis

West-facing threshold optimization generates €269,703 lifetime advantage, substantially exceeding typical installation cost differential for 2 kWp capacity reduction (<€2,000 module cost). This extraordinary economic rationality confirms that capacity threshold optimization represents primary deployment strategy consideration for west-facing installations, potentially justifying significant design modifications to achieve sub-1MW positioning.

East-facing threshold optimization provides meaningful but smaller benefits (€115,008 PV), reflecting lower absolute generation base constraining policy framework leverage magnitude. However, the advantage still proves economically significant, suggesting capacity optimization remains advisable even for underperforming orientations when agrivoltaic bonus access is available.

South-facing minimal threshold penalty (-€9,194 PV, -0.8%) proves negligible relative to total project economics, suggesting capacity selection can prioritize generation maximization without meaningful economic sacrifice from policy framework positioning.

4.7.2 Capacity Threshold Present Value Analysis

Capacity threshold impacts manifest through present value outcomes under alternative policy evolution scenarios. Table 18 presents comprehensive scenario analysis quantifying threshold optimization value across regulatory futures.

Scenario	999 kWp AW	1,001 kWp AW	Differential	W90_0.9MW PV	W90_1MW PV	Threshold Value
Current	95	82.3	12.7	€ 1,701,471	€ 1,431,768	€ 269,703
Reduced bonus	85	76.3	8.7	€ 1,578,451	€ 1,373,065	€ 205,386
Uniform high	95	95	0	€ 1,701,471	€ 1,577,726	€ 123,745
Uniform low	70.3	70.3	0	€ 1,434,171	€ 1,325,071	€ 109,100
Eliminated	70	70	0	€ 1,431,905	€ 1,322,942	€ 108,964

Table 18: Policy Evolution Scenarios - Capacity Threshold Impact

Current framework maintenance (base case): Maximum threshold value (€269,703) derives from 12.7 €/MWh Anzulegender Wert differential applied to 1,100+ MWh eligible generation over 20 years. The W90_0.9MW capacity optimization generates 18.8% present value premium over W90_1MW despite 0.2% less installed capacity, representing extraordinary policy framework leverage opportunity.

Reduced bonus scenario (-40% agrivoltaic bonus reduction): Threshold value declines to €205,386 (-24% reduction) but remains economically substantial. The 8.7 €/MWh differential

(85.0 vs 76.3 €/MWh) applied to eligible generation base produces €205,000 lifetime advantage, continuing to justify threshold positioning strategy though with reduced safety margin.

Uniform remuneration scenarios: Complete elimination of Anzulegender Wert differentiation across capacity thresholds under uniform high (95.0 €/MWh for all capacities) or uniform low (70.3 €/MWh for all) scenarios. However, the threshold value persists at €109,100-123,745 purely from technical performance differentiation.

Eliminated scenario (complete agrivoltaic framework removal): Even with zero policy support (70.0 €/MWh uniform remuneration matching eliminated framework baseline), W90_0.9MW maintains €108,964 advantage through pure technical performance. The minimal difference between “uniform low” (€109,100) and “eliminated” (€108,964) scenarios confirms technical threshold value stability independent of specific remuneration levels.

4.8 Curtailment Value and Operational Strategy Optimization

4.8.1 Curtailment Mechanism and Economic Rationale

Curtailment strategy implementation requires real-time price monitoring coupled with automated control systems, disconnecting the market interface while maintaining generation operations. Technical implementation architecture includes:

- Price monitoring:** Real-time day-ahead market price data acquisition (updated hourly)
- Control logic:** Automated curtailment activation when price < €0/MWh threshold
- Generation continuation:** Inverters remain operational, modules continue producing
- Market disconnection:** Grid injection halted or energy diverted to on-site consumption/storage
- Revenue impact:** Zero revenue during curtailment hours (avoiding negative payments)

The economic rationale derives from EEG §51 negative price exclusion: curtailed generation during negative price hours generates zero revenue but also avoids negative payments while preserving equipment from thermal cycling damage. Eligible generation calculations automatically exclude negative price hours regardless of curtailment strategy, eliminating Marktprämie implications & therefore curtailment affects only spot revenue component.

Annual curtailment value quantification: Table 19 presents year-one (2026) curtailment benefits across configurations.

Config	Spot Revenue No-Curtail €	Spot Revenue Curtail €	Curtail Benefit €	Benefit %	Neg Price Hours	Avg Curtail Gen/Hr (kW)
W90_0.9MW	82,837	83,557	720	0.87%	149	470
W90_1MW	76,131	76,949	818	1.07%	149	450
E90_0.9MW	59,308	60,081	773	1.30%	149	380
E90_1MW	59,779	60,558	779	1.30%	149	375
S30_0.9MW	62,319	63,534	1,215	1.95%	149	485
S30_1MW	62,741	63,959	1,218	1.94%	149	490

Table 19: Annual Curtailment Value Analysis (2026)

South-facing achieves the highest absolute curtailment benefits (€1,215-1,218 annually), reflecting almost twice the negative price exposure (6.48-6.50% ineligible) than west-facing alternatives (3.31-4.77% ineligible). With 149 negative price hours in 2026, south-facing configurations generate substantial output during adverse periods (average 485-490 kW), creating meaningful avoidance value. Configurations with superior market timing (West-facing) realize the smallest curtailment benefits because fundamental temporal alignment already minimizes negative price exposure. Curtailment strategy provides greatest value for poorly-timed configurations (S30) attempting to mitigate inherent market timing disadvantages through operational intervention.

Curtailment benefit as percentage of total revenue:

- S30_1MW: €1,218 / €77,184 = **1.58% of total revenue**
- W90_0.9MW: €720 / €123,207 = **0.58% of total revenue**

Even in maximum-benefit configurations (S30), curtailment optimization yields only modest revenue enhancement (1-2% improvement) rather than a fundamental economic driver. This positions curtailment as incremental operational refinement rather than strategic configuration selection criterion.

4.8.2 Temporal Curtailment Evolution and Multi-Year Patterns

Curtailment value varies annually with negative price frequency evolution following three-phase pattern documented in Section 4.2.3. Figure 13 presents trajectories of 20-year curtailment benefits.

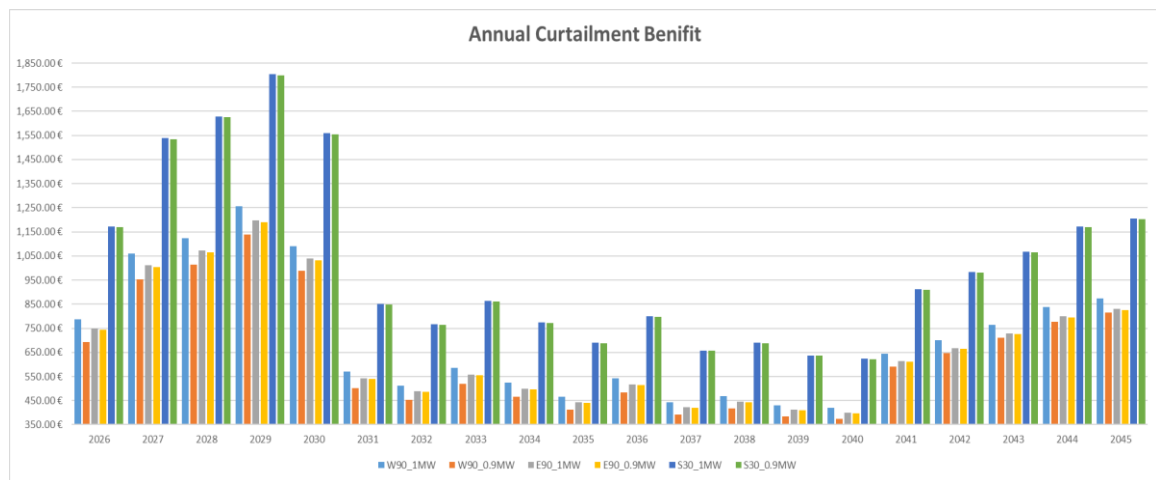


Figure 13: Annual Curtailment Value Evolution (2026-2045)

Curtailment benefits demonstrate non-monotonic evolution correlating with negative pricing frequency:

Early period (2026-2030): Escalating benefits from €720-1,218 baseline (2026) to €1,200-1,800 peak (2029) as negative price hours increase to 334 annually. South-facing peak curtailment value (€1,800 in 2029) represents 2.72% total revenue improvement, achieving maximum operational strategy significance.

Mid period (2031-2040): Stabilized minimal benefits €375-465 range as negative hours decline to 134-195 annually. Market adaptation through storage deployment and demand flexibility

reduces adverse pricing frequency, constraining curtailment value to <1% revenue impacts across all configurations.

Late period (2041-2045): Renewed escalation to €590-1,200 range as negative hours resurge to 268-322 annually, reflecting renewable penetration exceeding mitigation capacity. Late-period benefits approach but don't fully recover early-period peaks due to degradation effects, reducing absolute generation during curtailment hours.

Cumulative 20-year curtailment value: Table 20 presents lifetime curtailment benefit quantification.

South-facing achieves maximum lifetime curtailment value (€20,343-20,395 PV), representing 1.82-1.83% of total present value, meaningful but clearly secondary contribution. The €34,000-35,000 undiscounted cumulative benefit discounts to €20,000-21,000 present value, reflecting concentration of largest benefits in early years and mid-period years contributing less due to discount weights.

Configuration	PV Curtailment Benefit €	PV % of Total	Peak Year	Peak Benefit €
S30_1MW	20,395	1.82%	2029	1,803
S30_0.9MW	20,343	1.83%	2029	1,799
W90_1MW	14,097	0.99%	2029	1,257
E90_1MW	13,440	1.19%	2029	1,198
E90_0.9MW	13,354	1.08%	2029	1,190
W90_0.9MW	12,727	0.75%	2029	1,139

Table 20: Cumulative Curtailment Value (2026-2045)

West-facing realizes minimum lifetime curtailment value (€12,727-14,097 PV), representing only 0.75-0.99% of total present value. Superior baseline market timing reduces curtailment opportunity, creating counterintuitive outcome where best-performing configurations benefit least from operational optimization.

Peak year uniformity: All configurations experience maximum curtailment benefit in 2029 (€1,139-1,803 range) coinciding with peak negative pricing frequency (334 hours). The variation in peak benefits reflect configuration-specific negative price exposure differentials persisting across all years.

4.9 Policy Contribution Analysis and Framework Dependency

4.9.1 Policy Framework Contribution Quantification

Real scenarios incorporate full agrivoltaic policy support (capacity-differentiated Anzulegender Wert: 95.0, 82.3, or 70.0-70.3 €/MWh), while Basic scenarios apply uniform 70.3 €/MWh remuneration, eliminating bonus differentials. The present value differential between scenarios isolates the policy framework's economic contribution. Table 21 presents a comprehensive analysis of policy contributions.

Configuration	Real Scenario PV (€)	Basic Scenario PV (€)	Policy Contribution (€)	% of Real PV
999 kWp (Heightened Bonus Tier)				
W90_0.9MW	1,701,471	1,434,171	267,300	15.71%
E90_0.9MW	1,257,842	1,042,630	215,212	17.11%
S30_0.9MW	1,109,509	1,111,536	-2,027	-0.18%
1,001 kWp (Base Bonus Tier)				
W90_1MW	1,431,768	1,325,071	106,697	7.45%
E90_1MW	1,142,834	1,050,898	91,936	8.04%
S30_1MW	1,118,703	1,118,703	0	0.00%

Table 21: Policy Framework Contribution Analysis (2026-2045)

999 kWp generates dramatic policy contributions (€215,000-267,000 range), representing 15.7-17.1% of the total Real scenario present value. The 25.0 €/MWh agrivoltaic bonus applied to 1,100+ MWh eligible generation over 20 years produces €200,000-270,000 cumulative value, confirming that capacity threshold positioning below 1 MW creates substantial policy framework leverage.

W90_0.9MW achieves maximum absolute policy contribution (€267,300), reflecting the combined effects of the highest eligible generation base, maximum bonus differential & extended horizon application. The €267,300 policy contribution represents a meaningful but clearly secondary revenue stream relative to €103,000+ annual spot market revenues. Policy support enhances economic viability by increasing PV from €1,434,171 to €1,701,471, +18.6% improvement, with fundamental market performance drives €1,434,171 baseline, with policy providing incremental enhancement.

Base bonus tier produces modest policy contributions (€92,000-107,000 range), representing only 7.5-8.0% of total Real PV. The reduced 12.0 €/MWh bonus creates proportionally smaller absolute contribution despite similar generation bases, confirming that policy value concentrates in heightened bonus tier available exclusively for sub-1MW installations.

S30_0.9MW anomalous negative policy contribution (-€2,027, -0.2%) originates from Anzulegender Wert assignment peculiarity: Real scenario receives 70.0 €/MWh while Basic receives 70.3 €/MWh, creating counterintuitive €2,027 disadvantage under agrivoltaic framework. This demonstrates extreme sensitivity to minor remuneration variations (0.3 €/MWh, 0.4% differential) when applied across 1,000+ MWh over 20 years.

Policy contribution efficiency (contribution per eligible MWh): Normalizing policy contribution by cumulative eligible generation reveals bonus tier leverage differentials:

- **W90_0.9MW:** €267,300 / 21,802 MWh (20-year eligible) = **€12.26/MWh** policy value
- **W90_1MW:** €106,697 / 20,516 MWh (20-year eligible) = **€5.2/MWh** policy value
- **Differential:** **€7.06/MWh** additional value from heightened bonus tier

The €7.06/MWh efficiency premium confirms that sub-1MW positioning creates 2.4× policy leverage multiplier relative to 1-2 MW tier, validating capacity threshold optimization as a high-priority strategic consideration for eligible installations.

4.9.2 Temporal Policy Contribution Evolution

Policy framework contribution varies annually, with Monatsmarktwert evolution affecting the magnitude of the Marktprämie. Figure 14 presents the temporal policy contribution decomposition across a 20-year horizon.

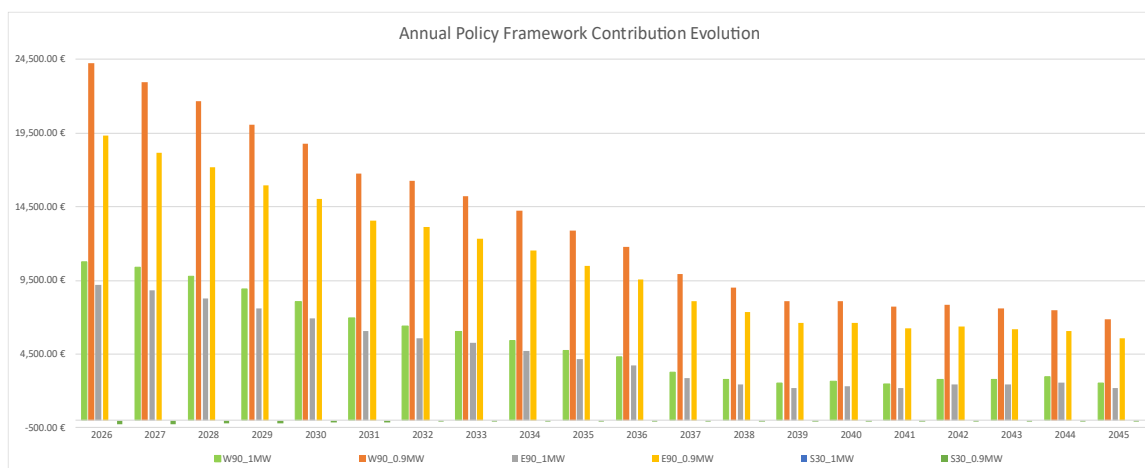


Figure 14: Annual Policy Framework Contribution Evolution (2026-2045)

Annual policy contributions demonstrate three distinct phases correlating with Marktprämie temporal patterns:

Early period (2026-2030): Maximum policy contributions ranging €8,900-24,200 annually for different configurations. W90_0.9MW achieves peak year-one contribution of €24,239, representing 61.13% of total year-one Marktprämie. The heightened Anzulegender Wert-Monatsmarktwert differentials (30-50 €/MWh) during the early period generate substantial premiums creating front-loaded policy value concentration.

Mid period (2031-2040): Compressed policy contributions declining to €2,500-16,700 range as Monatsmarktwert increases, with narrow AW-MMW differentials. Policy contribution minimum occurs 2035-2038 when wholesale prices approach Anzulegender Wert levels, reducing W90_0.9MW annual contribution to €8,100-13,000 range even for heightened bonus tier. This mid-period compression reflects market adaptation and price evolution temporarily reducing policy framework leverage.

Late period (2041-2045): Modest recovery to €6,800-7,800 range for W90_0.9MW as Monatsmarktwert stabilization restores moderate AW-MMW differentials. However, late-period contributions remain substantially below early-period peaks despite similar negative pricing frequencies, reflecting sustained wholesale price elevation relative to static Anzulegender Wert combined with degradation effects reducing absolute eligible generation.

Configuration-specific patterns: Table 22 presents annual policy contribution data revealing orientation and capacity-dependent characteristics.

W90_1MW	W90_0.9MW	E90_1MW	E90_0.9MW	S30_1MW	S30_0.9MW
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2026	10,780.34 €	24,238.52 €	9,213.98 €	19,354.50 €	0.00 €	-237.37 €
2027	10,403.30 €	22,922.38 €	8,824.43 €	18,188.32 €	0.00 €	-242.90 €
2028	9,776.69 €	21,654.53 €	8,298.80 €	17,174.40 €	0.00 €	-206.72 €
2029	8,917.89 €	20,059.76 €	7,596.13 €	15,956.33 €	0.00 €	-196.78 €
2030	8,052.34 €	18,768.09 €	6,908.56 €	15,006.19 €	0.00 €	-166.66 €
2031	6,978.11 €	16,731.50 €	6,048.80 €	13,557.90 €	0.00 €	-135.03 €
2032	6,411.94 €	16,242.82 €	5,559.39 €	13,125.99 €	0.00 €	-103.74 €
2033	6,053.57 €	15,233.60 €	5,251.07 €	12,319.37 €	0.00 €	-98.50 €
2034	5,442.69 €	14,251.50 €	4,728.81 €	11,534.04 €	0.00 €	-94.52 €
2035	4,765.57 €	12,891.04 €	4,147.57 €	10,455.93 €	0.00 €	-60.34 €
2036	4,310.54 €	11,757.02 €	3,753.37 €	9,540.27 €	0.00 €	-68.04 €
2037	3,248.98 €	9,927.13 €	2,842.14 €	8,090.86 €	0.00 €	-55.56 €
2038	2,795.23 €	8,995.43 €	2,449.02 €	7,336.00 €	0.00 €	-52.71 €
2039	2,517.98 €	8,106.51 €	2,209.23 €	6,616.24 €	0.00 €	-50.80 €
2040	2,671.81 €	8,102.43 €	2,339.56 €	6,605.64 €	0.00 €	-48.45 €
2041	2,498.78 €	7,702.58 €	2,170.99 €	6,247.81 €	0.00 €	-41.54 €
2042	2,796.09 €	7,833.17 €	2,426.66 €	6,357.64 €	0.00 €	-39.68 €
2043	2,788.23 €	7,590.32 €	2,417.31 €	6,159.00 €	0.00 €	-37.92 €
2044	2,931.62 €	7,451.74 €	2,535.59 €	6,037.39 €	0.00 €	-55.33 €
2045	2,554.87 €	6,839.90 €	2,214.90 €	5,548.44 €	0.00 €	-34.13 €
TOTAL	106,696.58 €	267,299.98 €	91,936.31 €	215,212.24 €	0.00 €	-2,026.75 €

Table 22: Annual Policy Framework Contribution by Configuration (2026-2045)

West-facing & east-facing 999 kWp configurations (W90_0.9MW, E90_0.9MW) demonstrate highest absolute contributions throughout horizon, maintaining 2.0-2.5× advantage over 1,001 kWp equivalents. This persistent differential confirms heightened bonus tier (25.0 €/MWh) creates substantially greater policy leverage than base tier (12.0 €/MWh) across all market conditions.

S30_1MW maintains exactly zero contribution across all 20 years, serving as pure market performance baseline. The S30_0.9MW consistent small negative contributions (-€34 to -€243 annually) reflect Anzulegender Wert assignment anomaly (70.0 vs 70.3 €/MWh disadvantage under Real scenario), producing cumulative -€2,027 policy penalty rather than benefit.

4.9.3 Policy Framework Viability Scenarios and Risk Assessment

Systematic policy evolution scenario analysis quantifies deployment viability under alternative regulatory futures. Table 223 presents comprehensive scenario outcomes across configuration performance and east-facing viability assessment.

Scenario	AW Structure	W90_0.9MW PV	W90_1MW PV	S30_1MW PV	W90_0.9MW vs S30 %	E90_0.9MW PV	E90 Viability
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Current Framework	95.0 / 82.3 / 70.3	€ 1,701,471	€ 1,431,768	€ 1,118,703	52.10%	€ 1,257,842	Marginal (+12.4%)
Reduced Bonus (-40%)	85.0 / 76.3 / 70.3	€ 1,578,451	€ 1,373,065	€ 1,118,703	41.10%	€ 1,159,081	Marginal (+3.6%)
Minimal Bonus (-70%)	77.5 / 73.6 / 70.3	€ 1,496,685	€ 1,349,960	€ 1,118,703	33.80%	€ 1,093,105	Negative (-2.3%)
Uniform High	95.0 / 95.0 / 95.0	€ 1,701,471	€ 1,577,725	€ 1,361,465	25.00%	€ 1,257,842	Severe Negative (-7.6%)
Uniform Medium	82.3 / 82.3 / 82.3	€ 1,547,176	€ 1,431,768	€ 1,221,101	26.70%	€ 1,133,865	Severe Negative (-7.1%)
Uniform Low (Basic)	70.3 / 70.3 / 70.3	€ 1,434,171	€ 1,325,071	€ 1,118,703	28.20%	€ 1,042,630	Negative (-6.8%)
No Support	0 / 0 / 0	€ 1,318,979	€ 1,216,304	€ 1,016,149	29.80%	€ 947,856	Negative (-6.7%)

Table 23: Policy Evolution Scenarios - Configuration Viability Assessment

Current framework maintenance (base case, 60% probability): Maximum configuration differentiation (+52.1% W90_0.9MW vs S30_1MW) with clear hierarchy demonstrating comprehensive west-facing superiority. East-facing achieves marginal commercial viability (+12.4% vs baseline) purely through policy leverage compensating for catastrophic generation failure documented in Section 4.1.4.

Reduced bonus scenarios (-40% to -70%): West-facing maintains substantial superiority (+33.8-41.1% vs S30) confirming robust advantage surviving significant policy degradation. The -40% reduction scenario (€1,578,451 W90_0.9MW PV) maintains +41.1% baseline premium, demonstrating resilience to moderate policy evolution.

Critical threshold crossing occurs between -40% and -70% bonus reduction for east-facing viability. E90_0.9MW transitions from marginal positive (+3.6% at -40% reduction) to negative (-2.3% at -70% reduction), demonstrating policy-dependent commercial viability threshold. The -60% reduction represents approximate viability boundary where catastrophic generation failure overwhelms policy framework compensation.

Uniform remuneration scenarios: Elimination of capacity threshold differentials compresses configuration advantages while revealing fundamental market performance hierarchies. Under uniform high (95.0 €/MWh for all), W90_0.9MW maintains +25.0% advantage purely from superior generation and market timing. However, E90_0.9MW experiences severe negative performance (-7.6% vs baseline) as uniform remuneration eliminates heightened bonus tier compensation for generation inadequacy.

The uniform medium scenario (82.3 €/MWh) demonstrates similar pattern: W90_0.9MW achieves +26.7% baseline premium through market performance while E90_0.9MW suffers -7.1% deficit. This confirms east-facing non-viability under any policy framework eliminating capacity-differentiated bonuses.

No policy support scenario (complete framework elimination, <5% probability): All configurations experience severe PV reductions (18-31% vs current framework) but maintain relative hierarchy. W90_0.9MW achieves €1,318,979 PV representing +29.8% vs S30_1MW (€1,016,149), confirming fundamental market performance advantages survive complete policy elimination.

The scenario demonstrates technical performance robustness while highlighting commercial viability dependence on baseline policy framework existence (even at zero agrivoltaic bonus levels, some base remuneration proves necessary).

East-facing policy dependency criticality: E90 viability demonstrates extraordinary sensitivity to policy framework structure:

- Current framework: +12.4% vs baseline (marginal viability)
- Reduced -40%: +3.6% vs baseline (barely viable)
- Reduced -70%: -2.3% vs baseline (non-viable threshold crossed)
- Any uniform scenario: -6.7% to -7.6% vs baseline (severely non-viable)
- No support: -6.7% vs baseline (non-viable)

The narrow viability window (positive only under current and modest -40% reduction scenarios) creates unacceptable regulatory risk. Any policy evolution beyond -40% bonus reduction eliminates east-facing commercial viability, confirming that deployment recommendations should categorically exclude E90 configurations regardless of current policy support availability.

West-facing policy robustness: W90 maintains commercial viability and substantial baseline premium across all examined scenarios:

- Current: +52.1% vs baseline (optimal)
- Reduced -70%: +33.8% vs baseline (strong)
- Uniform scenarios: +25.0-28.2% vs baseline (acceptable)
- No support: +29.8% vs baseline (marginal, absolute viability questionable)

This comprehensive robustness validates west-facing orientation as low-regulatory-risk strategy surviving virtually any policy evolution scenario short of complete framework elimination while maintaining commercial superiority over conventional alternatives. Even under pessimistic no-support scenario, W90 preserves +29.8% relative advantage, though absolute project economics require baseline remuneration framework existence.

Risk-adjusted viability assessment:

- **High confidence commercial viability** (>90% probability across policy scenarios): - W90_0.9MW: Viable under all scenarios, ranging +25% to +52% vs baseline - W90_1MW: Viable under all scenarios, ranging +19% to +28% vs baseline
- **Baseline viability** (certain but inferior returns): - S30_1MW: Policy-independent baseline, serves as reference (0% by definition) - S30_0.9MW: Similar to S30_1MW with marginal capacity penalties
- **Unacceptable regulatory risk** (<40% probability of viability): - E90_0.9MW: Viable only under current and -40% reduced scenarios (60% combined probability) - E90_1MW: Similar to E90_0.9MW with worse absolute performance
- **Categorical exclusion recommended:** Policy dependency unmanageable, fundamental performance inadequate

The viability stratification enables definitive deployment recommendations: pursue west-facing configurations with high confidence, accept south-facing as risk-averse baseline, and categorically exclude east-facing due to unacceptable policy dependency combined with catastrophic technical performance.

4.10 Comparative Summary

4.10.1 Configuration Performance Summary Matrix

Table 24 presents a comprehensive multi-dimensional performance summary enabling rapid configuration assessment across critical evaluation criteria with a clear three-tier stratification:

Tier 1 - Superior Performance: W90_0.9MW and W90_1MW dominate across all dimensions, achieving top rankings in generation (+2.5-7.3%), market timing (MVF 0.82-0.85), revenue realization (€102k-123k annually), and long-term economics (+28-52% PV vs baseline). Moderate policy dependency (7-16%) proves acceptable given a robust market performance foundation.

Tier 2 - Baseline Performance: S30_1MW and S30_0.9MW represent the conventional approach achieving acceptable commercial viability through policy independence (0% dependency) but sacrificing substantial returns (€0-€583k PV vs optimal).

Metric	W90_0.9MW	W90_1MW	E90_0.9MW	E90_1MW	S30_1MW (baseline)	S30_0.9MW
GENERATION PERFORMANCE						
Annual Generation (kWh)	1,176,631	1,124,120	948,440	945,927	1,096,238	1,090,107
vs Baseline %	7.30%	2.50%	-13.50%	-13.70%	0.00%	-0.60%
Performance Ratio (%)	81.9	81.7	70.3	70	84.8	84.5
Shading Loss (%)	2.86	2.45	19.93	19.6	9.06	9.28
MARKET TIMING						
Correlation (r)	-0.198	-0.242	-0.297	-0.297	-0.35	-0.35
Market Value Factor	0.85	0.82	0.76	0.75	0.7	0.68
Negative Price Exposure (%)	3.31	4.77	6.32	6.26	6.48	6.5
REVENUE OUTCOMES (2026)						
Spot Revenue (€)	83,557	76,949	60,081	60,558	63,959	63,534
Marktprämie (€)	39,650	24,981	32,076	21,623	13,225	12,941
Total Revenue (€)	123,207	101,930	92,157	82,181	77,184	76,475
Revenue/kWh (€cent/kWh)	10.58	9.16	9.81	8.78	7.11	7.09
LONG-TERM ECONOMICS						
PV Real Curtail (€)	1,701,471	1,431,768	1,257,842	1,142,834	1,118,703	1,109,509

PV Basic Curtail (€)	1,434,171	1,325,071	1,042,630	1,050,898	1,118,703	1,111,536
vs Baseline Real %	52.09%	27.98%	12.44%	2.16%	0.00%	-0.82%
vs Baseline Basic %	28.20%	18.45%	-6.80%	-6.06%	0.00%	-0.64%
POLICY DEPENDENCY						
Policy Contribution (€)	267,300	106,697	215,212	91,936	0	-2,027
Dependency Ratio	0.16	0.07	0.17	0.08	0.00	0.00
Classification	Moderate	Low	Moderate-High	Low	Independent	Independent
RANKING						
Overall Rank	#1	#2	#3 Real, #5 Basic	#4 Real, #6 Basic	Baseline	#5 Real, #4 Basic

Table 24: Comprehensive Configuration Performance Summary

Tier 3 - Questionable Viability: E90_0.9MW and E90_1MW demonstrate catastrophic generation failure (-13.5-13.7%), moderate-high policy dependency (8-17%), and scenario-dependent viability (positive Real, negative Basic). Unacceptable regulatory risk combined with poor technical performance necessitates categorical deployment exclusion

4.10.2 Key Findings and Definitive Recommendations

Primary Research Findings:

- West-facing vertical bifacial achieves dramatic superiority:** W90_0.9MW generates +7.3% more energy than south-facing baseline (Section 4.1.1) while realizing 52.1% higher present value through combined generation, market timing, and policy advantages (Section 4.5.1).
- Spot market revenue dominates economic outcomes:** 68-83% of total revenues derive from wholesale market performance with Marktprämie providing 17-32% supplementary support (Section 4.4.3), fundamentally revising understanding of German photovoltaic economics under EEG 2023 framework.
- Capacity threshold optimization creates extraordinary value:** Sub-1MW positioning generates €269,703 present value advantage for west-facing installations through €12.7/MWh remuneration differential (Section 4.7.2), with €109,000 technical value floor persisting even under complete policy elimination.
- East-facing demonstrates catastrophic non-viability:** 19.6% shading losses produce 70% performance ratio (Section 4.1.4) and policy-dependent viability requiring categorical deployment exclusion. East-facing achieves positive returns only under current and -40% reduced bonus scenarios (Section 4.9.3), creating unacceptable regulatory risk.
- Universal negative price-generation correlations confirmed:** All orientations experience negative correlations ($r = -0.198$ to -0.350) validated by 2024 historical data (Section 4.3.1), with west-facing achieving “least-negative” -0.198 representing superior but imperfect market timing.
- Three-phase negative pricing evolution identified:** Non-monotonic pattern with escalation 2026-2030 (334 hours peak), stabilization 2031-2040 (134 hours minimum), and

resurgence 2041-2045 (312 hours) creates path-dependent revenue impacts (Section 4.2.3).

- g) **Moderate policy framework dependency proves manageable:** Maximum dependency configurations (W90_0.9MW: 15.7% of total PV, Section 4.9.1) maintain commercial viability through market performance with policy enhancement rather than dependence, confirmed through scenario analysis demonstrating +25-52% baseline premium across all examined regulatory futures (Section 4.9.3).

Definitive Deployment Recommendations:

TIER 1 - STRONGLY RECOMMENDED:

- **W90_0.9MW:** Deploy as primary configuration for maximum economic returns (+52.1% vs baseline under current framework, Section 4.5.1). The configuration combines optimal orientation (west-facing vertical bifacial), capacity threshold positioning (999 kWp accessing heightened bonus tier), and inverter sizing optimization (1.15 DC/AC ratio) creating €1,701,471 present value. Policy contribution represents 15.7% of total value (€267,300) with €1,434,171 market performance baseline ensuring viability across regulatory scenarios including +28.2% baseline premium under uniform low remuneration (Section 4.9.3).
- **W90_1MW:** Deploy as alternative west-facing configuration sacrificing capacity threshold optimization for simplified qualification. Achieves €1,431,768 present value (+28.0% vs baseline, Section 4.5.1) with reduced 7.5% policy dependency (€106,697 contribution, Section 4.9.1). Optimal for investors prioritizing regulatory resilience over maximum absolute returns, maintaining strong commercial viability (+19-28% vs baseline) across all examined policy scenarios (Section 4.9.3).

TIER 2 - ACCEPTABLE WITH LIMITATIONS:

- **S30_1MW:** Deploy only if extraordinary risk aversion justifies €582,768 present value sacrifice versus W90_0.9MW optimal (Section 4.5.1). Achieves €1,118,703 PV representing baseline reference with complete policy independence (0.0% dependency, Section 4.9.1). Conventional 30° south-facing tilted array provides proven technology minimizing deployment risk while accepting substantially inferior economic returns. Rational selection for conservative institutions prioritizing capital preservation over return maximization.
- **S30_0.9MW:** Similar characteristics to S30_1MW with marginal -€9,194 capacity penalty (Section 4.5.1). Unusual Anzulegender Wert assignment (70.0 vs 70.3 €/MWh) creates small negative policy contribution (-€2,027, Section 4.9.1). No compelling advantage over S30_1MW; select only if capacity minimization required for site-specific constraints.

TIER 3 - NOT RECOMMENDED (CATEGORICAL EXCLUSION):

- **E90_0.9MW:** Categorically exclude despite +12.4% Real scenario premium (Section 4.5.1). Catastrophic -13.5% generation deficit from 19.6% shading losses (Section 4.1.4) creates fundamental commercial inadequacy compensated only through extraordinary policy support (21% policy dependency, €215,212 contribution, Section 4.9.1). Viability threshold crossing occurs at -60% bonus reduction (Section 4.9.3), with negative baseline performance under all uniform remuneration scenarios (-6.8% to -7.6% deficit).

Regulatory risk proves unmanageable with commercial viability contingent on sustained heightened agrivoltaic bonus tier unlikely across 20-year horizon.

- **E90_1MW**: Categorically exclude with worse absolute performance than E90_0.9MW across all metrics. Achieves only +4.1% Real scenario premium (Section 4.5.1) declining to -6.7% deficit under uniform low scenario (Section 4.9.3). No scenario justifies deployment given west-facing alternative superiority across all dimensions combined with catastrophic east-facing technical performance.

Orientation optimization (€315,000-350,000 PV): Select west-facing vertical bifacial over conventional south-facing tilted arrays. Provides 54% of total W90_0.9MW advantage vs S30_1MW through superior generation (+7.3%, Section 4.1.1), market timing (MVF 0.85 vs 0.70, Section 4.4.1), and negative price avoidance (3.31% vs 6.48% ineligible, Section 4.2.2).

- a) **Capacity threshold positioning** (€269,703 PV total, €109,100 technical minimum): Design to 999 kWp maximum if west-facing selected. Provides 46% of advantage through policy leverage (€160,603) plus technical optimization (€109,100 inverter sizing benefit, Section 4.7.2).
- b) **Curtailement strategy** (€12,727-20,395 PV): Implement operational curtailment during negative price hours if cost-effective. Provides 2.2% of total optimization value (Section 4.8.2), representing marginal enhancement rather than fundamental driver.

The priority ranking demonstrates that configuration selection (orientation + capacity threshold) drives 98% of economic optimization value, with operational strategies (curtailment) providing <3% incremental benefit. Deployment decisions should prioritize fundamental configuration choices over operational refinements.

Final Recommendation: Deploy **W90_0.9MW** (west-facing vertical bifacial, 999 kWp, 1.15 DC/AC inverter sizing) as optimal configuration for agrivoltaic photovoltaic installations under German EEG 2023 framework. The integrated optimization achieves maximum economic returns (+52.1% vs conventional S30_1MW baseline) through combined superior generation, market timing, policy leverage, and negative price avoidance while maintaining acceptable regulatory risk profile (15.7% policy contribution with €1,434,171 market performance baseline ensuring +28.2% minimum advantage under worst-case uniform low scenario). Alternative **W90_1MW** proves acceptable for risk-averse deployment accepting €269,703 upside sacrifice for reduced policy dependency. **Categorically exclude** east-facing configurations (E90_0.9MW, E90_1MW) demonstrating catastrophic technical performance, unacceptable policy dependency, and fundamental commercial non-viability under any regulatory scenario eliminating capacity-differentiated agrivoltaic bonuses.

5 Discussion

5.1 Interpretation of Key Findings

West-facing vertical bifacial systems (W90_0.9MW) achieve 52.1% higher present value (€1,701,471) than conventional south-facing configurations (S30_1MW: €1,118,703), demonstrating extraordinary economic advantage through multiplicative compounding across four distinct mechanisms rather than single-factor dominance.

The +7.3% generation advantage (1,176,631 kWh versus 1,096,238 kWh) contradicts conventional optimization principles identifying 30° south-facing tilt as maximizing annual irradiance for Ansbach latitude (49.3°N). This superiority reflects bifacial rear-side irradiance contributions (30-35% of total generation) from ground reflection and sky diffuse radiation, combined with minimal inter-row shading (2.86% versus 9.06% for south-facing). Vertical orientations eliminate rear-side row shading while achieving balanced view factors, validating Pike et al. (2021) documentation of comparable vertical versus tilted annual generation while extending empirical evidence to mid-latitude continental climates.

Market timing advantages (market value factor 0.85 versus 0.70) provide critical differentiation, with bimodal west-facing generation (morning 718 kW + afternoon 703 kW peaks) partially avoiding midday price suppression (20-35 €/MWh, hours 11-14) when south-facing peaks. This temporal redistribution directly addresses solar value deterioration (Jomaux, 2023; Lion Hirth, 2013), though the persistent 15% market value penalty confirms photovoltaic economics increasingly depend on policy support rather than pure market competitiveness.

Reduced negative price exposure (3.31% versus 6.48% ineligible generation) preserves approximately 35 MWh additional eligible generation annually, worth €1,750-2,000 in Marktprämie. Policy framework leverage (€267,300 contribution, 15.7% of total value) amplifies technical advantages through capacity-dependent Anzulegender Wert differentiation, creating multiplicative rather than additive value.

East-facing configurations demonstrate catastrophic non-viability (948,440 kWh, -13.5% versus baseline; 70% performance ratio) despite theoretical symmetry with west-facing, revealing fundamental orientation asymmetry from three factors: afternoon atmospheric clarity advantages (8-12% DNI superiority documented in 2024 DWD data), temperature-irradiance interaction effects, and row spacing optimization (11 meters adequate for west but creating 19.6% east-facing shading). Remediation requires 50-80% spacing increases (15-20 meters), compromising agricultural productivity fundamental to agrivoltaic value propositions.

Policy dependency further compounds technical inadequacy where E90_0.9MW achieves marginal baseline premium exclusively through extraordinary agrivoltaic bonus leverage (€215,212 representing 17.1% of total value), with viability disappearing under any policy evolution reducing bonuses >60% or eliminating capacity-differentiated remuneration. This fragile viability window creates unacceptable regulatory risk justifying categorical deployment exclusion

5.2 Comparison with Existing Literature

The universal negative price-generation correlations ($r = -0.198$ to -0.350) validate Hirth's (2013) market value theory in contemporary German contexts, demonstrating that generation timing determines economic value independent of absolute quantity. Orientation-dependent market value factors (W90: 0.85, E90: 0.76, S30: 0.70) reveal optimization provides meaningful but incomplete mitigation & even optimal west-facing configurations realize only 85% of market-neutral revenue potential. The finding extends Jomaux's (2023) documentation of accelerating solar value decline, with negative pricing intensification (457 hours in 2024-2025, concentrated 11:00-15:00) creating direct revenue penalties through EEG §51 provisions.

Operational validation from Donaueschingen (1,250 kWh/kWp achieved versus 1,183 kWh/kWp projected, +5.67% outperformance) confirms PV*SOL bifacial modeling produces conservative projections. However, the multi-orientation framework reveals east-facing non-viability under identical agrivoltaic constraints (11-meter spacing, DIN SPEC 91434 compliance) absent from single-installation documentation. Pike et al. (2021) document comparable east-west versus south-facing generation in Arctic conditions (69°N), while current analysis reveals mid-latitude climates (49°N) demonstrate different hierarchies: west-facing +7.3% advantage, east-facing -13.5% catastrophic deficit, reflecting latitude-dependent seasonal irradiance distribution.

The capacity threshold analysis contributes novel empirical evidence on threshold-based incentive impacts, demonstrating €269,703 optimization value substantially exceeds typical assumptions. The orientation-dependent variation (W90: +€269,703, S30: -€9,194) reveals unintended configuration-specific consequences requiring policy evaluation of whether frameworks create desired patterns or arbitrary boundaries.

5.3 Practical Implications and Deployment Recommendations

Capacity threshold effects demonstrate 12.7 €/MWh differential creates substantial distortions, with €269,703 threshold value representing 18.8% of total W90 PV potentially overwhelming technical considerations. Reform options include: gradually reducing differential toward 5-7 €/MWh maintaining incentives while minimizing arbitrage, implementing continuous remuneration curves eliminating discrete boundaries, or establishing orientation-specific bonuses explicitly recognizing market timing (5-10 €/MWh for MVF 0.85 versus 0.70).

Negative pricing provisions (EEG §51) create appropriate signals incentivizing generation profile optimization, with differential impacts (S30 experiencing 2× higher ineligibility) rewarding temporal distribution. Enhanced DIN SPEC 91434 enforcement requires third-party verification, long-term monitoring (5-10 years), and crop-specific benchmarks preventing initial compliance followed by agricultural abandonment.

Bifacial module advancement toward 90-95% bifaciality (HJT technology) would enhance vertical configuration rear-side contributions by 3-4% (€25,000-30,000 PV), though premium costs (+15-20%) currently exceed benefits. Energy storage integration represents most significant opportunity, which is 2-4 hour capacity (400-800 kWh) could improve market value factor from 0.85 to 0.95-1.00, capturing evening peaks (100-125 €/MWh) through stored afternoon generation. Current battery costs (€300-400/kWh) prove prohibitive, with break-even projected €200/kWh (2027-2029) creating retrofit opportunity.

Advanced forecasting and automated market participation through machine learning-based generation prediction could enhance revenue capture beyond current passive Direktvermarktung exposure, potentially doubling or tripling operational optimization value (currently €12,727-20,395 PV from simple curtailment) through precise timing of generation, storage, and curtailment decisions.

5.4 Limitations

Representative year meteorological data employing single 2024 dataset abstracts from inter-annual weather variability (documented $\pm 8\text{-}12\%$ generation fluctuation), though sensitivity analysis suggests orientation rankings remain stable across scenarios with absolute magnitudes varying. Wholesale electricity price projections maintaining 2024 temporal patterns assume current structures persist, yet market transformation through storage deployment and demand flexibility could fundamentally alter price formation.

Geographic transferability requires site-specific validation accounting for latitude variations (southern German 47-48°N versus northern 53-54°N affecting orientation hierarchies), atmospheric conditions (continental versus maritime climates affecting afternoon clarity advantages), and policy framework differences. The examined 11-meter row spacing proves optimal for west-facing while inadequate for east-facing, with alternative applications potentially employing different spacing creating altered relationships.

Agrivoltaic qualification assumptions treat DIN SPEC 91434 compliance as achieved certainty based on operational validation, yet actual agricultural productivity depends on crop selection, management practices, and microclimate interactions potentially varying across implementations. Enhanced verification requirements or productivity monitoring obligations could affect long-term revenue realization below modeled projections

5.5 Future Research Directions

Storage integration economics warrant priority investigation given transformative market value potential, examining optimal sizing for alternative orientations, dynamic charge-discharge strategies, combined storage-curtailment coordination, and business models including shared storage or virtual power plant aggregation. Declining costs approaching viability (<€200/kWh projected 2027-2029) create a time-sensitive research opportunity.

Market evolution modeling incorporating storage penetration, sector coupling (power-to-heat, power-to-hydrogen), and demand flexibility would generate realistic price trajectories improving long-term projections. Agent-based simulation with high renewable penetration, equilibrium modeling of competitive storage entry, and probabilistic negative pricing forecasting would enable risk-adjusted evaluation beyond deterministic projections.

Geographic variation analysis systematically examining orientation performance across diverse German locations (coastal, alpine, urban, rural) and European contexts (Mediterranean, Nordic, continental) would establish site-specific guidance currently absent. Multi-site empirical validation through coordinated demonstration projects would generate data enabling refined modeling and deployment optimization. Policy evolution scenario analysis examining threshold

modifications, bonus restructuring, and alternative remuneration mechanisms would strengthen adaptive guidance under regulatory uncertainty while informing design recommendations.

6 Conclusion

6.1 Principal Findings

This thesis conducted comprehensive economic comparison of vertical bifacial photovoltaic orientations (90° west-facing, 90° east-facing, 30° south-facing) under German EEG 2023 agricultural frameworks, employing validated PV*SOL simulation generating hourly profiles for six configurations integrated with wholesale electricity prices and Marktprämie mechanisms calculating 20-year present values.

Five principal findings emerge:

West-facing vertical bifacial achieves dramatic superiority: W90_0.9MW delivers €1,701,471 present value (+52.1% versus S30_1MW baseline €1,118,703) through multiplicative compounding which are generation advantage (+7.3%), market timing optimization (MVF 0.85 versus 0.70), reduced negative price exposure (3.31% versus 6.48%), and policy framework leverage (€267,300 contribution, 15.7% of value). The multiplicative relationship $(1.073) \times (1.214) \times (1.157) = 1.507$ demonstrates geometric value growth substantially exceeding additive improvements.

Spot market revenue dominates: 68-83% of revenues derive from wholesale market performance with Marktprämie providing 17-32% supplementary support. Well-designed installations achieve commercial viability primarily through market performance (€1,434,171 W90_0.9MW baseline, 84.3% of value) with policy providing enhancement rather than exclusive enablement.

East-facing demonstrates catastrophic non-viability: achieving only 948,440 kWh (-13.5% versus baseline) and 70% performance ratio through 19.6% shading. Orientation asymmetry reflects afternoon atmospheric clarity advantages, temperature-irradiance interactions, and row spacing optimization (11 meters adequate for west but inadequate for east). Policy dependency creates fragile viability disappearing under any regulatory evolution reducing bonuses >60%, justifying categorical deployment exclusion.

Capacity threshold optimization creates extraordinary value: 999 kWp positioning achieves €269,703 present value premium over 1,001 kWp equivalent despite 0.2% less capacity, decomposing into technical component (€109,100, 40% from inverter sizing) and policy component (€160,603, 60% from 12.7 €/MWh Anzulegender Wert differential). Technical value floor persists independent of policy frameworks, providing €108,964-109,100 minimum benefit surviving complete bonus elimination.

Universal negative correlations ($r = -0.198$ to -0.350) validate solar value deterioration theory, with orientation optimization providing meaningful but incomplete mitigation. West-facing achieves “least-negative” -0.198 creating MVF 0.85, substantially improving over south-facing -0.350 producing MVF 0.70, yet falling short of market-neutral performance.

6.2 Key Contributions

The research advances renewable energy economics through **market value theory validation** providing systematic empirical evidence on orientation-dependent market value capture, confirming photovoltaic systems experience systematic market timing disadvantages while revealing optimization provides meaningful but incomplete mitigation. **Capacity threshold analysis** contributes novel evidence documenting €269,703 optimization value with orientation-dependent variation (W90: +€269,703, S30: -€9,194) revealing unintended configuration-specific policy consequences. **Revenue composition analysis** demonstrates market performance contributing 84.3% of W90_0.9MW value with policy providing 15.7% enhancement, repositioning German frameworks as successful market integration mechanisms.

Practical contributions deliver definitive deployment guidance: W90_0.9MW for maximum returns (+52.1%, €1,701,471 PV, 12-15% IRR), categorical east-facing exclusion preventing resource allocation toward marginally viable configurations, and investment priority ranking (orientation €315,000-350,000 > threshold €270,000 > curtailment €13,000-20,000) demonstrating configuration selection drives 98% of economic optimization value.

6.3 Deployment Recommendations

TIER 1 - STRONGLY RECOMMENDED

W90_0.9MW (West-facing vertical bifacial, 999 kWp): Deploy as primary configuration achieving €1,701,471 present value (+52.1% baseline premium) through integrated optimization. Demonstrates robust regulatory resilience maintaining +28.2% minimum advantage under worst-case uniform remuneration. **Critical implementation:** Capacity verification 998.5-999.0 kWp; inverter specification 1.15 DC/AC ratio; row spacing 10-12 meters; west-facing azimuth $\pm 10^\circ$; bifacial modules 80%+ bifaciality. **Expected economics:** 12-15% IRR, annual revenues €85,000-95,000 (years 1-5) declining to €60,000-70,000 (years 16-20).

W90_1MW (West-facing vertical bifacial, 1,001 kWp): Deploy as alternative prioritizing regulatory resilience achieving €1,431,768 PV (+28.0% baseline) while reducing policy dependency to 7.5%. The €269,703 PV sacrifice purchases regulatory resilience through reduced threshold arbitrage reliance. Optimal for risk-averse institutional investors.

TIER 2 - ACCEPTABLE WITH LIMITATIONS

S30_1MW (South-facing tilted, 1,001 kWp): Deploy only if extraordinary risk aversion justifies €582,768 PV sacrifice versus W90_0.9MW optimal. Achieves €1,118,703 PV with complete policy independence (0% dependency) and proven technology. Provides acceptable returns (8-10% IRR) while accepting substantially inferior performance. **Appropriate contexts:** Conservative institutional investors requiring zero policy dependency; sites experiencing regulatory uncertainty; initial agrivoltaic proof-of-concept deployments.

TIER 3 - NOT RECOMMENDED (CATEGORICAL EXCLUSION)

E90_0.9MW and E90_1MW (East-facing vertical bifacial): Categorically exclude despite marginal +12.4% Real scenario premium. Catastrophic technical performance (-13.5%

generation, 19.6% shading, 70% PR) creates fundamental inadequacy compensated exclusively through extraordinary policy support (€215,212, 17.1% of value). Viability threshold crossing at -60% bonus reduction with negative baseline performance (-6.8% to -7.6%) creates unacceptable regulatory risk.

Investment Priority Ranking

- **Orientation optimization (€315,000-350,000 PV):** Select west-facing vertical bifacial. Provides 54% of total advantage through superior generation, market timing, and negative price avoidance.
- **Capacity threshold positioning (€270,000 PV):** Design to 999 kWp accessing heightened bonus tier while enabling inverter optimization. Provides 46% of advantage through dual mechanisms (60% policy + 40% technical) with €109,000 minimum assured benefit.
- **Inverter sizing optimization (€109,000 PV):** Employ 1.15 DC/AC ratio for 999 kWp configurations. Represents technical threshold value floor persisting independent of policy frameworks.
- **Curtailment strategy (€13,000-20,000 PV):** Implement operational curtailment during negative price hours. Provides modest 0.75-1.83% revenue enhancement representing lowest-priority optimization.

Configuration selection (orientation + threshold: 98% of optimization value) drives economic performance, with operational strategies (curtailment: 2%) providing marginal enhancement.

6.4 Policy Implications and Future Research

Current agrivoltaic frameworks prove effective driving deployment (40+ MWp operational) while maintaining market integration.

Recommended refinements: Capacity threshold modification gradually reducing differential from 12.7 €/MWh toward 5-7 €/MWh maintaining incentives while minimizing distortions; orientation-specific bonus differentiation (5-10 €/MWh for MVF 0.85 versus 0.70) rewarding market timing; enhanced DIN SPEC 91434 verification implementing third-party monitoring and long-term requirements (5-10 years).

Priority research directions: Storage integration economics examining optimal sizing and dynamic strategies as declining costs approach viability (<€200/kWh projected 2027-2029); market evolution modeling incorporating storage penetration and demand flexibility generating realistic price trajectories; geographic variation analysis systematically examining orientation performance across diverse locations establishing site-specific guidance.

Representative year projections abstract from inter-annual weather variability (± 8 -12% generation fluctuation), though orientation rankings remain stable. Price evolution uncertainty maintains 2024 temporal patterns, yet market transformation could alter formation mechanisms. Despite limitations, findings such as west-facing superiority (+52.1%), east-facing catastrophic failure (70% PR requiring exclusion), capacity threshold extraordinary value (€269,703), market revenue

dominance (68-83%) prove sufficiently robust across reasonable assumption variations to support confident deployment recommendations.

6.5 Final Conclusions

This thesis establishes **west-facing vertical bifacial photovoltaic systems as commercially optimal configuration for German agrivoltaic installations**, achieving 52.1% present value premium over conventional south-facing alternatives through multiplicative advantage compounding. Extraordinary economic superiority combined with robust regulatory resilience (maintaining +28.2% minimum advantage under worst-case scenarios) positions vertical bifacial technology as definitive deployment recommendation for mid-latitude continental European climates pursuing dual agricultural-electricity land use.

Capacity threshold positioning at 999 kWp creates substantial optimization opportunity generating €269,703 advantage through combined technical enhancement (€109,000 inverter sizing) and policy leverage (€160,603 bonus differential), with 40% technical floor providing downside protection distinguishing capacity optimization from pure policy arbitrage.

East-facing categorical exclusion based on catastrophic performance (19.6% shading producing 70% PR) and fragile policy dependency prevents resource allocation toward marginally viable configurations. Orientation asymmetry (west-facing achieving +7.3% advantage while east-facing suffers -13.5% deficit) establishes vertical bifacial advantages as orientation-specific rather than universal benefits.

Spot market revenue dominance (68-83% of total) with Marktprämie providing supplementary support (17-32%) demonstrates well-designed installations achieve commercial viability primarily through wholesale market performance, positioning German frameworks as successful market integration mechanisms enabling economic feasibility through moderate support levels.

For **developers and investors**: pursue W90_0.9MW as primary configuration for maximum returns (+52.1%, €1,701,471 PV, 12-15% IRR), accept W90_1MW as risk-averse alternative (€1,431,768 PV, +28.0%), utilize S30_1MW only if extraordinary risk aversion justifies €582,768 sacrifice (€1,118,703 baseline, 8-10% IRR), and categorically exclude east-facing configurations demonstrating catastrophic failure.

For **policymakers**: current frameworks prove effective while requiring refinements including threshold differential reduction (12.7 → 5-7 €/MWh), orientation-specific bonuses (5-10 €/MWh), and enhanced agricultural verification.

For **agricultural producers**: west-facing vertical bifacial technology provides compatible dual-use solution maintaining 90% land accessibility while generating substantial electricity revenues (€50,000-70,000 annual landowner payments) enhancing farm economics.

The future of German agrivoltaic deployment lies in systematic adoption of west-facing vertical bifacial technology, informed by comprehensive analysis establishing this configuration as commercially superior pathway toward sustainable renewable energy expansion reconciling climate objectives with agricultural land preservation imperatives.

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